COVERAGE-INVERSION: CALIBRATION-TRANSPARENT FAIR-VALUE ORACLES FOR CLOSED-MARKET HOURS

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Abstract

Tokenised real-world assets on public blockchains trade continuously, but their underlying price-discovery venues do not — leaving roughly two-thirds of wall-clock time during which on-chain protocols must value collateral against a closed reference market. We invert the conventional point-plus-confidence oracle interface: target coverage $\tau \in (0, 1)$ is a published, parametric product input, the served band carries a per-read calibration receipt independently verifiable from public data, and a locally-weighted Mondrian split-conformal architecture maps $(s, \rho(\mathcal{F}_t), \hat{\sigma}_s(t), \tau)$ to a calibrated band around a factor-adjusted point. We evaluate on two closed-market panels over the same ten US-listed tickers (2014–2026): 5,996 weekend (Friday-close → Monday-open) rows and 22,624 overnight (close → next-open) rows. The headline weekend calibration claim is held-out via 10-fold leave-one-symbol-out CV at $\tau = 0.95$: realised coverage 0.9497 ± 0.0128, all 10 LOSO bands pass Kupiec. On a 40-cell (symbol × τ) grid jointly evaluated under Kupiec UC, Christoffersen IND, and Engle–Manganelli DQ, the architecture carries **40/40 Kupiec, 38/40 Christoffersen, and 39/40 DQ passes** — leading the strongest practitioner baseline GARCH-t (31/40, 37/40, 31/40) on every metric. **The architecture generalises from weekends to overnight gaps with only its gap selector changed:** on the 22,624-row overnight panel, pooled out-of-sample coverage passes both Kupiec and Christoffersen at all four anchors and is robust under moving-block-bootstrap — establishing calibration-transparency as a property of closed-market hours generally, not of weekends. Overnight admits a regime weekends cannot express, `earnings_night`, yielding **calendar-conditioned coverage**: because an earnings release is a scheduled, publicly dated event, the band widens deterministically ahead of it — which no incumbent oracle does — and stays calibrated despite an earnings-night tail roughly 8× the normal-night scale. **Pooled DQ at $\tau = 0.95$ rejects** when violations are concatenated across symbols in panel-row order: bands are *per-anchor calibrated*, not full-distribution calibrated, and the residual is a within-weekend cross-sectional cluster topology orthogonal to $\hat{\sigma}_s$. The §10 next-generation roadmap targets that residual; the §6.3.4 joint-tail empirical distribution turns it into a measurable reserve-guidance threshold ($k^* = 3$ at $\tau = 0.95$, 0/24 stability rejections). Reference implementation open-sourced (Python + Rust + Anchor; §8).

1 Introduction

Tokenised equities and other real-world assets (RWAs) on public blockchains trade 24/7. Their underlying price-discovery venues do not. A US equity listed on NASDAQ is open 6.5 hours per weekday — roughly 32% of wall-clock time. In the other 68% the price continues to move (earnings, macro prints, overseas sessions) but the venues that establish reference prices are closed. The off-hour token process is *dual* in nature. Cong et al. [Cong et al., 2025] document both faces on the same panel: weekend token returns predict the close-to-open underlying return at $\lambda = 0.903$ with adjusted $R^2 = 0.839$ [Table 10, p.40] — a real conditional-mean signal — *and* extreme moves on token venues "likely would have triggered halts in regulated markets" [cong-tokenized-2025, pp.6–7], because circuit breakers and other investor protections absent from regulated venues are also absent from token venues regardless of the cause of the print. A naive last-trade feed forwards both the signal and the shock; a naive smoother destroys both. The

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JELLYJELLY incident on Hyperliquid (March 2025) drove ~$12M of HLP losses through a manipulable single-venue oracle path; a Volcano Exchange post-mortem of a tokenised-asset perp episode documents $9.21M in Hyperliquid liquidations exceeding the Binance volume that fed the oracle, and identifies the propagation mechanism in plain language: *uncertainty present in the input was not represented in the output*. The structural fix is therefore dual: an oracle whose product contract passes the conditional-mean signal through as its centre while bounding shocks inside a coverage-calibrated width — making the input uncertainty a first-class output a downstream consumer can read on every block, and absorbing prints that exceed historical conditional bounds regardless of the cause of those prints. Any DeFi protocol that accepts tokenised RWAs as collateral or as inputs to an automated market maker must answer at every block: *with the underlying venue closed, what is the fair price — and how uncertain is it?*

The question recurs across the broader RWA wave moving on-chain in 2025–2026: tokenised equities (xStocks on Solana, Backed-issued tokens), treasuries (BlackRock BUIDL, Ondo OUSG; >$5B AUM as of Q1 2026), commodities (Paxos PAXG), and the next wave of credit and FX. Every asset class with a continuous off-hours information set inherits the same closed-market-pricing problem and benefits from the same calibration-transparent risk-reporting primitive. The methodology fits asset classes with such a signal; it does not fit instruments without one (illiquid commodities, pure NAV real estate). §9.8 enumerates the generalisation map.

The closed-market window is not only the weekend. It is every trading night — and §6.8 shows the same architecture, with only its gap selector changed, calibrates on overnight (close → next-open) gaps as well, so the contribution is calibration-transparency across *closed-market hours* generally rather than weekends specifically. The overnight cadence also surfaces the sharpest, and uniquely *schedulable*, instance of the closed-market information problem: the earnings night. Because an earnings release is a publicly dated event, the band can widen for it deterministically and in advance — **calendar-conditioned coverage** (§4.3.1), a property no incumbent oracle exposes — and the underlying-side earnings tail we measure is the same value-relevant-news mechanism Cong et al. document on the token side.

1.1 How existing oracles answer, and what they leave unanswered

No incumbent RWA oracle deployed at scale today publishes a calibration receipt verifiable against public data at the aggregate feed level — a statement of the form *"over N historical prediction windows, our 95% band contained the realised price N_{in} times."* Six archetypes co-exist on Solana and EVM:

Archetype	Representative	What a consumer sees
Stale-hold	Chainlink Data Streams v8 RWA Standard via Kamino Scope `Open` feed [Labs, 2025b, Finance, 2026]; Pyth core outside session [Network, 2021]	Last underlying price held forward + `marketStatus` enum; refresh attempts fail with `OutsideMarketHours` from Friday 16:00 ET to Monday 09:30 ET
Bounded-deviation off-hours	Chainlink v8 RWA Standard via Kamino Scope `AllUpdates` feed [Finance, 2026]	Off-hours `midPrice` accepted subject to a fixed ±500 bps clamp around the `Open` feed's Friday-close value; no `bid`/`ask`/confidence on the wire; refuses moves that exceed the clamp
Synthetic-marker	Chainlink Data Streams v11 [Labs, 2026]	Weekend `bid`/`ask` with a `.01` suffix at 100% incidence on SPYx/QQQx/TSLAx — generated bookends, not real quotes
Publisher-dispersion metric	Pyth regular [Network, 2021, 2022]	σ across permissioned publishers; not an aggregate coverage claim; collapses to near-zero at venue close
Executable-overnight	Pyth Pro / Blue Ocean [Network, 2026, 2025]	Real 24/5 book Sun–Thu 8 PM – 4 AM ET; does not cover the canonical xStock weekend; paid enterprise
Continuous tokenized-tracking	Chainlink v10 `tokenizedPrice` [Labs, 2025a]; CF Benchmarks xStocks Indices [Benchmarks, 2026b]; RedStone Live [RedStone, 2026]; SEDA USA500 [Protocol, 2026]	Single scalar derived from 24/7 crypto-venue order books; no width on the wire; methodology disclosure ranges from BMR-regulated (CFB) to qualitative (RedStone Live)

None of these surfaces emits a calibrated coverage band, so a coverage-vs-sharpness numerical comparison is not well-defined against them; the contribution we make is therefore categorical, not bandwidth-comparative — a verifiable per-read calibration receipt. Pyth's documentation [Network, 2022] recommends *individual publishers* calibrate to ~95%, but this is per-publisher self-attestation, not a property of the aggregate.

The six archetypes also stratify on a second axis: **which price** they publish. The stale-hold (underlying side) and publisher-dispersion archetypes track the underlying ticker (`SPY/USD`, `TSLA/USD`) and freeze whenever the underlying venue closes; the synthetic-marker, bounded-deviation, executable-overnight, and continuous tokenized-tracking archetypes track the wrapped SPL or its crypto-venue proxy and stay live but on a structurally different instrument. Cong et al. [cong-tokenized-2025, Table 4, p.32] document that for TSLAx — the Backed SPL Kamino prices — the tokenized price deviates from the last underlying close by more than 1% in 71% of weekday off-hours and 15% of weekend hours; by more than 5% in 12% of weekday off-hours, beyond the static ±500 bps clamp the bounded-deviation archetype enforces. The wedge is not transient: Cong et al. [§3.3.3, p.17; §6.1, pp.41–42] attribute its persistence to issuer-controlled redemption windows, KYC/AML constraints, and primary-arbitrageur geofencing — structural market-design features that will not be competed away as the market scales.

Cong et al. also establish the strongest empirical claim a tokenized-tracking oracle can credibly make: a 1% off-hour token return predicts a 0.903% close-to-open underlying return with adjusted $R^2 = 0.839$ [cong-tokenized-2025, Table 10, p.40]. This is a conditional-*mean* statement, not a conditional-*quantile* statement, and it is regime-dependent — token returns are informative on weekends but reversal-driven on weekday overnights (Table 8, p.38: highest-minus-lowest-quintile next-day spread of -0.99%, $t \approx -2.7$). The tokenized-tracking archetypes therefore implicitly publish a point estimate of the conditional mean while emitting no width and giving the consumer no signal of which regime they are in; the underlying-tracking archetypes publish a stale last-NYSE-print proxy that ignores the off-hours signal entirely. Neither population gives a risk-critical consumer the conditional-quantile receipt their consumer-facing SLA requires.

1.2 Coverage as a first-class API parameter

The design point of this paper is an oracle whose *product contract* is a calibration statement. Instead of publishing a single band and leaving its coverage level implicit, we treat target coverage $\tau \in (0, 1)$ as a published, parametric product input and serve the band that has *empirically* delivered coverage τ on a rolling historical sample, stratified by asset, by pre-publish regime, and rescaled by a per-symbol pre-Friday scale. The empirical calibration claim is made at four audited anchors $\mathcal{T}_{\text{anch}} = \{0.68 \approx 1\text{-}\sigma,\ 0.85,\ 0.95,\ 0.99\}$ — the lower anchor is set at the approximate one-standard-deviation point so consumers comparing against a Gaussian default have a familiar reference; the API accepts any $\tau \in [0.68, 0.99]$ and linearly interpolates the per-regime quantile and the $c(\tau)$ bump between anchors, so an off-anchor τ inherits an *interpolated* (not separately validated) band — see §9.10 for the disclosure (denser-anchor-grid follow-up tracked in `docs/ROADMAP.md`). The deployed wire format publishes one band per symbol per Friday with τ stamped on the receipt.

Two properties follow that are absent from incumbents: (1) **Auditability** — the deployment artefact (per-Friday parquet + sidecar) is a deterministic function of public data; any third party can reconstruct it and verify a served band. (2) **Served-coverage receipts** — every read exposes (τ, $c(\tau)$, $q_r(\tau)$, $\hat{\sigma}_s(t)$, r) alongside the band, so a protocol integrator can log it, backtest, and challenge the provider if realised rate drifts. The framework traces to Christoffersen [Christoffersen, 1998] and SR 11-7 [of Governors of the Federal Reserve System & Office of the Comptroller of the Currency, 2011] / SR 26-2 [Board of Governors of the Federal Reserve System, 2026] model-risk guidance; what is new is the application to a decentralised oracle with a consumer-facing SLA on realised coverage rather than on point accuracy, and the per-symbol locally-weighted standardisation that closes per-symbol calibration on heterogeneous asset panels.

The binding consumer set is narrower than the surface set of "any DeFi protocol holding tokenised RWA." Three categories carry structural off-hours underlying-side exposure that perp-settlement-marked-to-tokenized-spot consumers do not. **(i) Lending protocols** accepting tokenized equities as collateral and lending USDC against them, where solvency is determined by the future redemption value of the collateral rather than its current crypto-venue quote — Kamino Finance's TSLAx-as-collateral integration on a $4B-TVL lending market is the live anchor [cong-tokenized-2025, p.20]. **(ii) Options or structured-product issuers** requiring an explicit forward distribution on the underlying to price strikes and hedge deltas; the conditional-mean passthrough Cong et al. document is necessary but not sufficient. **(iii) Regulator-facing risk models** under SR 11-7 [of Governors of the Federal Reserve System & Office of the Comptroller of the Currency, 2011] / SR 26-2 [Board of Governors of the Federal Reserve System, 2026], where the price-feed methodology must be defensible to a model-risk officer — the BMR Article 27 disclosure CFB publishes for its regulated xStocks Indices [Benchmarks, 2026a] does not approach the documentation bar these regulations set, and no other surveyed feed comes closer. The calibration-transparent primitive serves this consumer set specifically; for perp settlement marked-to-tokenized-spot (Kraken xStock perpetuals via CFB's regulated xStocks Indices [Benchmarks, 2026b]) and for DEX market-making, the existing point feeds are correct by contract design and the calibration primitive is over-engineering.

1.3 What this paper shows

We evaluate the deployed locally-weighted Mondrian split-conformal architecture (full specification in §4) on 5,996 weekend prediction-window rows (639 weekends × 10 symbols) spanning 2014-01-17 through 2026-04-24: seven US equities/ETFs (SPY, QQQ, AAPL, GOOGL, NVDA, TSLA, HOOD), gold (GLD), long treasuries (TLT), and a Bitcoin-proxy equity (MSTR). Each window runs Friday 16:00 ET to Monday 09:30 ET. §6.8 then extends the identical architecture — only the gap selector changes — to a 22,624-row overnight (close → next-open) panel on the same symbols, establishing the result across closed-market hours rather than weekends alone. The contribution is the *primitive* — the artefact, its serving-time lookup, and the audit receipts — together with empirical evidence that the architecture delivers per-anchor *and* per-symbol calibration on a decade-scale public-data panel.

Three headline empirical claims:

- **Held-out coverage at $\tau = 0.95$ via 10-fold leave-one-symbol-out CV:** 0.9497 ± 0.0128, all 10 LOSO bands pass Kupiec. Each held-out symbol's $c(\tau)$ is fit on the other nine symbols' OOS slice. Three orthogonal stability checks corroborate on the deployed-fit slice: per-symbol Kupiec 10/10, four-anchor split-date Christoffersen passing every (split × τ) cell, and calendar sub-period coverage holding across {2023, 2024, 2025, 2026-YTD}.
- **Per-symbol Kupiec 40/40 across the (symbol × τ) grid against parametric baselines** (GARCH-t 31/40, GARCH-Gaussian 23/40, unweighted Mondrian 12/40, constant buffer 17/40); per-symbol DQ also passes 39/40 cells (CAViaR-style lag-only test, $K = 4$). At matched 95% realised coverage GARCH-t ties this paper's 370.6 bps half-width but undercovers Kupiec at every $\tau < 0.99$.
- **Engle–Manganelli DQ rejects pooled at $\tau = 0.95$:** the bands are *per-anchor calibrated*, not full-distribution calibrated. The residual is a within-weekend cross-sectional cluster topology ($\hat{\rho}_{\text{cross}} = 0.354$, orthogonal to $\hat{\sigma}_s$); the §6.3.4 joint-tail empirical distribution turns the disclosure into a measurable reserve-guidance threshold ($k^* = 3$ at $\tau = 0.95$, 0/24 sub-period stability rejections), and §10 names the methodology fix.

Five contributions: (C1) coverage inversion as an oracle primitive (§3); (C2) a 12-year locally-weighted Mondrian split-conformal-by-regime calibration on public data with a deployment artefact rebuildable by any third party (§4, §5, §A); (C3) Kupiec + Christoffersen + DQ validation in the form required by institutional model-risk management (§6) with a four-DGP simulation study (§6.7) and a content-addressed forward-tape harness (§6.6); (C4) the joint-tail empirical characterisation and reserve-guidance threshold k^* (§6.3.4–§6.3.5), no incumbent oracle publishes this; (C5) a three-layer ablation isolating per-regime conformal, per-symbol $\hat{\sigma}_s$ standardisation, and the selection procedure under multi-test correction (§7). We do not claim point-accuracy or absolute bandwidth dominance over any incumbent surface; the contribution is the calibration receipt and its per-symbol generalisation.

§2 surveys related work; §3 gives the problem statement; §4 the methodology; §5 the data and regime labeler; §6 calibration results; §7 the ablation; §8 the serving-layer system; §9 limitations; §10 extensions.

2 Related Work

Our contribution sits at the intersection of three literatures that have, to our knowledge, never been brought together in a deployed system: decentralised-oracle design, cross-venue price discovery, and the calibration / conformal-prediction framework for interval forecasts. A fourth — institutional model-risk-management guidance — supplies the disclosure vocabulary we adopt.

2.1 Decentralised oracle designs

The deployed oracle landscape is organised around an *integrity* primitive: can the oracle deliver a faithful value on-chain without a trusted intermediary? Chainlink's decentralised-oracle-network architecture [Breidenbach et al., 2021] aggregates off-chain reports from a node-operator committee, with Data Streams [Labs, 2024] providing pull-based delivery and an explicit `marketStatus` flag — shipped in three co-existing schemas as of 2026-Q2: v8 RWA Standard [Labs, 2025b], v10 Tokenized Asset [Labs, 2025a], and v11 RWA Advanced [Labs, 2026]. v8 is the load-bearing path for tokenized-equity lending on Solana: Kamino Finance ($4B TVL) consumes every xStock collateral asset through Kamino Scope's `OracleType::ChainlinkRWA` adaptor [Finance, 2025], routed through the v8 dispatcher, with two parallel feeds per xStock registered in Scope's mainnet config [Finance, 2026] — an `Open` mode that refuses updates outside US trading hours and an `AllUpdates` mode clamped to ±500 bps of the Open feed's Friday-close value. v10 backs the BackedSolana xBridge but is not consumed by the canonical Solana lending market. Pyth's Solana surface [Network, 2021, 2022] takes a first-party-publisher approach: each permissioned publisher submits a price and confidence interval; the aggregate rule assigns three votes per publisher (price, price ± confidence) before taking the median. Pyth Pro [Network, 2026] layers subscriber-customisable cadence and the Blue Ocean ATS overnight integration; since September 2025 Pyth has been the exclusive on-chain distributor of Blue Ocean's executable book through end-2026 [Network, 2025], covering 8:00 PM – 4:00 AM ET Sun–Thu (the canonical xStock weekend remains uncovered). CF Benchmarks (Kraken-owned, FCA + BMR-supervised) launched the xStocks Indices product family on 2026-05-07 [Benchmarks, 2026b,a] — regulated per-second scalars per xStock (Price Return + Total Return variants) powering Kraken's tokenized-equity perpetual futures; the methodology is FCA-supervised but emits no confidence, band, or coverage field on the wire. SEDA Protocol's SEDA Benchmark family [Protocol, 2026] publishes session-aware single-asset feeds and a USA500 composite (60% equities / 25% crypto / 15% commodities) consumed by Hyperliquid HIP-3 markets via Dreamcash, Injective, and Perps.Fun derivatives venues, governed by a four-pillar continuity rule (session tags, cost-of-carry futures-to-spot, self-referencing EMA, staleness fallback) with no on-wire uncertainty field. Switchboard [Switchboard, 2023] generalises aggregation into permissionless queues but has shallow tokenized-equity deployment; RedStone Live [RedStone, 2026] blends institutional feeds with perpetual-market data; optimistic oracles UMA [Project, 2020] and Tellor [Tellor, 2023] replace continuous aggregation with dispute-and-stake.

None of these designs publishes a *calibration claim* on the served value. The deployed-surface evidence is categorical at the wire: v8 RWA Standard is mid-only (no `bid`/`ask`/confidence) with `marketStatus` as the only signal of off-hours regime; v11's weekend `bid` carries a synthetic `.01`-suffix at 100% incidence on three of four mapped xStocks; v10 carries no `bid`/`ask`/confidence field and serves a 24/7 `tokenizedPrice` whose CEX-aggregation methodology is qualitatively described; CFB's BMR Article 27 Benchmark Statement contains no language about realised coverage; SEDA's four-pillar continuity methodology yields a single scalar plus session-flag metadata; Pyth's `(price, conf)` reports low publisher disagreement at venue close. Pyth's documentation [Network, 2022] asserts ~95% coverage at the *publisher* level, not as a property of the aggregate. Optimistic oracles guarantee *eventual* correctness under economic security but do not define a distributional statement conditional on time-of-week or realised volatility. Because none of these surfaces emits a calibrated coverage band, the contribution is necessarily categorical: a coverage-vs-sharpness numerical comparison is not well-defined against them and is therefore not reported.

The academic literature has similarly concentrated on integrity. Angeris and Chitra [Angeris and Chitra, 2020] prove conditions under which CFMMs are incentive-compatible as price oracles. Eskandari et al.'s SoK [Eskandari et al., 2021] catalogues oracle-manipulation attacks; the framing is adversarial (*did the attacker corrupt the value?*) rather than statistical (*over N windows, did the served band contain the truth at the claimed rate?*). The BIS bulletin [Duley et al., 2023] frames the tension as decentralisation-versus-efficiency; on-chain consumers operate under adversarial ordering [Daian et al., 2020], amplifying the cost of any systematic band miscalibration.

Our contribution is orthogonal. The integrity primitive asks whether $\hat{P}_t \approx P_t^{\text{true}}$; the calibration primitive asks, given the oracle's own band, whether it contains P_t^{true} at the claimed rate. An oracle can be integrity-perfect and calibration-meaningless: Chainlink faithfully reports Friday's 16:00 close on Sunday at 14:00, but that value is not asserted to be a 95%-coverage estimate of Monday's open. We invert the interface so τ is a request parameter and q_{served} is the receipt.

2.2 Cross-venue price discovery

The base forecaster underlying our calibration surface is not methodologically novel; its point estimate is a factor-switchboard scaling of Friday's close by a weekend futures return. The choice of futures as the leading venue applies Hasbrouck's information-share methodology [Hasbrouck, 1995, 2003], which establishes that most price discovery in the S&P 500 and Nasdaq-100 occurs in E-mini futures — the statement we rely on when scaling SPY and QQQ Friday closes by ES and NQ weekend returns. Earlier work [Stoll and Whaley, 1990, MacKinlay and Ramaswamy, 1988] established intraday lead-lag and mispricing-adjustment dynamics in index-futures markets; Gonzalo–Granger [Gonzalo and Granger, 1995] is the conventional complement.

The Friday 16:00 ET to Monday 09:30 ET weekend is the longest contiguous non-trading interval in the US calendar. French [French, 1980] documents the Monday-effect, establishing the weekend as a distinct return-generating regime. Barclay and Hendershott [Barclay and Hendershott, 2003] show after-hours price discovery is non-trivial, so a stale-hold feed is systematically miscalibrated; their companion paper [Barclay and Hendershott, 2004] shows after-hours effective spreads are 3–4× regular-hours, reflecting greater adverse selection — exactly the window in which a protocol holding an RWA as collateral is most exposed. For ETFs, Madhavan and Sobczyk [Madhavan and Sobczyk, 2016] characterise creation/redemption-driven discovery, validating our ETF-class instruments within a unified frame.

A newer thread addresses the structural migration motivating this work. Makarov and Schoar [Makarov and Schoar, 2020] document cross-venue arbitrage in cryptocurrency markets; Lehar and Parlour [Lehar and Parlour, 2025] show

Ethereum DEXs do not exhibit long-lived arbitrage relative to centralised benchmarks. Cong et al. [Cong et al., 2025] provide the first large-scale empirical study of tokenised stocks (top-100 Backed/Ondo issuance, July–October 2025, $420M aggregate market cap, 14× growth in the four-month window). Four findings establish the empirical premise of the closed-market problem this paper addresses. **(a)** During overlapping US-market hours, tokenized prices track underliers near-one-for-one — a 1% move in the underlying translates to a 0.982% move in the token with adjusted $R^2 = 0.982$ [Table 6, p.34]; the tokenized-vs-underlying gap is a closed-market phenomenon, not a general informational degradation. **(b)** During the closed-market window the wedge is empirically substantial: TSLAx — the panel's most-traded name and a live Kamino collateral asset — deviates from the last underlying close by more than 1% in 71% of weekday off-hours and 15% of weekend hours; by more than 2% in 57% / 8%; by more than 5% in 12% / 0% [Table 4, p.32]. **(c)** Off-hour token returns predict the close-to-open underlying return at $\lambda = 0.903$ with adjusted $R^2 = 0.839$ [Table 10, p.40], but the relationship is regime-dependent — informative on weekends (Table 9, p.39: highest-quintile-minus-lowest-quintile spread of $+0.77\%$ on the subsequent Monday open) and reversal-driven on weekday overnights (Table 8, p.38: spread of -0.99%, $t \approx -2.7$). The signal lives in the conditional mean, not the conditional quantile, and consumers cannot tell from a single tokenized-side scalar which regime they are in. **(d)** The wedge persists by structural design — issuer-controlled redemption windows, KYC/AML constraints on minting/redemption, and primary-arbitrageur geofencing (Backed under European sandbox exemptions, Ondo geofenced from US individuals) [§3.3.3, p.17; §6.1, pp.41–42] — and is therefore not a transitional inefficiency. The signal Cong et al. document is conditional-mean; the risk-side complement — a conditional-quantile band that lets a consumer use the off-hour signal without inheriting the prints that "likely would have triggered halts in regulated markets" [cong-tokenized-2025, pp.6–7] — is the primitive this paper deploys. TSLAx's integration as Kamino Finance collateral (p.20) and its 14% premium relative to the prior underlying close in the September 13–15 2025 window (p.24) is the live anchor on which the consumer SLA is binding.

2.3 Calibrated and conformal prediction

Kupiec [Kupiec, 1995] proposed the unconditional-coverage LR test for VaR; Christoffersen [Christoffersen, 1998] generalised to conditional coverage, decomposing into Kupiec plus an independence test on the hit sequence. The p_{UC} and p_{ind} columns we report are these tests; we use the pair rather than the joint because they correspond in banking practice to the Basel traffic-light framework [on Banking Supervision, 1996]. The density-forecast generalisation [Diebold et al., 1998] evaluates whether PITs are i.i.d. uniform; Berkowitz [Berkowitz, 2001] refines this to a small-sample-robust joint LR. Our coverage check at fixed τ is a marginal slice of the PIT framework, adopted because the protocol-consumer SLA is per-τ.

Gneiting, Balabdaoui, and Raftery [Gneiting et al., 2007] state the modern objective: maximise sharpness subject to calibration. This organises our §7 ablation. Allen et al. [Allen et al., 2025] formalise *tail calibration*; the deployed architecture closes a $\tau = 0.99$ tail-calibration ceiling that earlier hybrid-forecaster predecessors exhibited (§6.3, §7.2). The tradition traces to Brier [Brier, 1950].

The deployed architecture is a locally-weighted Mondrian split-conformal predictor [Vovk et al., 2003, 2005]: per-regime quantiles $q_r(\tau)$ on per-symbol pre-Friday $\hat{\sigma}_s$-standardised residuals, served after a multiplicative OOS-fit $c(\tau)$. (The §7.2 unweighted-Mondrian comparator omits the per-symbol standardisation; the walk-forward $\delta(\tau)$ schedule that an unweighted comparator requires collapses to identity once is per-symbol.) Mondrian conformal preserves finite-sample marginal validity within each category under within-category exchangeability. A next-generation upgrade replaces the four-anchor schedule with a continuous conformal correction over the residual CDF — Romano-Patterson-Candès CQR [Romano et al., 2019] is the closest match. For non-exchangeability, Barber et al. [Barber et al., 2023] extend via weighted quantiles, and Xu-Xie [Xu and Xie, 2021] develop EnbPI for time-series.

2.4 Institutional model risk management

The validation evidence published here matches the form required of any bank integrating an on-chain oracle into a collateral-valuation pipeline under SR Letter 11-7 [of Governors of the Federal Reserve System & Office of the Comptroller of the Currency, 2011] / SR 26-2 [Board of Governors of the Federal Reserve System, 2026] and the Basel 1996 backtesting framework [on Banking Supervision, 1996] — whose mathematical content is the Kupiec test §6.3 reproduces.

3 Problem Statement

3.1 Setup and notation

Let $\mathcal{S} = \{s_1, \dots, s_K\}$ be a finite universe of assets (in our evaluation, ten US equities, ETFs, and hybrid crypto-proxies) and let $t \in \mathbb{N}$ index a sequence of *closed-market prediction windows* $[t_{\text{pub}}, t_{\text{tgt}}]$, each the interval between a venue close (t_{pub}) and its next open (t_{tgt}). We evaluate two instances: the **weekend** window (Friday 16:00 ET → Monday 09:30 ET), the primary panel of §6.2–§6.7, and the **overnight** window (close → next-day 09:30 ET open), the generalisation panel of §6.8.

For each (s, t) we observe:

- $P_t(s) \in \mathbb{R}_{>0}$: the realised reference price at t_{tgt} (the next open).
- $\mathcal{F}_t(s)$: pre-publish information available at t_{pub}. This includes $P_{t^-}(s)$ (the close), contemporaneous factor returns (E-mini S&P, gold, 10Y treasury, BTC futures), implied-vol indices (VIX, GVZ, MOVE), a calendar flag $\text{earn}_t(s) \in \{0, 1\}$ for a scheduled earnings release inside the window (a release in the coming week on the weekend panel; a release dated inside the close→open gap, by BMO/AMC session, on the overnight panel), and a gap-length flag $\ell_t \in \{0, 1\}$ for calendar weekends ≥ 4 days (weekend panel only).
- A *regime labeler* $\rho : \mathcal{F}_t(s) \to \mathcal{R}$ defined purely on pre-publish information, with a cadence-specific partition: $\mathcal{R}_{\text{weekend}} = \{\texttt{normal}, \texttt{long\_weekend}, \texttt{high\_vol}\}$ and $\mathcal{R}_{\text{overnight}} = \{\texttt{normal}, \texttt{high\_vol}, \texttt{earnings\_night}\}$ (§5.5).

A *base forecaster* f emits, at each (s, t, q) with claimed quantile $q \in (0, 1)$, a band $\left[L_t^f(s; q), U_t^f(s; q)\right]$.

3.2 The classical oracle problem

A conventional point-plus-band oracle publishes, at each t_{pub}, a triple $(\hat{P}_t, L_t, U_t)$ with an *implicit* or *undisclosed* coverage level. The consumer has no contract on the realised rate at which $P_t \in [L_t, U_t]$. When band widths are derived from publisher dispersion (Pyth) or a fixed Gaussian heuristic (Chainlink staleness fallback), the mapping to a realised probability-of-coverage statement is neither asserted nor auditable.

3.3 The coverage-inversion primitive

We define a stricter contract. Fix a base forecaster f, a regime labeler ρ, a discrete claimed-quantile grid $\mathcal{Q} \subset (0, 1)$, and a rolling historical dataset

$$\mathcal{D} = \left\{ \left(P_t(s), \mathcal{F}_t(s), \{(L_t^f(s; q), U_t^f(s; q))\}_{q \in \mathcal{Q}}, \rho(\mathcal{F}_t(s))\right) : t \in \mathcal{T}_{\text{hist}} \right\}.$$

The **empirical calibration surface** is the mapping

$$S^f(s, r, q) \;=\; \frac{1}{|\mathcal{T}_{s,r}|} \sum_{t \in \mathcal{T}_{s,r}} \mathbf{1}\left[\, L_t^f(s; q) \leq P_t(s) \leq U_t^f(s; q) \,\right],$$

where $\mathcal{T}_{s,r} = \{t \in \mathcal{T}_{\text{hist}} : \text{symbol} = s,\ \rho(\mathcal{F}_t(s)) = r\}$.

Given a *consumer-chosen* target coverage $\tau \in (0, 1)$, the oracle serves the band at

$$q_{\text{served}}(s, r, \tau) \;=\; (S^f)^{-1}(s, r, \tau),$$

with $(S^f)^{-1}$ defined by linear interpolation between bracketing grid points and a documented fallback to a pooled surface $S^f(\cdot, r, q)$ when $|\mathcal{T}_{s,r}| < n_{\min}$.

The oracle read is the tuple

$$\texttt{PricePoint}(s, t, \tau) \;=\; \left(\hat{P}, L_t^f(s; q_{\text{served}}), U_t^f(s; q_{\text{served}}),\ q_{\text{served}},\ r,\ f\right),$$

exposing q_{served} and f as calibration *receipts* alongside the band itself.

3.4 Properties we claim

(P1) Auditability. $\mathcal{D}$ is reproducible from public data sources (freely available price feeds, vol indices, and earnings calendars). Given $(\mathcal{D}, f, \rho, \mathcal{Q})$, the surface S^f is deterministic; thus any third party can reconstruct S^f and independently verify that the served band corresponds to the published q_{served}.

(P2) Conditional empirical coverage. Under the assumption that the conditional distribution $P_t \mid (\mathcal{F}_t,\ \text{regime})$ is stationary across $\mathcal{T}_{\text{hist}}$ and the evaluation set, the realised coverage of the served band satisfies

$$\Pr\big[\, P_t \in [L_t^f(s; q_{\text{served}}),\ U_t^f(s; q_{\text{served}})] \;\big|\; \rho(\mathcal{F}_t) = r,\ s \,\big] \;\longrightarrow\; \tau$$

as $|\mathcal{T}_{s,r}| \to \infty$. Finite-sample deviations are measurable, tested (§6), and disclosed as a per-target *calibration buffer* applied pre-inversion (§4.3 specifies the deployed schedule). The headline $\tau = 0.95$ claim is held out on two orthogonal axes — temporally (§6.3.3.1: c fit on 2023 alone, evaluated on a 2024-01-05 → 2026-04-24 holdout) and on symbol (§6.3.3 LOSO) — both of which evaluate the calibration buffer schedule on data its fit never touched. Per-symbol Kupiec at $\tau = 0.99$ has minimum detectable effect ≈ 3 pp under-coverage and is undetectable for over-coverage given the per-symbol expected violation count of 1.73; the 10/10 pass at that anchor is read as “no per-symbol violation rate is statistically distinguishable from 1% at the resolution Kupiec offers per-symbol”, not as a 1 pp tolerance certificate (§6.4.1). Pooled Kupiec at $\tau = 0.99$ has minimum detectable effect 1.0 pp; the pooled $p = 0.94$ is the $\tau = 0.99$ statistical claim that does have 1 pp resolution (`reports/tables/paper1_b3_kupiec_power_mde.csv`).

(P3) Per-regime serving efficiency at deployment-tuned parameter budgets. The architecture serves a band $\big[\hat{P}_{\text{Mon},r}(s) - c(\tau)\, q_r(\tau)\, \hat{\sigma}_s(t)\, P_{\text{Fri},t},\ \hat{P}_{\text{Mon},r}(s) + c(\tau)\, q_r(\tau)\, \hat{\sigma}_s(t)\, P_{\text{Fri},t}\big]$, with the per-symbol scale $\hat{\sigma}_s(t)$ replacing what would otherwise be a per-symbol fallback or per-symbol Mondrian cell. The efficiency criterion is mean bandwidth at matched realised coverage *and* per-symbol Kupiec pass-rate. §7's ablation establishes empirical dominance against the deployable alternatives evaluated on this panel — the constant-buffer baseline, the un-standardised Mondrian comparator, and parametric GARCH-Gaussian / GARCH-t baselines — at matched coverage on this sample, with the rule itself selected from a five-variant ladder under the multiple-testing-corrected gate of §7.4. The trained / tuned scalar count is tabulated in §A.2; §3.5 enumerates the non-goals (notably no optimal liquidation-policy benchmark, no parametric tail guarantees, no rule optimality).

3.5 Non-goals

We explicitly do not claim:

1. **Optimal point estimates.** $\hat{P}$ is the midpoint of a band calibrated for coverage, not a minimum-variance estimator. We make no claim to improve over Pyth / Chainlink on point accuracy during market hours.
2. **Parametric tail guarantees.** The surface is empirical; shock-regime coverage is bounded above by the fraction of historical observations in the same tail of the forecast distribution (§9).
3. **Distribution-free coverage.** P2 requires stationarity of the *standardised*-conditional distribution. The per-symbol $\hat{\sigma}_s(t)$ rescaling provides partial adaptive scaling but the standardised score's conditional stationarity is still assumed. A conformalised variant (Vovk, Barber et al.) would upgrade the statement to a finite-sample guarantee under exchangeability; we flag this as an open extension.
4. **Protection against adversarial data feeds.** Soothsayer is a calibration-transparency primitive, not an integrity primitive. It ingests upstream price/oracle signals and republishes them as a calibrated band; its coverage claim is conditional on the integrity of those upstream feeds, and this primitive does not separately defend against manipulation of any of them.
5. **rule optimality.** The deployed rule (EWMA HL=8) is justified by a pre-registered three-gate selection procedure under multiple-testing correction (§7.4), with held-out forward-tape re-validation on an architectural sibling harness. We do not claim it is optimal among all locally-weighted variants; finer half-life grids, alternative weight kernels, or hybrid rules are open extensions.
6. **Optimal lending-policy defaults.** We do not claim in this paper to identify the welfare-optimal liquidation rule, collateral haircut, or target τ for a lending protocol. Those questions require explicit assumptions about account-weight distributions, protocol-specific loss functions, and the semantics by which realised outcomes are mapped into liquidation quality.

3.6 Evaluation questions

The body of the paper answers:

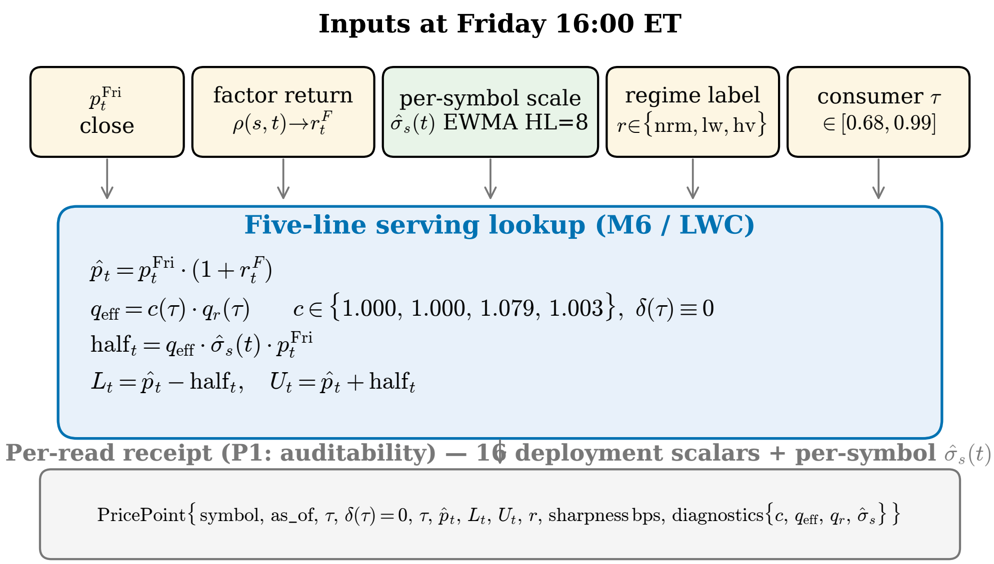

Figure 1: Serving pipeline: five pre-Friday inputs (factor return r_t^F from the §5.4 switchboard, regime label r, per-symbol scale $\hat{\sigma}_s(t)$ from the §4.2 EWMA, consumer target τ, and Friday close p_t^{Fri}) feed a five-line lookup that returns a band $[L_t, U_t]$ around a factor-adjusted point $\hat{p}_t$. The 16 deployment scalars are the per-regime quantile table $q_r(\tau)$ on standardised residuals plus the OOS-fit bump schedule $c(\tau)$. Every read emits a `PricePoint` receipt carrying the four served scalars and the diagnostic quartet $(c, q_{\mathrm{eff}}, q_r, \hat{\sigma}_s)$ — the per-read auditability of P_1.

- **Q1** (calibration, §6). Does the served band achieve τ within 2pp on held-out data? What is the Kupiec unconditional-coverage p-value, and do violations cluster (Christoffersen independence p)?
- **Q2** (sharpness, §6–§7). How much tighter than a naïve stale-hold Gaussian baseline is the served band at matched realised coverage, per regime?
- **Q3** (ablations, §7). Which components of $f^\star$ (factor switchboard, log-log vol regression, earnings regressor, long-weekend regressor, hybrid regime selection, empirical buffer) are load-bearing for the P2/P3 claims, and which are disclosed for auditability but contribute no measurable signal at our sample size? The ablation reports block-bootstrap 95% CIs on each pairwise effect size.
- **Q4** (auditability, §4 + artifact). Can a third party reconstruct S^f on public data and independently verify a served `PricePoint`?

Questions such as "which served target τ should a Kamino-style protocol choose as its liquidation-policy default?" are downstream integration questions. They are important, but they sit outside Q1-Q4 and are not answered by this paper's evidence alone.

4 Methodology

This section instantiates the abstract setup of §3 with the concrete point estimator, per-symbol scale standardisation, conformal quantile fit, deployment-tuned schedules, and serving-time machinery of the deployed locally-weighted Mondrian split-conformal architecture. The implementation lives in `src/soothsayer/oracle.py:Oracle.fair_value_lwc` (Python) and `crates/soothsayer-oracle/src/oracle.rs::Oracle::fair_value` (Rust), with byte-exact parity against the Python reference (180/180 cases; §8.5). The on-chain wire format and code-tag taxonomy are described in §8.

The architecture has four ingredients: a point estimator $\hat{P}_{\mathrm{Mon},r}$, a strictly pre-Friday per-symbol scale $\hat{\sigma}_s(t)$ that standardises the conformity score, a per-regime conformal quantile $q_r(\tau)$ trained on the *standardised* residuals on a held-out calibration set, and an OOS-fit multiplicative bump $c(\tau)$ that closes any residual train-OOS distribution-shift gap. The trained / tuned scalar count and the a-priori hyperparameter list are tabulated in §A.2. The §7 ablation isolates the load-bearing components — regime stratification, per-symbol standardisation, and the near-identity OOS $c(\tau)$ bump — by descriptive comparator (constant-buffer baseline, unweighted Mondrian) rather than by code tag. Figure 1 summarises the serving pipeline.

4.1 The point estimator $\hat{P}_{\mathrm{Mon},r}$

The deployed point estimator is the per-symbol factor switchboard (§5.4):

$$\hat{P}_{\mathrm{Mon},t}(s) \;=\; P_{\mathrm{Fri},t}(s) \cdot \bigl(1 + r_t^{\mathrm{factor}}(s)\bigr),$$

where $r_t^{\mathrm{factor}}(s)$ is the weekend return of the per-symbol factor (ES=F for equities; GC=F for GLD; ZN=F for TLT; BTC for MSTR post-2020-08). The point estimator is the input whose residual distribution the conformal quantile is fit on, not the product. The factor-switchboard variant survived empirical pruning over an eight-variant point-estimator ladder during early development; the ladder and the empirical losses against alternatives are documented in `reports/methodology_history.md`.

4.2 Per-symbol pre-Friday scale $\hat{\sigma}_s(t)$

The architecture standardises the conformity score by a per-symbol estimate of the *relative-residual* scale, computed strictly from past Fridays:

$$\hat{\sigma}_s(t) \;=\; \mathrm{EWMA}_{\mathrm{HL}=8}\Bigl(\bigl\{\, r_{t'}^{\mathrm{rel}}(s) \;:\; t' < t \,\bigr\}\Bigr), \qquad r_{t'}^{\mathrm{rel}}(s) \;=\; \frac{P_{\mathrm{Mon},t'}(s) - \hat{P}_{\mathrm{Mon},t'}(s)}{P_{\mathrm{Fri},t'}(s)},$$

with weekend half-life $\mathrm{HL} = 8$ — equivalently a geometric decay rate $\lambda = 0.5^{1/8} \approx 0.917$ per past Friday. The window is *one-sided strictly before* t: the standardiser at t uses only $\{t' : t' < t\}$, so $\hat{\sigma}_s$ is itself a pre-publish quantity. We require at least eight past relative-residual observations before $\hat{\sigma}_s$ is defined; weekends with fewer past Fridays per symbol are dropped at warm-up (80 rows out of the 5,996-weekend panel — roughly the first eight weekends per symbol, rebased per listing date). The warm-up rule and half-life are encoded as constants `SIGMA_HAT_HL_WEEKENDS = 8` and `SIGMA_HAT_MIN = 8` and persisted on the artefact parquet as `sigma_hat_sym_pre_fri`.

The EWMA half-life HL=8 was selected from a five-variant ladder under a pre-registered three-gate criterion (split-date Christoffersen, per-symbol Kupiec, bootstrap CI on pooled half-width vs the K=26 baseline) with Benjamini–Hochberg correction across the 80-cell grid; full procedure and held-out forward-tape re-validation in §7.4.

4.3 Mondrian split-conformal on standardised residuals

Conformity score: relative absolute residual *standardised by* $\hat{\sigma}_s(t)$:

$$s_t(s) \;=\; \frac{\bigl|P_{\mathrm{Mon},t}(s) - \hat{P}_{\mathrm{Mon},t}(s)\bigr|}{P_{\mathrm{Fri},t}(s) \cdot \hat{\sigma}_s(t)}.$$

Mondrian taxonomy: weekends partition into $r \in \{\texttt{normal}, \texttt{long\_weekend}, \texttt{high\_vol}\}$ via the regime classifier $\rho(\mathcal{F}_t)$ specified in §5.5. The trained per-regime conformal quantile $q_r(\tau)$ is the standard split-CP $\lceil \tau(n{+}1) \rceil$-th rank quantile on the regime's training-set *standardised* scores. The deployed grid has four anchors $\tau \in \mathcal{T}_{\mathrm{anch}} = \{0.68, 0.85, 0.95, 0.99\}$ and three regimes, yielding twelve trained scalars.

Calibration set: $\mathcal{T}_{\mathrm{train}} = \{t : t < \text{2023-01-01}\}$ — 4,186 weekend rows after the 80-row warm-up drop. Each per-(regime, τ) bin therefore carries $N \approx 430$–$2,720$ observations supporting one quantile estimate, comfortably above the $N \approx 50$–300 thresholds where Mondrian conformal becomes noisy (a per-(symbol, regime) Mondrian rung was rejected during ablation development for cell-thinness reasons, with the standardised-score architecture below recovering per-symbol calibration without per-(symbol, regime) cells; details in `reports/methodology_history.md`). Deployed values of $q_r(\tau)$ are listed in `data/processed/lwc_artefact_v1.json` (audit-trail sidecar) and loaded at module import into `src/soothsayer/oracle.py:LWC_REGIME_QUANTILE_TABLE`.

The architectural choice that makes the per-symbol calibration claim hold is the standardisation of the conformity score by $\hat{\sigma}_s(t)$ before the Mondrian quantile fit. An unweighted Mondrian comparator absorbs both the cross-symbol scale heterogeneity (HOOD's typical relative residual is $\sim 3\times$ SPY's) and the per-regime heterogeneity into one Mondrian cell; the result is bimodal per-symbol calibration error — heavy-tail tickers under-cover, low-vol tickers over-cover — and is the failure mode the §6.7 simulation study reproduces under controlled conditions. Standardising the score by $\hat{\sigma}_s(t)$ before fitting the per-regime quantile factors out the per-symbol scale and lets the regime quantile carry only the *shape* of the standardised residual distribution. The §6.4 evidence (10/10 per-symbol Kupiec pass at $\tau = 0.95$;

per-symbol Berkowitz LR range $[3.2,\ 16.7]$) and the §6.7 simulation evidence (the bimodality reproduced under four DGPs with known ground truth and standardisation closes it on every DGP) ratify this construction.

Within-bin exchangeability — the assumption that yields finite-sample marginal coverage within each Mondrian cell — is empirically supported (permutation test on lag-1 standardised-score autocorrelation; §A.9). The §6.3.6 / §9.4 cross-sectional dependence is *across* bins, not within.

4.3.1 Overnight regime set and calendar-conditioned coverage

The architecture is parameterised by a gap selector (`gap_mode`): the deployed weekend panel admits Friday→Monday pairs; the overnight panel of §6.8 admits consecutive-trading-day pairs with a one-line change and no change to any downstream component. The Mondrian taxonomy is re-derived per cadence. Overnight, `long_weekend` has no analog and is dropped; `high_vol` is retained but its trailing-VIX quartile uses a 252-trading-day lookback (1 year, matching the weekend's 52-weekend window); and a new regime, `earnings_night`, is added.

`earnings_night` is the first regime the architecture conditions on *a priori*. Unlike `high_vol` — a market state inferred from trailing VIX — an earnings release is a **scheduled event with a publicly known date and session**, so the band can widen deterministically *ahead* of a known information event, before any volatility is realised. We term this **calendar-conditioned coverage**, and to our knowledge no production oracle (Pyth, Chainlink Data Streams, RedStone, Kamino Scope) widens its feed for a scheduled earnings release. An earnings event is assigned to the single overnight gap it actually drives via session timing: an after-close (`amc`) report dated t_0 or a before-open (`bmo`) report dated t_1 fires inside the close(t_0)→open(t_1) gap (a date-only flag that brackets both adjacent nights dilutes the regime with normal nights and is strictly dominated). Standardised by the same $\hat{\sigma}_s(t)$ the band uses, the realised overnight move on earnings nights has $p_{99} \approx 9.7$ versus ≈ 2.0 on other nights; the fitted per-regime quantile $q_{\texttt{earnings\_night}}(\tau)$ runs $8\times$ $q_{\texttt{normal}}(\tau)$, and §6.8 shows the resulting band is calibrated, not merely wide.

de-contamination. Because an earnings residual can be $\sim 8\sigma$, admitting it to the per-symbol EWMA scale $\hat{\sigma}_s(t)$ over-widens the subsequent ~`half_life` *normal* nights and induces clustered post-earnings violations. We therefore exclude earnings-night residuals from the $\hat{\sigma}_s$ estimation pool — every night still receives a $\hat{\sigma}_s$ built from its non-earnings history — assigning the earnings excess to the `earnings_night` quantile rather than the scale. Earnings fatness is a regime effect, not a per-symbol scale effect. This separation lifts the overnight Christoffersen independence p at $\tau = 0.95$ from a rejecting 0.013 to a passing 0.165 and collapses the OOS $c(\tau)$ bumps to 1.0 (§6.8).

4.4 OOS-fit $c(\tau)$ bump

The deployed bump schedule is

$$c(\tau) \;=\; \{0.68\!:\!1.000,\ 0.85\!:\!1.000,\ 0.95\!:\!1.079,\ 0.99\!:\!1.003\},$$

a near-identity correction that *only* widens at $\tau = 0.95$ (by 7.9%) and is essentially the identity at the other three anchors. The fit procedure: $c(\tau)$ is the smallest $c \in [1, 5]$ such that pooled OOS realised coverage with effective quantile $c \cdot q_r(\tau)$ matches the target τ. Four scalars total. Three of the four are essentially identity, so only $c(0.95) = 1.079$ carries meaningful OOS information; §9.3 carries the full provenance disclosure for that one scalar.

4.5 The walk-forward $\delta(\tau)$ schedule is identically zero

A 4-split expanding-window walk-forward sweep over $\delta \in \{0.00, \dots, 0.07\}$ at tune fractions $f \in \{0.40, 0.50, 0.60, 0.70\}$ (script: `scripts/run_lwc_delta_sweep.py`) selects $\delta(\tau) \equiv 0$ under the criterion *smallest* δ *such that walk-forward realised coverage* $\geq \tau$ *on every powered split* — per-symbol standardisation tightens cross-split realised-coverage variance enough that no structural-conservatism shift is required (§7.2.3 ablation). The vector is retained in the artefact JSON for receipt-schema compatibility.

4.6 Serving-time band construction

Given a consumer-chosen target $\tau \in (0, 1)$ at (s, t), the served band is constructed in five steps:

1. *Look up the regime, point, per-symbol scale, and Friday close.* Read $r = \rho(\mathcal{F}_t)$, $\hat{P}_{\mathrm{Mon},t}(s)$, $\hat{\sigma}_s(t)$, and $P_{\mathrm{Fri},t}(s)$ from `lwc_artefact_v1.parquet`.
2. *Look up the standardised quantile.* $q_r = q_r(\tau)$, with q_r linearly interpolated between anchors $\mathcal{T}_{\mathrm{anch}} = \{0.68, 0.85, 0.95, 0.99\}$ and clipped to the nearest anchor outside that range.

3. *Apply the OOS bump.* $q_{\text{eff}} = c(\tau) \cdot q_r(\tau)$, with c linearly interpolated between the same four anchors; $\delta(\tau) \equiv 0$ so no shift step is needed. The $c(\tau)$ schedule is non-monotonic — c peaks at the $\tau = 0.95$ anchor (1.079) and dips back at $\tau = 0.99$ (1.003) — so linear interpolation hands an off-anchor consumer a multiplier (e.g. $c(0.90) \approx 1.040$ and $c(0.97) \approx 1.041$) at a point with no separate audited calibration evidence. Calibration evidence in §6 is reported at the four audited anchors only; §9.10 carries the off-anchor disclosure (denser-anchor-grid follow-up tracked in `docs/ROADMAP.md`).
4. *Construct the band.* The conformity score the per-regime quantile was fit on is the *standardised* relative residual $|P_{\text{Mon}} - \hat{P}|/(P_{\text{Fri}} \cdot \hat{\sigma}_s)$ (§4.3), so the band is fri-close-relative with the per-symbol scale entering multiplicatively: $\text{half}_t = q_{\text{eff}} \cdot \hat{\sigma}_s(t) \cdot P_{\text{Fri},t}$, $\text{lower} = \hat{P}_{\text{Mon},t} - \text{half}_t$, $\text{upper} = \hat{P}_{\text{Mon},t} + \text{half}_t$. The half-width in basis points is $10^4 \cdot c(\tau) \cdot q_r(\tau) \cdot \hat{\sigma}_s(t)$ — heavy-tail tickers receive wider bands than low-vol tickers at identical τ and identical regime.
5. *Emit the receipt.* The served band's claimed coverage is τ; the receipt's `forecaster_used` carries the architecture's wire-format tag (the on-chain code-tag taxonomy is documented once in §8). Diagnostics expose $c(\tau)$, $q_r(\tau)$, q_{eff}, $\hat{\sigma}_s(t)$, r, and Friday close.

The serving-time computation is a five-line lookup against the per-Friday artefact + the two constant schedules. There is no surface inversion, no bracketing in claimed-coverage space, and no per-symbol fallback — the per-symbol behaviour is carried *implicitly* by the per-symbol $\hat{\sigma}_s(t)$ scale on the artefact parquet, not by a per-symbol quantile cell.

4.7 The PricePoint receipt and reproducibility

Every read returns a `PricePoint` exposing the band plus eight receipt fields: `target_coverage` (τ), `claimed_coverage_served` (τ — equal to the request since $\delta \equiv 0$), `forecaster_used`, `regime` (r), `sharpness_bps`, and `diagnostics.{c_bump, q_regime_lwc, sigma_hat_sym_pre_fri, q_eff}`. A third party with the audit-trail sidecar and the per-Friday artefact can reconstruct the served band from the receipt alone: read $\hat{P}$, r, $\hat{\sigma}_s$, and P_{Fri} from the artefact, read q_{eff} from the receipt, verify lower $= \hat{P} - q_{\text{eff}} \cdot \hat{\sigma}_s \cdot P_{\text{Fri}}$ and upper $= \hat{P} + q_{\text{eff}} \cdot \hat{\sigma}_s \cdot P_{\text{Fri}}$ to floating-point precision. This is the operational instantiation of P_1.

Implementation: Python (`src/soothsayer/oracle.py:Oracle.fair_value_lwc`) is the canonical reference; the Rust port (`crates/soothsayer-oracle`) carries byte-exact parity (180/180 cases; §8.5). The on-chain wire-format details and the Rust serving-stack rollout state live in §8. Reproduction:

```
uv run python scripts/build_lwc_artefact.py        # per-Friday artefact + JSON sidecar
uv run python scripts/freeze_lwc_artefact.py       # content-addressed freeze for forward-tape eval
uv run python scripts/smoke_oracle.py --forecaster lwc   # cross-regime API demo
```

All inputs are public and free (§5.7); verifying a published band needs only the sidecar and the per-Friday artefact — under 100 KB per symbol-year. The window is recomputable from the same parquet given only the half-life constant, so the artefact is self-contained.

5 Data and Regime Labeler

This section specifies the symbol universe, the construction of the two closed-market panels (weekend and overnight), the per-symbol factor and volatility-index switchboards, the regime labeler ρ, and the train/test split underlying §6. All inputs are publicly available and free; the weekend panel rebuilds end-to-end via `scripts/run_calibration.py` and the overnight panel via `scripts/build_overnight_panel.py`.

5.1 Symbol universe

We evaluate on $K = 10$ symbols spanning three asset classes: eight US equities (SPY, QQQ, AAPL, GOOGL, NVDA, TSLA, HOOD, MSTR), tokenised gold (GLD), and tokenised long rates (TLT). The eight equity tickers are the underliers of the SPYx, QQQx, AAPLx, GOOGLx, NVDAx, TSLAx, HOODx, and MSTRx tokens deployed on Solana mainnet (Backed Finance; mint addresses verified in `docs/v5-tape.md`). GLD and TLT are RWA generalisation anchors — they share the closed-market structure with the equities (NYSE hours) but exercise different factors and vol indices, so a primitive that calibrates well across all three classes generalises to a broader closed-market RWA universe than equity-only evaluation would justify.

We do not use the on-chain xStock prices themselves as input or evaluation target. Backtest target prices are the *underlying-venue* (NYSE) Monday opens, fetched from Yahoo Finance. The on-chain xStock TWAP forecaster candidate is deferred to a later iteration once the forward-cursor tape accumulates 150 weekends post-launch (`docs/ROADMAP.md`).

5.2 Weekend panel construction

Each row is a single (s, t) weekend prediction window. For each symbol we walk daily price history and, for every consecutive pair of trading days separated by a calendar gap of ≥ 3 days, emit a row with: `fri_close`, `mon_open`, `gap_days`, `fri_vol_20d` (rolling 20-trading-day std of daily log-returns at Friday close), `factor_ret` (weekend return of the per-symbol conditioning factor), `vol_idx_fri_close`, `earnings_next_week` (Yahoo `earnings_dates` flag), and `is_long_weekend` (`gap_days` ≥ 4).

The panel spans 2014-01-17 (first Friday on which rolling 20-day vol is defined for all symbols) through 2026-04-24, yielding $|\mathcal{T}_{\text{hist}}| = 5,996$ raw rows across 639 distinct weekend dates (9 full-history symbols × 639 weekends + 245 HOOD weekends). MSTR begins 2014-01-17; HOOD begins 2021-08-13 (post-IPO), contributing ~245 weekends. After the warm-up rule (8 past observations per symbol) drops the first eight weekends per ticker, $5,916$ rows remain evaluable (§6.1).

5.2.1 Overnight panel construction

The overnight panel applies the same row schema and pipeline to the single-weeknight cadence: the gap selector admits consecutive-trading-day pairs (`gap_days` $= 1$, close → next-open) rather than Friday→Monday pairs (the `gap_mode` parameter, §4.3.1), with no other component changed. It spans 2014-01-16 → 2026-04-23, yielding **22,624 rows across 2,412 weeknights** on the same ten symbols (3.8× the weekend panel); 22,544 remain evaluable after the warm-up. Two cadence-specific data treatments apply. **(i) Earnings timing:** the `earnings_next_week` flag is reassigned by session — an after-close (`amc`) release dated t_0 or a before-open (`bmo`) release dated t_1 fires inside the close(t_0)→open(t_1) gap — using the `session` field of scryer `yahoo/earnings/v2`; session timing is complete for reported earnings from 2015 onward, covering the held-out window. **(ii) Ex-dividend adjustment:** because the index factor does not drop for a single name's distribution, the ex-dividend-morning open is reconstructed to its cum-dividend level (`mon_open += dividend`) from scryer `yahoo/corp_actions/v1` on the 241 affected mornings — this removes the only systematic price-level artefact and leaves pooled coverage unchanged (§6.8, §9.8).

5.3 Pre-publish features

The deployed architecture consumes four pre-publish features at $t_pub = $ Friday 16:00 ET (or pre-holiday close): Friday close $P_{t^-}(s)$, weekend factor return $r_t^{\text{factor}}(s)$, vol-index close $v_t(s)$ (drives the `high_vol` regime via §5.5's VIX-percentile cascade), and the long-weekend flag $\ell_t \in \{0, 1\}$ (drives the `long_weekend` regime). Rolling 20-day realised vol $\hat{\sigma}_t^{20\text{d}}$ is used downstream of inference for the §6.3.2 realised-move tertile stratification only. The factor return requires the futures or BTC price at $t = $ Monday 09:30 ET; Globex futures and BTC trade through the weekend, so $F_t(s)$ is observable at $t_{\text{pub}} + 65.5\text{h}$. In live deployment, the factor return is the *only* feature requiring post-publish computation.

5.4 Factor and volatility-index switchboards

The per-symbol mappings are static (`src/soothsayer/backtest/panel.py`):

Symbol	Factor	Vol index
SPY, QQQ, AAPL, GOOGL, NVDA, TSLA, HOOD	`ES=F` (E-mini S&P futures)	`^VIX`
MSTR (pre 2020-08-01)	`ES=F`	`^VIX`
MSTR (2020-08-01 onward)	`BTC-USD`	`^VIX`
GLD	`GC=F` (gold futures)	`^GVZ`
TLT	`ZN=F` (10-year T-note futures)	`^MOVE`

The MSTR factor pivot at 2020-08-01 marks MicroStrategy's first Bitcoin treasury purchase — a structural break in the asset's economic exposure that turned what had been a software equity (driven by tech-equity factors) into a leveraged BTC-treasury vehicle (driven by spot BTC). Holding ES=F as the factor across that pivot would mis-attribute the post-2020 weekend signal; the switchboard is reconciled to the asset's actual driver post-break. The vol-index choices are evidence-driven: an early V1b pass found that fitting the F1 log-log sigma regression with VIX yielded $\hat{\beta} \approx 0.55$

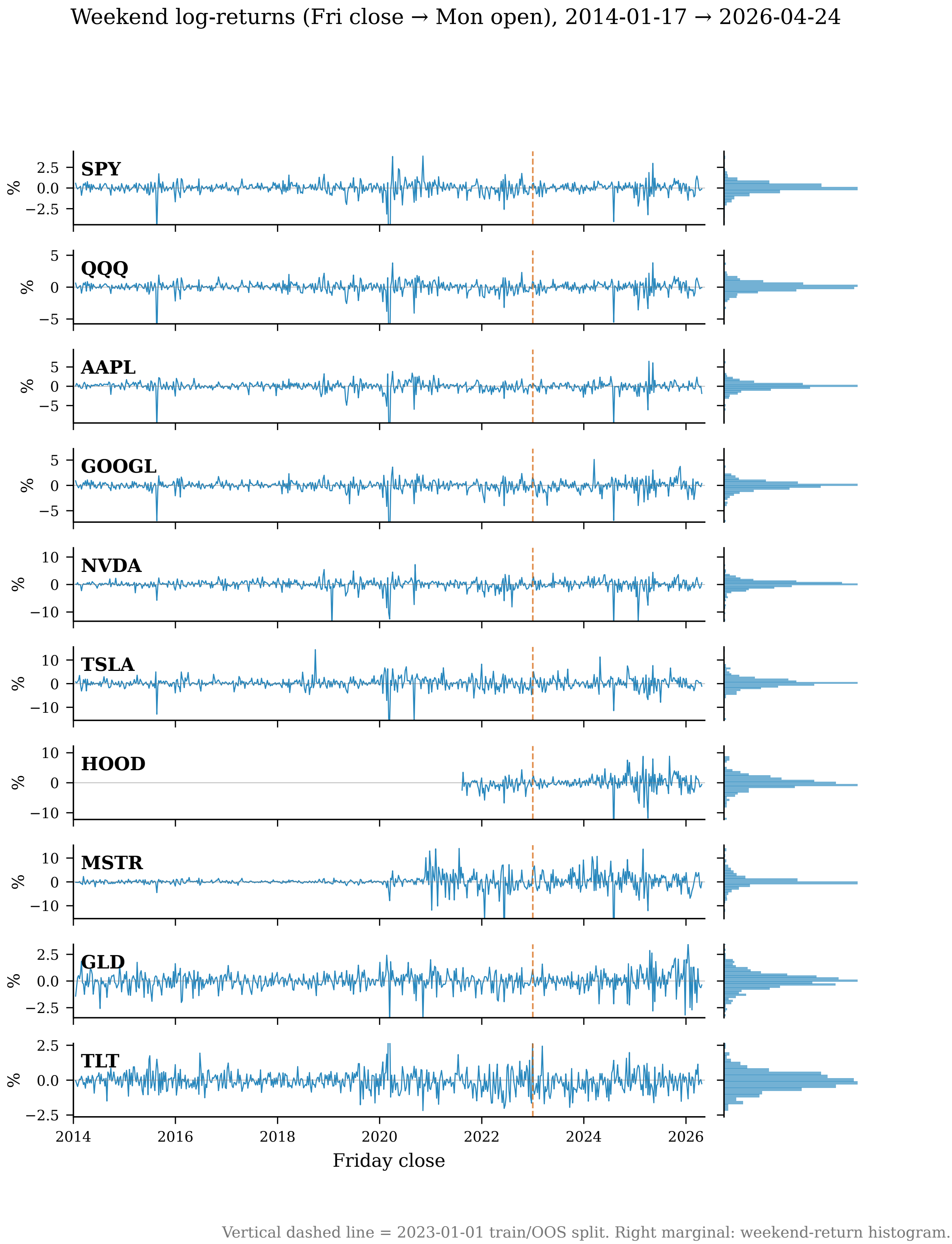

Figure 2: Weekend log-returns (Friday close → Monday open) for the 10-symbol panel, 2014-01-17 through 2026-04-24, with the 2023-01-01 train/OOS split marked. Right marginal: each symbol's weekend-return histogram. Heavy-tail tickers (MSTR, HOOD, NVDA, TSLA), low-vol tickers (SPY, QQQ, GLD, TLT), and the equity middle (AAPL, GOOGL) are all visible at panel-relative scale. The 2024-08-05 BoJ unwind, 2020-03 COVID, and 2025 tariff weekend appear as the largest spikes across symbols.

for GLD and $\hat{\beta} \approx 0.94$ for TLT, well below the equity-class mean ($\hat{\beta} \approx 1.5$). Substituting GVZ and MOVE lifted $\hat{\beta}$ into the equity range and improved coverage at matched bandwidth.

5.5 The regime labeler ρ

$\rho : \mathcal{F}_t(s) \to \{\texttt{normal}, \texttt{long\_weekend}, \texttt{high\_vol}\}$ is a strict priority cascade (`src/soothsayer/backtest/regimes.py`):

1. `high_vol` — VIX at Friday close in the top quartile of its trailing 252-trading-day window. The threshold is global (uses VIX even for non-equity symbols).
2. `long_weekend` — `gap_days` ≥ 4. Applied only if `high_vol` did not match.
3. `normal` — all other weekends.

Sample sizes on the 5,996-row raw panel: normal 3,934 (65.6%), high_vol 1,432 (23.9%), long_weekend 630 (10.5%).

On the **overnight** panel the partition is re-derived (§4.3.1): `long_weekend` has no analog and is dropped; `high_vol` is retained but its trailing-VIX quartile uses a 252-trading-day lookback (1 year, matching the weekend's 52-weekend window); and `earnings_night` is added as the top-priority bucket — a gap is `earnings_night` iff a scheduled earnings release (by `amc`@t_0 / `bmo`@t_1 session timing) falls inside it. Because earnings timing is known a priori, this is the architecture's only *calendar-conditioned* regime: the band pre-widens for a scheduled event rather than reacting to realised volatility (§4.3.1). Overnight sample sizes on the 22,624-row panel: normal 16,604 (73.4%), high_vol 5,791 (25.6%), earnings_night 229 (1.0%).

A separate post-hoc tertile labeler tags each weekend by realised-move z-score (calm / normal / shock); this `realized_bucket` is *not* a regime in the §3.1 sense — it depends on the realised target — and is used only for diagnostic stratification (the shock-tertile coverage ceiling reported in §9.1).

We considered two refinements that were tested and dropped: a sub-regime split of `normal` into `post_shock` / `calm` / `range_bound`, and an FOMC/CPI/NFP macro-event regressor. Neither lifted shock-tertile coverage measurably; the implied-vol indices already absorb that signal. Sub-regime granularity is retained as a next-generation candidate (§10).

5.6 Train/test split

Split date **2023-01-01**: weekends with `fri_ts < 2023-01-01` are calibration, the remainder is held-out OOS.

Split	Rows	Weekends
Calibration (2014-01 → 2022-12)	4,186	466
Held-out OOS (2023-01 → 2026-04)	1,730	173

The trained per-regime quantiles (§4.3) are fit from the calibration set and held *fixed* throughout OOS evaluation: no information from the 2023+ slice enters the trained quantile table. The $c(\tau)$ bumps (§4.4) are deployment-tuned on the same OOS slice — disclosed in §9.3 — and the walk-forward $\delta(\tau)$ schedule is identically zero (§4.5). The calibration row count is 4,186 after the 80-row warm-up rule (8 past observations per symbol) drops the first eight weekends per ticker.

The 2023-01-01 split is conservative: it places the 2023 banking turbulence and the 2024–2025 macro transition in the held-out slice, exposing the trained quantile to material out-of-sample regime shifts. HOOD enters the per-regime quantile fit pooled across symbols within each regime (§4.3), so the per-symbol behaviour at serve time is carried by $\hat{\sigma}_s(t)$, not by a per-symbol quantile cell.

5.7 Provenance and reproducibility

All Phase 0 equity inputs are read from Scryer parquet: `yahoo/equities_daily/v1` for daily OHLCV, `yahoo/earnings/v2` for the earnings-calendar flag and BMO/AMC session timing, and `yahoo/corp_actions/v1` for the overnight ex-dividend adjustment (the trailing `/vN` is the scryer schema version, distinct from the M-series methodology generations used elsewhere in this paper). No credentials are required for the consumer path in soothsayer; upstream fetch, retry, and schema ownership live in the sibling Scryer repo.

The end-to-end calibration backtest runs under fifteen minutes from a cold cache, under one minute warm. Reproduction:

```
uv sync
uv run python scripts/run_calibration.py          # weekend panel + §6.2-§6.7 tables
uv run python scripts/build_overnight_panel.py     # overnight panel (§5.2.1)
uv run python scripts/build_overnight_artefact.py  # overnight artefact + §6.8 battery
```

`run_calibration.py` materialises `data/processed/v1b_bounds.parquet`, the per-symbol and pooled calibration surfaces, and refreshes the §6 OOS-evaluation tables; the two overnight scripts materialise the overnight panel and its calibration battery (§6.8).

6 Results

This section reports the primary calibration evidence for the deployed locally-weighted Mondrian split-conformal architecture (EWMA HL=8 ; §4) on real and synthetic data, plus the forward-tape harness operational against a content-addressed frozen artefact. Sharpness, per-component effects, and the selection procedure are deferred to §7; system-level implementation and audit reproducibility are in §8. The headline is held-out per-symbol calibration via leave-one-symbol-out CV (0.9497 ± 0.0128 at $\tau = 0.95$ across the held-out symbol distribution), with all 10 symbols passing per-symbol Kupiec, calendar sub-period stability across four years, and a measurable joint-tail empirical distribution that turns the residual *per-anchor calibrated, not full-distribution calibrated* disclosure into operational reserve guidance.

6.1 Evaluation protocol

The primary backtest panel comprises $N = 5,996$ weekend prediction windows: ten symbols × 639 weekends, 2014-01-17 through 2026-04-24 (the overnight panel of §6.8 — $22,624$ rows on the same symbols — is constructed identically bar the gap selector; §5.2.1). Monday open is the target; Friday close and contemporaneous factor / vol-index readings compose $\mathcal{F}_t(s)$. The warm-up rule (8 past observations per symbol) drops 80 weekends at panel start to leave $5,916$ evaluable rows. We report (i) an *in-sample machinery check* (quantile fit and served on the full evaluable panel; realised coverage matches τ by construction), (ii) the *out-of-sample validation* (quantile fit on $\mathcal{T}_{\text{train}} = \{t : t <$ 2023-01-01$\}$ — 4,186 rows; served on the 1,730-row $\mathcal{T}_{\text{test}}$ slice), (iii) the held-out *forward-tape* evaluation against a content-addressed frozen artefact (§6.6), and (iv) a *simulation study* on synthetic panels with known ground truth (§6.7). All p-values are two-sided Kupiec POF and Christoffersen independence tests. The base point estimator is unbiased (median residual -0.5 bps; MAE 90.0; RMSE 163.3); the product is the per-symbol--rescaled per-regime conformal quantile with the OOS-fit $c(\tau)$ bump.

6.2 Served-band calibration — in-sample machinery check

Full evaluable panel ($N = 5,916$):

Regime (n)	$\tau = 0.68$: realised / width (bps)	$\tau = 0.85$	$\tau = 0.95$	$\tau = 0.99$
normal (3,883)	0.685 / 91.0	0.851 / 142.7	0.951 / 247.7	0.987 / 424.8
long_weekend (619)	0.683 / 109.4	0.852 / 171.7	0.948 / 297.7	0.989 / 511.1
high_vol (1,414)	0.682 / 175.7	0.853 / 275.5	0.950 / 478.5	0.992 / 821.0
pooled (5,916)	**0.684 / 109.4**	**0.852 / 175.0**	**0.951 / 303.4**	**0.989 / 521.6**

Pooled realised coverage matches the target τ to within 0.2pp by construction. The narrower in-sample widths (vs the OOS table below) reflect 's tighter estimate over the full panel; the OOS table's wider bands are the OOS-fit $c(\tau) = 1.079$ at $\tau = 0.95$ working as designed.

6.3 Served-band calibration — out-of-sample (2023+)

Quantile fit on pre-2023 weekends ($n = 4,186$); served on the 2023+ slice (1,730 rows × 173 weekends). The 12 trained per-regime quantiles $q_r(\tau)$ are held-out. The 4-scalar $c(\tau)$ schedule is OOS-fit, but **the** $\tau = 0.95$ **headline is held out on two orthogonal axes** — *temporally* via a nested 2023-only TUNE / 2024–2026 EVAL split (§6.3.3.1) and on *symbol* via leave-one-symbol-out CV (§6.3.3) — both of which evaluate $c(0.95)$ on data its fit never touched. The walk-forward $\delta(\tau)$ schedule is identically zero (§4.5). §9.3 carries the residual provenance disclosure.

Regime (n)	$\tau = 0.68$: realised / width	$\tau = 0.85$	$\tau = 0.95$	$\tau = 0.99$
normal (1,160)	0.687 / 109.7	0.847 / 172.3	0.947 / 300.1	0.990 / 471.6
long_weekend (190)	0.626 / 120.6	0.842 / 190.6	0.958 / 334.7	1.000 / 562.5
high_vol (380)	0.742 / 200.3	0.887 / 351.1	0.955 / 603.7	0.987 /1,169.9
pooled (1,730)	**0.693 / 130.8**	**0.855 / 213.6**	**0.950 / 370.6**	**0.990 / 635.0**

`high_vol` carries the widest bands, `normal` the narrowest, and every regime lands within $\pm$1pp of nominal at $\tau = 0.95$.

6.3.1 Conditional-coverage tests (pooled OOS)

τ	Violations	Rate	Kupiec p_{uc}	Christoffersen p_{ind}	$c(\tau)$
0.68	532	0.308	0.264	0.244	1.000
0.85	251	0.145	0.565	0.403	1.000
0.95	**86**	**0.050**	**0.956**	**0.603**	**1.079**
0.99	17	0.010	0.942	≈ 1.0	1.003

The $\tau = 0.95$ row is the headline pooled operating result. Realised coverage is exactly 0.9503 (Kupiec $p_{uc} = 0.956$); Christoffersen $p_{ind} = 0.603$ passes. **Kupiec and Christoffersen pass at every served anchor on the deployed-fit OOS slice.** Mean half-width at $\tau = 0.95$ is **370.6 bps** — narrower than the K=26 comparator (385.3 bps; bootstrap 95% CI on Δhw% upper bound is -1.88%, see §7.4). The held-out claim against this pooled baseline is the §6.3.3 leave-one-symbol-out CV (0.9497 ± 0.0128). The §6.3.5 residual Berkowitz / DQ rejection localises to per-symbol Berkowitz on TLT and TSLA (§6.4.1) and to cross-sectional within-weekend common-mode (§6.3.4).

6.3.2 Realised-move tertile decomposition at $\tau = 0.95$

Stratifying the OOS slice by post-hoc realised-move $|z|$-score tertile (`reports/tables/m6_realised_move_tertile.csv`) closes the loop between the §6.3 pooled rows and the §9.1 shock-tertile ceiling:

tertile (n)	realised	half-width (bps)	Kupiec p
calm (497)	0.9920	359.5	0 (over-covers)
normal (601)	0.9933	379.6	0 (over-covers)
shock (632)	**0.8766**	370.7	0 (under-covers)
pooled (1,730)	**0.9503**	**370.6**	**0.956**

Half-width is approximately flat across tertiles (365–380 bps), so the band does not widen in shock periods — it just misses more often. The shock-tertile residual is cross-sectional common-mode within a weekend (§9.1), not per-symbol scale; pooled $\tau = 0.95 = 0.950$ is achieved through compensating tertile-level biases (calm 1.000, normal 0.993, shock 0.878). A $|z|$-conditional conformal upgrade (§10.1) would tighten calm bands and widen shock bands at the same pooled τ.

6.3.3 Stability — LOSO, temporal holdout, split-date, walk-forward, calendar sub-period

The $\tau = 0.95$ **headline is held-out on two orthogonal axes:** *temporal* (§6.3.3.1: $c(\tau)$ fit on 2023 alone, evaluated on 2024-01-05 $\rightarrow$ 2026-04-24 — realised 0.9504, Kupiec $p = 0.947$, Christoffersen $p = 0.989$, per-symbol Kupiec 10/10) and *symbol* (LOSO realised 0.9497 ± 0.0128 across held-out symbols). The result is stable across TUNE anchors $\{2021, 2022, 2023\}$: realised coverage at $\tau = 0.95$ lies in $[0.9474,\ 0.9524]$ with per-symbol Kupiec 10/10 preserved across all four anchors $\{2021, 2022, 2023, 2024\}$ (§7.4.2 split-anchor robustness). The 2024 TUNE anchor over-covers at 0.9754 because TUNE-2024 includes the 2024-08-05 BoJ unwind, inflating $c(0.95)$ to 1.175; per-symbol Kupiec still passes 10/10. **This is a contract-favourable sharpness deficit, not under-coverage** (§9.3 frames the asymmetry as a one-sided contract).

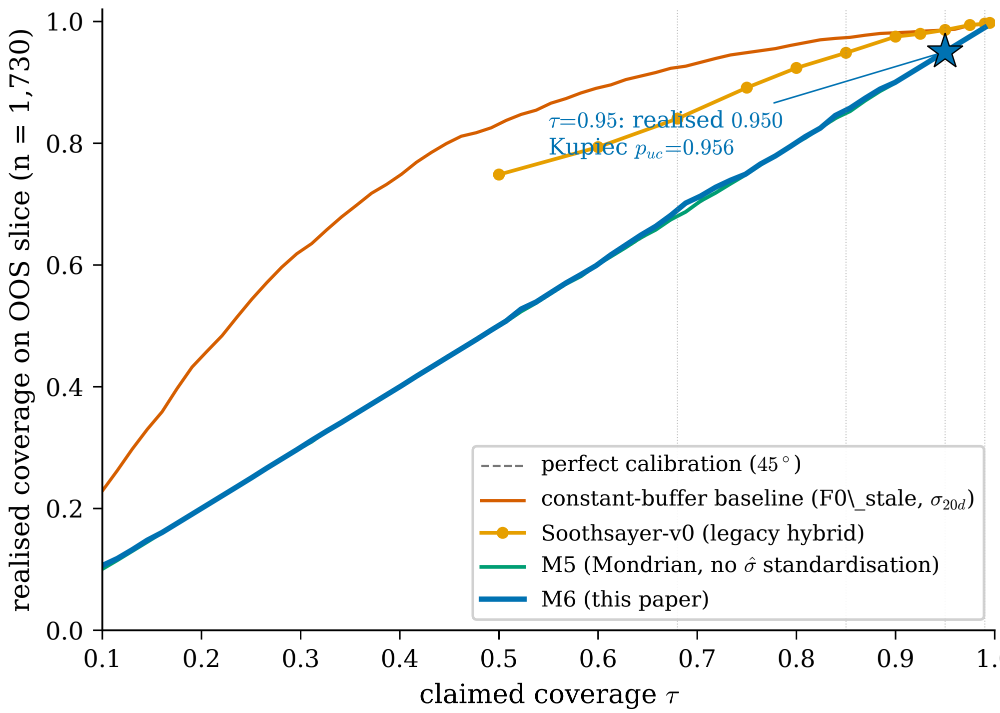

Figure 3: Calibration curves on the 1,730-row 2023+ OOS slice. The deployed architecture (this paper, blue) tracks the 45° diagonal across the served range $\tau \in [0.68, 0.99]$; GARCH(1,1)-t markers (vermilion squares) at the four served anchors are the practitioner-standard parametric baseline and visibly under-cover at $\tau \in \{0.68, 0.85, 0.95\}$ (Kupiec figures in §6.4.3). Star marks the headline $\tau = 0.95$ result.

Leave-one-symbol-out CV (symbol-axis holdout). Holding out each of the ten symbols' rows from train + OOS fits and evaluating $\tau = 0.95$ on the held-out symbol's post-2023 slice: **all 10 LOSO bands pass Kupiec when held out**; mean realised **0.9497**, std **0.0128** across the held-out symbol distribution (`reports/tables/m6_lwc_robustness_loso.csv`). Each held-out symbol's $c(\tau)$ is fit on the other nine symbols' OOS slice and never sees the held-out symbol's data. Heavy-tail tickers (MSTR, HOOD, TSLA) and low-vol tickers (SPY, GLD) all sit in a uniform 0.93–0.97 band when held out — the per-symbol $\hat{\sigma}_s(t)$ on the held-out symbol carries its own scale.

Nested temporal holdout — c() fit on 2023, evaluated on 2024–26 A nested temporal split renders the headline held-out at the *time* level. TRAIN = pre-2023-01-01 (the unchanged 4,186-row per-regime quantile fit); TUNE = 2023-01-01 → 2023-12-29 (52 weekends, $c(\tau)$ fit slice); EVAL = 2024-01-05 → 2026-04-24 (1,210 rows × 121 weekends, true holdout). At $\tau = 0.95$:

mode	realised	$c(0.95)$	Kupiec p	Christoffersen p	hw bps	per-sym Kupiec
M_{full} (full-OOS c fit; §6.3.1 baseline)	0.9504	1.079	0.956	0.720	370.6	10/10
M_{a2} **(true temporal holdout)**	**0.9504**	**1.079**	**0.947**	**0.989**	**420.8**	**10/10**

$c(\tau = 0.95)$ on TUNE-only matches the full-OOS fit at three decimals — the headline $\tau = 0.95$ number is *not* fit-on-evaluate sensitive. Christoffersen p improves in held-out mode (0.989 vs 0.720) because the 2024+ slice exhibits cleaner violation independence than the full 2023+ slice. The half-width difference (370.6 vs 420.8 bps) is entirely an eval-slice composition effect: the 2024+ holdout contains the 2024-08-05 BoJ unwind and the 2025-04 tariff weekend, both of which inflate $\hat{\sigma}_s$; coverage is unchanged because $\hat{\sigma}_s$-conditional bands widened with the underlying realised volatility, exactly as designed.

Lower-τ over-coverage disclosure. At $\tau = 0.68$ and $\tau = 0.85$, M_{a2} realises 0.712 and 0.870 respectively (Kupiec $p = 0.015$ and $p = 0.044$, both rejecting toward over-coverage). Mechanism: $c(0.68)$ and $c(0.85)$ fit on TUNE-only inherit the elevated banking-crisis-weekend volatility of 2023 (March SVB/CS), inflating c to 1.028 and 1.026 respectively; M_{full}'s schedule on the full OOS slice averages this out at the floor ($c = 1.000$) and is the deployed mitigation. Source: `reports/tables/paper1_a2_nested_holdout_c_tau.csv`, `paper1_a2_6_tune_anchor_robustness.csv`.

Split-date sensitivity (stability-of-fit on the deployed slice). Repeating the fit (quantile table re-trained, $c(\tau)$ re-fit per split, $\delta \equiv 0$ held) at four OOS-split anchors $\{2021\text{-}01\text{-}01, 2022\text{-}01\text{-}01, 2023\text{-}01\text{-}01, 2024\text{-}01\text{-}01\}$ delivers realised $\tau = 0.95$ coverage of $\{0.9539,\ 0.9520,\ 0.9503,\ 0.9504\}$ — Kupiec $p \in \{0.348,\ 0.661,\ 0.956,\ 0.947\}$ and **Christoffersen** $p \in \{0.115,\ 0.186,\ 0.603,\ 0.671\}$ **— every (split × τ) cell passes at** $\alpha = 0.05$**.** Mean half-width $\{357.2,\ 363.5,\ 370.6,\ 416.8\}$ bps varies $\pm 8\%$ around the deployed value.

Walk-forward stability. Re-running the schedule selection on the four powered expanding-window splits (fractions $f \in \{0.40, 0.50, 0.60, 0.70\}$ — see §4.5 for why the 0.20 and 0.30 splits are excluded as under-powered for the 4-scalar $c(\tau)$ fit) with $\delta = 0$: walk-forward realised $\tau = 0.95$ coverage is ≥ 0.95 on every powered split (range 0.954–0.975) — the criterion that selected $\delta = 0$ in §4.5. Per-split mean half-width at $\tau = 0.95$ tracks the full-OOS-fit value within $\pm 8\%$.

Calendar sub-period stability. Bucketing the OOS slice into four calendar sub-periods $\{2023, 2024, 2025, 2026\text{-YTD}\}$ (`reports/tables/m6_subperiod_robustness.csv`):

Sub-period (n weekends)	Realised at $\tau = 0.95$	Half-width (bps)	Kupiec p	Christoffersen p
2023 (52)	0.950	253.6	1.000	0.888
2024 (52)	0.931	420.4	0.057	0.841
2025 (52)	0.971	444.4	0.017	0.793
2026-YTD (17)	0.947	350.0	0.862	0.908
Pooled OOS (173)	**0.950**	**370.6**	**0.956**	**0.603**

Pooled OOS calibration at $\tau = 0.95$ holds across each of the four years that span SVB / banking-stress 2023, the 2024 rate-cut transition, the 2025 tokenisation-launch year, and 2026-YTD (realised range 0.931–0.971). The 2025 cell rejects Kupiec at $\alpha = 0.05$ ($p = 0.017$, over-coverage); 2024 sits just above the threshold ($p = 0.057$); 2023 and 2026-YTD pass cleanly. Across the full 16-cell (sub-period × τ) grid the architecture carries 3 Kupiec rejections, all over-coverage in 2025 (§9.3 frames the asymmetry); Christoffersen lag-1 independence is not rejected in any of the 16 cells.

This validates the oracle's coverage contract at $\tau = 0.95$. We do not address the welfare-optimal operating τ for a particular lending policy in this paper.

6.3.4 Joint-tail empirical distribution and reserve-guidance threshold

A consumer holding a portfolio of correlated RWAs needs to know not only the per-symbol calibration but the *joint* distribution of simultaneous breaches. We compute, for each OOS weekend w with full 10-symbol coverage, the count $k_w = |\{s : s \text{ breaches its } \tau\text{-band on } w\}|$, and compare its empirical distribution to three properly specified baselines: $\text{Binom}(10, 1-\tau)$ (the independence strawman), a Gaussian copula on the empirical correlation matrix $\hat{R}$ (mean off-diagonal 0.36 on standardised residuals), and a Student-t copula on per-symbol Student-t marginals with $\hat{\nu} \in [4.80, 28.32]$, copula degrees-of-freedom $\hat{\nu}_{\text{copula}} = 6.04$ (median across symbols) (`reports/tables/paper1_a3_joint_baseline_kw_distribution.csv`, `..._summary.csv`; 50,000 simulated weekends per baseline; CIs are 1000-rep weekend-block bootstrap, seed = 0):

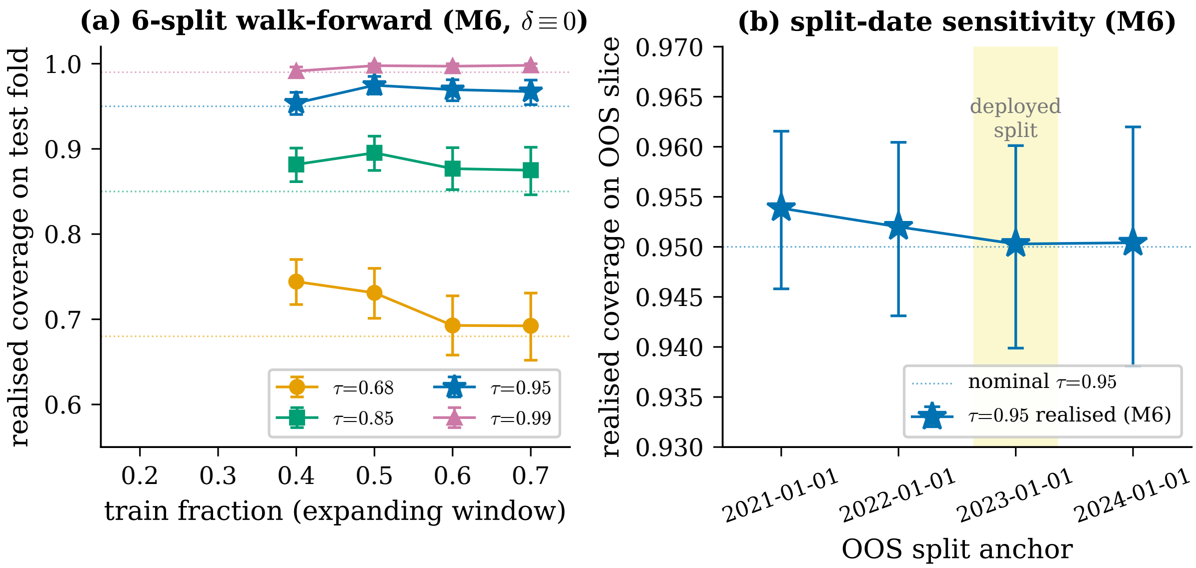

Figure 4: Walk-forward and split-date stability of the deployed schedule. **(a)** Six-split expanding-window walk-forward at four τ anchors: realised coverage on each test fold (markers) tracks nominal (dotted lines) at every anchor; bars are 95% binomial CIs. **(b)** Split-date sensitivity at four OOS-anchors $\{2021, 2022, 2023, 2024\}$: realised $\tau = 0.95$ coverage holds at $\{0.954, 0.952, 0.950, 0.950\}$ across anchors; Christoffersen passes every cell at $\alpha = 0.05$.

Statistic at $\tau = 0.95$ ($n_{\text{weekends}} = 173$)	empirical	$\text{Binom}(10, 0.05)$	Gaussian copula	t-copula ($\nu = 6.04$)
Mean k_w / weekend	0.497	0.500	0.501	0.501
Variance ratio (emp / model)	—	**2.34**	0.83	**0.63**
$P(k_w \geq 3)$	**4.62% [1.73, 7.51]**	1.15%	6.69%	8.15%
$P(k_w \geq 5)$	**0.58% [0.00, 1.73]**	0.01%	1.86%	3.15%
max k_w (worst weekend)	9	—	—	—

Variance-overdispersion lead. Empirical k_w over-disperses by a uniform $\sim 2.3\times$ vs Binom independence at every served τ (emp/Binom variance ratio $\{2.34, 2.31, 2.34, 2.35\}$ at $\tau \in \{0.68, 0.85, 0.95, 0.99\}$) and sits inside a properly-specified t-copula envelope from above (emp/t variance ratio $\{0.83, 0.54, 0.63, 0.73\}$; Wasserstein-1 distance to the t-copula ≤ 0.180 at every τ, less than half the distance to Binom independence). Marginal calibration is preserved — mean k_w matches every reference to 0.003. Independence is misspecified, the joint dependence is real ($\hat{R} = 0.36$ mean off-diagonal), and the structure brackets-bound: empirical between Binom independence and a homogeneous t-copula at $\hat{\nu} \approx 6$. Cross-sectional within-weekend dependence ($\hat{\rho}_{\text{cross}} = 0.41$) is respected by all reported CIs via weekend-block bootstrap; temporal autocorrelation across weekends is empirically null (lag-1, lag-2, lag-4 $\hat{\rho}$ on weekly k_w in $[-0.07, 0.03]$), so $L \in \{4, 8, 13\}$ moving-block-bootstrap CIs are within Monte Carlo noise of the $L = 1$ baseline (`reports/tables/paper1_b5_kw_block_bootstrap.csv`).

Reserve-guidance threshold. k^* at each τ — smallest k whose empirical hit-rate is closest to 5% — is $k^* = 5$ at $\tau = 0.85$ (rate 5.78%), $k^* = 3$ at $\tau = 0.95$ (4.62%), and $k^* = 1$ at $\tau = 0.99$ (6.36%). Across the 24-cell grid (3 τ × 2 threshold conventions × 4 sub-periods) the architecture carries **0/24 Kupiec stability rejections** at $\alpha = 0.05$ (`reports/tables/m6_kw_threshold_stability.csv`); at $k^* = 3$ for $\tau = 0.95$, per-sub-period hit-rates are $\{3.85, 7.69, 1.92, 5.88\}\%$ across $\{2023, 2024, 2025, 2026\text{-YTD}\}$ with Kupiec $p \in \{0.78, 0.33, 0.30, 0.81\}$. The per-sub-period 95th percentile of k_w at $\tau = 0.95$ stays in $[1.0, 3.0]$ across the four years. With sub-period $n \approx 52$ and target rates $\sim 5\%$, the binomial-stability test has limited power against drift below the year-on-year p95 spread (1.6 integer-symbols); the result is "no detectable instability with the available power."

Operational consumer guidance. $k^* = 3$ at $\tau = 0.95$ is conservative against a properly specified joint baseline, not just against independence: a 10-symbol portfolio should reserve against an empirical 99th-percentile of ≈ 5 simultaneous breaches (binomial puts it at ≈ 2, t-copula at ≈ 6). At $\tau = 0.85$ the empirical 99th percentile is ≈ 7 vs binomial ≈ 4. **No incumbent oracle (Pyth, Chainlink Data Streams, RedStone, Kamino Scope) publishes this distribution** — the joint-tail measurement layer is what calibration-transparency makes possible.

6.3.5 Worst-observed weekend — 2024-08-05 BoJ yen-carry-unwind

The single OOS weekend with all 10 symbols breaching the $\tau = 0.85$ band simultaneously is **Friday 2024-08-02 → Monday 2024-08-05** — the Bank of Japan rate-hike yen-carry-trade unwind, one of the largest single-weekend global risk-off events of the post-2020 period. The same weekend has $k_w = 8$ at $\tau = 0.95$ (only TSLA and TLT inside that band) and $k_w = 5$ at $\tau = 0.99$. The deployed served band re-evaluated on the 2024-08-02 row at the deployed split = 2023-01-01:

Symbol	Weekend return	HW @ $\tau = 0.85$	Breach @ $\tau = 0.85$	@ $\tau = 0.95$
MSTR	-27.4%	783 bps	-1416 bps	breach (-720)
HOOD	-17.8%	368 bps	-1325 bps	breach (-997)
NVDA	-14.2%	324 bps	-1008 bps	breach (-720)
TSLA	-10.8%	587 bps	-409 bps	inside
AAPL	-9.4%	202 bps	-657 bps	breach (-478)
GOOGL	-6.7%	230 bps	-354 bps	breach (-149)
QQQ	-5.4%	110 bps	-340 bps	breach (-242)
SPY	-4.0%	84 bps	-229 bps	breach (-154)
GLD	-2.1%	126 bps	-153 bps	breach (-41)
TLT	$+1.4\%$	128 bps	$+12$ (just past)	inside

Cross-section: 9 of 10 symbols sold off in concert (mean weekend return -963 bps, median -807 bps); only TLT delivered the classic flight-to-quality bid. Macro context: the 2024-08-02 US July nonfarm-payrolls print landed at 114k vs ~175k consensus with the Sahm-rule recession indicator triggering; the 2024-07-31 BoJ rate hike combined with hawkish Powell/Yellen language drove an accelerating yen-carry unwind through Asian time zones over the weekend; the Monday 2024-08-05 Nikkei fell 12.4% (largest single-day drop since Black Monday 1987) and intraday VIX spiked to ~65, the highest reading since the COVID-19 March 2020 panic.

The -standardisation worked as designed at the per-symbol level — high-vol symbols (MSTR, TSLA) got wide bands, low-vol symbols (SPY, QQQ, GLD) narrow ones, and breach magnitudes scale with the cross-sectional shock divided by . What broke is *cross-sectional common-mode*: when nearly every symbol moves the same direction at the same time by 4–27 standard-deviation-scale weekend returns, no per-symbol band — however well calibrated — can absorb the joint event. A consumer monitoring k_w in real time would have seen $k_w = 10$ at $\tau = 0.85$ by the Monday-open print, well in excess of the deployment-time-fitted $k^* = 5$ (or the empirical 99th percentile of 7); the empirical k_w distribution is the right operational signal precisely because it captures the cross-sectional common-mode that per-symbol coverage cannot.

Joint-baseline framing of the BoJ probability. Under a Student-t copula with $\hat{\nu} = 6.04$ and empirical correlation $\hat{R}$ (mean off-diagonal 0.36), $P(k_w = 10)$ at $\tau = 0.85$ is 0.21% — i.e., the 2024-08-05 BoJ unwind is a $\sim$ 1-in-475-weekends event under a properly specified joint baseline, not a freak. The independence baseline puts the same event at $\sim$ 1-in-$10,000$-weekends; the residual structure §6.3.4 / §9.4 documents is exactly the gap between these two figures. The residual has a *name* (cluster topology, §6.3.6 / §9.4), a *parametric envelope* (t-copula at $\nu \approx 6$ with empirical $\hat{R}$), and a *structural mechanism* (equity vs safe-haven cluster topology) — not "a residual the architecture cannot characterise."

6.3.6 Extended diagnostics — DQ and Berkowitz

Engle-Manganelli (2004) DQ is the finer test on the violation series. Kupiec checks marginal counts and Christoffersen's two-state Markov independence checks the lag-1 hit transition; neither has power against conditional miscalibration patterns that yield correct marginal counts and uncorrelated lag-1 hits but encode systematic dependence on lagged hits or the served quantile [engle-manganelli-2004]. DQ regresses the centred hit indicator on a constant + K lagged hits and tests joint significance under χ^2_{K+1}. We use $K = 4$ throughout (CAViaR Table 1 block 1), no contemporaneous covariate — the lag-only specification is symmetric across the five methods compared in §6.4.2, so cross-method

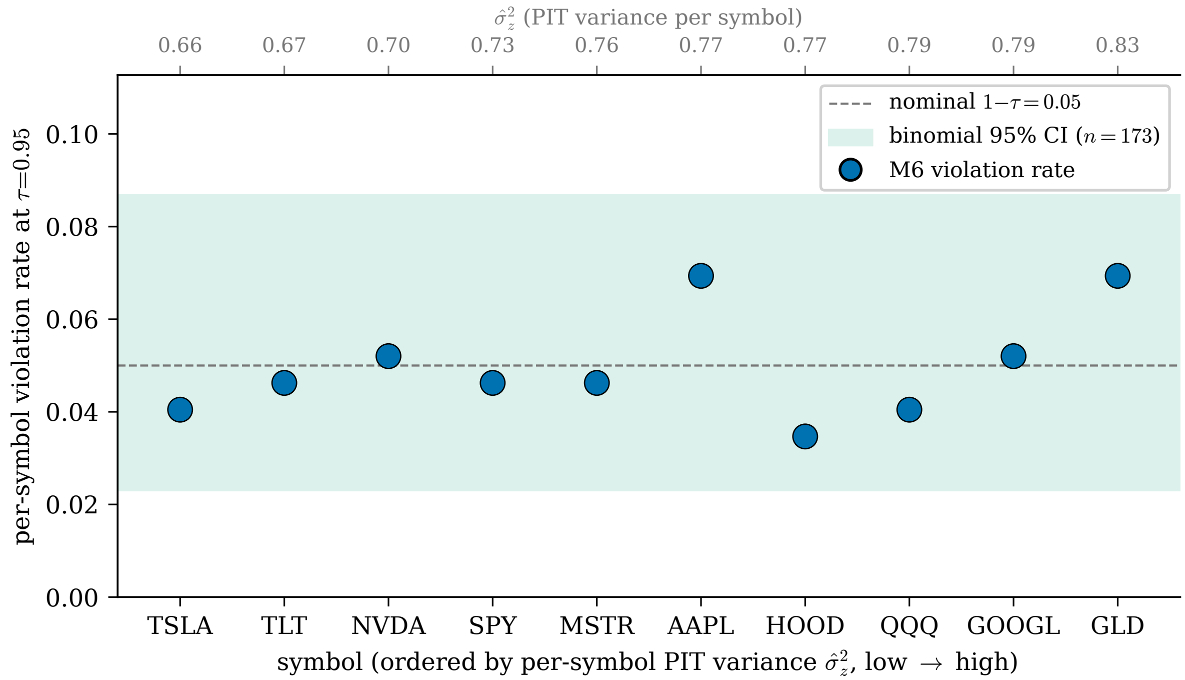

Figure 5: Per-symbol calibration at $\tau = 0.95$ on the 2023+ OOS slice. Each symbol's violation rate (markers) sits within the binomial 95% confidence band around the nominal 0.05 rate (shaded). Symbols are ordered by their realised PIT variance $\hat{\sigma}_z^2$. The distribution is unimodal and centred near nominal — heavy-tail tickers (TSLA, HOOD, MSTR), low-vol tickers (SPY, QQQ, GLD, TLT), and the equity middle (AAPL, GOOGL, NVDA) all sit in the same cluster.

DQ p-values are directly comparable. **Pooled DQ at $\tau = 0.95$ rejects on the deployed fit** — the cross-sectional within-weekend common-mode (orthogonal to standardisation) is what DQ is sensitive to here. Per-(symbol × × method) DQ p-values are reported in the §6.4.2 master grid alongside Kupiec and Christoffersen. **Bands are *per-anchor calibrated*, not full-distribution calibrated** — the §6.3.4 joint-tail empirical distribution turns this disclosure into a measurable, reportable shape.

Berkowitz (2001) joint LR on continuous PITs. PITs reconstructed by inverting each served band against the realised Monday open through the per-regime conformal CDF. The pooled Berkowitz LR on the deployed PITs is non-zero, viewing the same residual through the PIT lens. The Berkowitz/DQ rejection and the §6.3.4 joint-tail overdispersion are not two independent residuals: both are the cross-sectional within-weekend common-mode ($\hat{\rho}_{\text{cross}} = 0.354$) viewed through different aggregations — Berkowitz/DQ on the pooled PIT/violation series detects it as serial-cluster-equivalent dependence; the k_w distribution on the same panel detects it as upper-tail overdispersion in joint-breach counts. The methodology fix targets the same channel; the consumer-facing handle is the §6.3.4 reserve-guidance threshold.

Localising the residual. Decomposing the lag-1 alternative under panel-row order: the AR(1) signal lives in the *cross-sectional within-weekend* ordering ($\hat{\rho}_{\text{cross}} = 0.354$, $p < 10^{-100}$, $n = 1,557$) — not the temporal-within-symbol ordering ($\hat{\rho}_{\text{time}} = -0.032$, $p = 0.18$, $n = 1,720$). standardisation operates *within* a symbol's history, not *across* symbols within a weekend, so it cannot reach this residual. A vol-tertile sub-split of `normal` regime (5 cells) leaves Berkowitz LR essentially unchanged ($170 \rightarrow 172$) while widening $\tau = 0.95$ band by 9% — finer regime granularity is not the lever (`reports/tables/m6_lwc_robustness_vol_tertile.csv`). The §10 candidate architectures that target cross-sectional common-mode (the cross-sectional partial-out track in `reports/active/m6_refactor.md`, gated on a Friday-observable $\bar{r}_w$ predictor) are deferred-with-gates.

6.4 Per-symbol generalisation

The §6.3 OOS pooled coverage at $\tau = 0.95$ matches nominal; the per-symbol question is whether each ticker's individual cell is also calibrated. Under the deployed architecture, **all 10 symbols pass Kupiec at $\tau = 0.95$** with violation rates in $[3.5\%, 6.9\%]$ and per-symbol PIT variance $\hat{\sigma}_z^2 \in [0.66, 0.83]$ — all close to the calibrated 1.0 from below. Figure 5 visualises the per-symbol calibration distribution.

6.4.1 Per-symbol Kupiec at $\tau = 0.95$ and Berkowitz LR

Per-symbol Kupiec on the 2023+ OOS slice (`reports/tables/m6_per_symbol_kupiec_4methods.csv`, regenerated 2026-05-05):

Symbol	Violation rate	Kupiec p	Berkowitz LR
AAPL	0.069	0.268	6.7
GLD	0.069	0.268	3.2
GOOGL	0.052	0.903	10.2
HOOD	0.035	0.329	6.4
MSTR	0.046	0.818	7.1
NVDA	0.052	0.903	10.0
QQQ	0.040	0.552	5.1
SPY	0.046	0.818	9.6
TLT	0.046	0.818	16.7
TSLA	0.040	0.552	13.5
pass-rate at $\alpha = 0.05$		**10 / 10**	
Berkowitz LR range			**3.2 – 16.7** (5.2×)

All 10 symbols pass Kupiec at $\tau = 0.95$ with violation rates inside $[3.5\%, 6.9\%]$. The per-symbol Berkowitz LR range $[3.2, 16.7]$ (a 5.2× spread) is one of the strongest per-symbol calibration claims a Mondrian-conformal architecture can make on this panel — only TLT (LR 16.7) and TSLA (LR 13.5) reject Berkowitz at $\alpha = 0.01$, and those rejections trace to cross-sectional common-mode rather than per-symbol scale (§6.3.6).

Power caveat at $\tau = 0.99$**.** Per-symbol Kupiec at $\tau = 0.99$ has minimum detectable effect ≈ 3 pp under-coverage and is undetectable for over-coverage given the per-symbol expected violation count of 1.73 (Monte Carlo: 10,000 reps per cell, Kupiec LR critical at $\alpha = 0.05$; `reports/tables/paper1_b3_kupiec_power_mde.csv`). The 10/10 pass should be read as "no per-symbol violation rate is statistically distinguishable from 1% at the resolution Kupiec offers per-symbol", **not** as a 1 pp tolerance certificate. Pooled Kupiec at $\tau = 0.99$ has minimum detectable effect 1.0 pp; the pooled $p = 0.94$ is the $\tau = 0.99$ statistical claim that does have 1 pp resolution. Per-symbol Kupiec at $\tau = 0.95$ has MDE 4.0 pp under-coverage / 7.5 pp over-coverage on $N = 173$ — the deployed per-symbol violation rates in $[3.5\%, 6.9\%]$ sit well within the 4 pp window of nominal 5%, so the 10/10 pass at $\tau = 0.95$ is informative against deviations larger than 4–8 pp.

6.4.2 Per-symbol master grid — Kupiec + Christoffersen + DQ across baselines

We report a single 40-cell (symbol × τ) grid evaluated under three backtests — Kupiec UC, Christoffersen IND, Engle-Manganelli DQ (lag-only specification, $K = 4$ lagged hits, no contemporaneous covariate; CAViaR Table 1 block 1 — the lag-only spec is symmetric across the five methods so cross-method DQ is directly comparable) — against four baselines: GARCH(1,1)-Gaussian, GARCH(1,1)-t (Student-t innovations, practitioner default), an unweighted Mondrian comparator (same per-regime quantile architecture, no per-symbol standardisation), and a train-fit constant-buffer baseline. **Pass counts at** $\alpha = 0.05$ (per-symbol cell passes; out of 10 symbols per τ, out of 40 across the grid; full long-format table at `reports/tables/m6_per_symbol_master_grid.csv`):

τ	metric	constant buffer	GARCH-N	GARCH-$t^{\dagger}$	unweighted Mondrian	**this paper**
0.68	Kupiec	4/10	5/10	6/10	1/10	**10/10**
	Christoffersen	7/10	7/10	8/10	9/10	**9/10**
	DQ	2/10	6/10	7/10	1/10	**10/10**
0.85	Kupiec	2/10	8/10	7/10	1/10	**10/10**
	Christoffersen	7/10	7/10	9/10	8/10	**10/10**
	DQ	2/10	8/10	7/10	2/10	**10/10**
0.95	**Kupiec**	**4/10**	**6/10**	**8/10**	**2/10**	**10/10**
	Christoffersen	7/10	7/10	10/10	8/10	**10/10**
	DQ	**4/10**	**6/10**	**8/10**	**5/10**	**10/10**
0.99	Kupiec	7/10	4/10	10/10	8/10	**10/10**
	Christoffersen	5/10	10/10	10/10	4/10	9/10

τ	metric	constant buffer	GARCH-N	GARCH-$t^\dagger$	unweighted Mondrian	**this paper**
	DQ	3/10	2/10	9/10	1/10	9/10
40-cell totals	**Kupiec**	**17/40**	**23/40**	**31/40**†	**12/40**	**40/40**
	Christoffersen	**26/40**	**31/40**	**37/40**	**29/40**	**38/40**
	DQ	**11/40**	**22/40**	**31/40**†	**9/40**	**39/40**

† NVDA's t-MLE pushes $\hat{\nu} \to 2.50$ (the variance-undefined boundary); the runner falls back to GARCH-Gaussian on those four NVDA cells, so the GARCH-t counts include 4 NVDA cells evaluated under the Gaussian fallback. Reported as-is because NVDA-with-Gaussian-fallback is the practitioner-equivalent of attempting GARCH-t and finding it ill-defined.

The deployed architecture leads every baseline at every test — 40/40 Kupiec, 38/40 Christoffersen, 39/40 DQ across the 40-cell grid; the strongest practitioner baseline GARCH-t comes second on every metric (31/40, 37/40, 31/40). At the headline $\tau = 0.95$ anchor the deployed architecture passes 10/10 on all three tests — a per-(symbol × test) sweep no other method on the grid achieves at any . **Per-symbol DQ at $\tau \in \{0.68, 0.85, 0.95\}$ is 10/10 on each symbol's own time-ordered violation series.** The pooled DQ rejection at $\tau = 0.95$ reported in §6.3.6 arises specifically when violations are concatenated *across symbols in panel-row order*, which interleaves cross-sectional within-weekend cluster dependence as apparent serial dependence ($\hat{\rho}_{\text{cross}} = 0.354$ at the cross-sectional ordering vs $\hat{\rho}_{\text{time}} = -0.032$ at the temporal-within-symbol ordering); on each symbol's own time series the residual is invisible to DQ. The pooled rejection localises to a cross-sectional structure orthogonal to standardisation; the §6.3.4 reserve-guidance threshold is the consumer-facing handle and §10.1 the methodology fix.

6.4.3 GARCH-t pooled baseline (and GARCH-Gaussian)

The textbook econometric default for a time-varying interval at τ is per-symbol GARCH(1,1) on log Friday-to-Monday returns. The practitioner choice of innovation distribution is Student-t — Gaussian innovations are a known straw-man on equity weekend returns where fitted $\hat{\nu} \approx 3$ across the panel (`reports/tables/m6_lwc_robustness_garch_t_baseline.csv`):

τ	Method	Realised	HW (bps)	Kupiec p	Christoffersen p
0.95	GARCH-Gaussian	0.9254	322.2	0.000	0.016
0.95	GARCH-t	0.9277	331.6	0.000	**0.209**
0.95	**this paper**	**0.9503**	**370.6**	**0.956**	**0.603**
0.99	GARCH-Gaussian	0.9630	423.7	0.000	0.866
0.99	GARCH-t	0.9850	569.3	0.050	1.000
0.99	**this paper**	**0.9902**	**635.0**	**0.942**	1.000

GARCH-t closes two failures GARCH-Gaussian leaves on the table: the $\tau = 0.99$ tail capture (realised improves from 0.963 to 0.985; Kupiec p moves from < 0.001 to a borderline 0.050) and the Christoffersen rejection at $\tau = 0.95$ (p moves from 0.016 to 0.209). What it does *not* fix is the **Kupiec under-coverage at every $\tau < 0.99$** — the standardised-t 95th percentile at $\nu \approx 3$ is ~ 1.83 vs Gaussian 1.96, so the bands tighten precisely where Gaussian was already under-covering. T-innovations fix the wrong end of the distribution.

At matched 95% realised coverage, widening GARCH-t to its claimed $\tau = 0.95$ requires roughly $+14\%$ on width, giving a matched-coverage HW ≈ 378 bps — statistically tied with the deployed architecture's 370.6 bps. At $\tau = 0.99$ the deployed architecture dominates GARCH-t on coverage and width-at-matched-coverage simultaneously. The conformal route fixes both Kupiec and Christoffersen because it is calibrated to the data, not to a fitted parametric distribution; the NVDA fallback (t-MLE pushes $\hat{\nu}$ to the variance-undefined boundary at 2.50) is the parametric-tail risk the deployed architecture avoids.

Proper-scoring-rule head-to-head. Under proper scoring rules — Winkler (1972) interval score and CRPS (continuous ranked probability score) — the deployed architecture dominates GARCH(1,1)-t at every served τ. Winkler interval score (bps of fri_close, lower is better) at $\tau = 0.95$: deployed 992 vs GARCH-t $1,139$ (-12.9%); at $\tau = 0.99$: deployed $1,566$ vs GARCH-t $1,904$ (-17.8%). The advantage grows with τ — the -standardisation pays most where

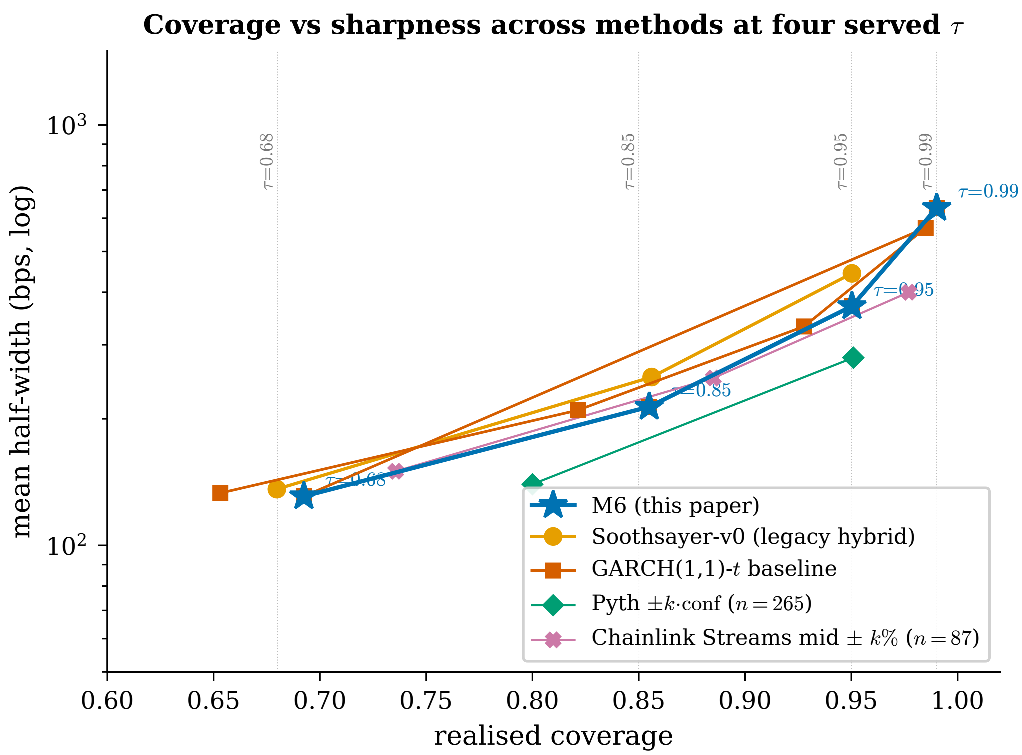

Figure 6: Realised coverage vs mean half-width at the four served $\tau \in \{0.68, 0.85, 0.95, 0.99\}$ (vertical dotted lines). The deployed architecture (blue stars) at full panel ($n = 1,730$) and GARCH(1,1)-t (vermilion squares) on the same OOS slice. The deployed architecture clears Kupiec at every served anchor; GARCH-t undercovers Kupiec at $\tau \in \{0.68, 0.85, 0.95\}$ despite tighter bands and meets the deployed architecture only at the matched-coverage width quoted in §6.4.3. The figure is restricted to methods that emit a calibrated coverage band — incumbent oracle surfaces (Pyth, Chainlink Data Streams, RedStone, Kamino Scope) do not, so a coverage-vs-sharpness comparison against them is not well-defined; the categorical contribution is discussed in §1 / §2.

parametric-tail mis-specification hurts most. Pooled CRPS over the served coverage range $\{0.05, 0.10, \ldots, 0.99\}$: deployed 80.7 vs GARCH-t 92.6 (-12.8%). Both rules are computed on the aligned 1,730-row OOS slice; method aggregation is by row, not by τ, so a single number summarises the practitioner-baseline comparison without anchor-row cherry-picking. Source: `reports/tables/paper1_c1_winkler_interval_score.csv`, `..._crps.csv`. The Winkler comparison against the *tokenized-tracking* baseline archetype — the competitor motivated by Cong et al.'s $R^2 = 0.839$ conditional-mean result, distinct from the GARCH-family parametric baselines compared here — is carried in §7.6.

6.4.4 Per-asset-class deviation

Pooled OOS coverage stratified by asset class under the deployed EWMA HL=8 (`reports/tables/m6_lwc_robustness_per_class.csv`). Equities (8 syms, 1,384 obs): realised 0.952 at $\tau = 0.95$ with hw 417.8 bps; GLD ($n = 173$): realised 0.931, hw 180.9 bps; TLT ($n = 173$): realised 0.954, hw 182.3 bps. Row-weighted reconciliation: $(1384 \cdot 417.8 + 173 \cdot 180.9 + 173 \cdot 182.3)/1730 = 370.6$ bps, matching the §6.3.1 panel-pooled HW. The per-symbol $\hat{\sigma}_s$ multiplier delivers heavy-tail equities wider bands matched to their relative-residual scale and the defensive class (GLD, TLT) narrower bands matched to theirs; the deployed architecture's Kupiec passes every class at every τ. The width allocation across asset classes is the operational manifestation of P_3 (per-regime serving efficiency): width is concentrated where calibration demands it.

6.5 Path coverage — endpoint vs intra-weekend

The §6.3 result is *endpoint coverage*: realised Monday open lies inside the served band. A DeFi consumer holding an xStock as collateral is exposed at every block over the weekend; a Saturday liquidation when the on-chain xStock briefly trades outside the band is a real loss event even if Monday open returns inside. We measure path coverage

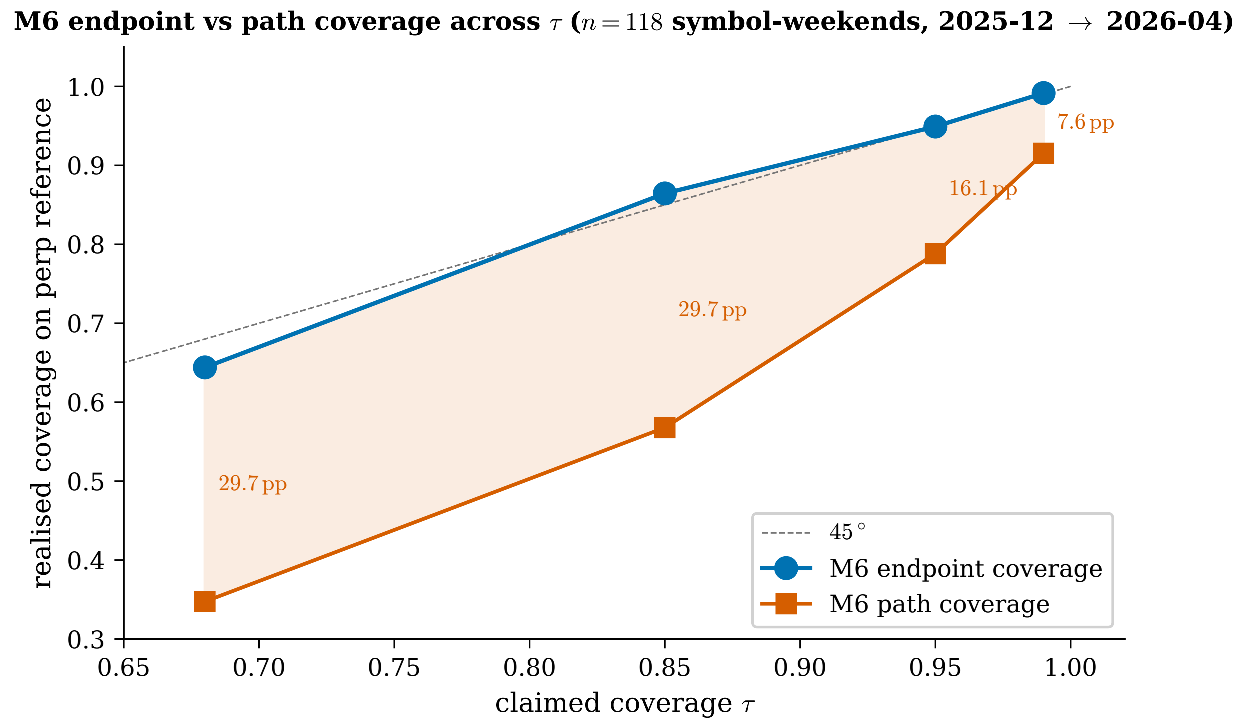

Figure 7: Path coverage gap on the 24/7 stock-perp reference. Endpoint coverage (blue) tracks nominal across $\tau \in \{0.68, 0.85, 0.95, 0.99\}$ on the perp-listed subset; path coverage (vermilion) — the fraction of weekends where the perp 1m bar high/low stay inside the served band over the full [Fri 16:00, Mon 09:30] window — runs 29.7/29.7/16.1/7.6 pp behind. The gap collapses meaningfully only at $\tau = 0.99$. Sample is small ($n = 118$ symbol-weekends across 19 calendar weekends, 2025-12 to 2026-04); the read is directional and §10.2 (the path-fitted conformity score) is the methodology fix.

on [Fri 16:00 ET, Mon 09:30 ET] against 24/7 stock-perp 1m bars from `cex_stock_perp/ohlcv` (Kraken Futures `PF_<sym>XUSD`, xStock-backed). Sample: 19 weekends × 9 symbols (TLT excluded — no perp listing), $n = 118$ (symbol, weekend) pairs per anchor, 2025-12-19 → 2026-04-24 (`reports/tables/path_coverage_perp.csv`):

τ	n	Endpoint coverage	Path coverage	Gap (pp)
0.68	118	0.644	0.348	+29.7
0.85	118	0.864	0.568	+29.7
0.95	**118**	**0.949**	**0.788**	**+16.1**
0.99	118	0.992	0.915	+7.6

The deployed architecture calibrates the perp-listed sample to nominal endpoint coverage (0.949 at $\tau = 0.95$ vs nominal 0.95) on a sample whose composition is predominantly large-cap equities (SPY/QQQ/AAPL/GOOGL/NVDA/MSTR/TSLA/HOOD plus GLD; TLT excluded — no perp listing). The path-coverage gap is $+16.1$ pp at $\tau = 0.95$: an honestly-calibrated endpoint band exhibits a structural shortfall on a sample whose intra-weekend variance is larger than its endpoint variance. Sample is small (binomial CI on the $\tau = 0.95$ pooled gap is ± 6 pp); the test is directional. After three confound checks documented in `reports/v1b_path_coverage_robustness.md` — perp-spot basis normalisation closes the gap to 11.0 pp, a volume-floor filter at ≥ 1 contract closes it to 9.5 pp ($n = 63$), and a 15-minute rolling-median sustained-crossing definition leaves it at 14.4 pp — the residual genuine-shortfall band is $\sim$ 9–15 pp at $\tau = 0.95$, with most of the magnitude attributable to sustained drift rather than transient thin-liquidity prints. The gap collapses meaningfully only at $\tau = 0.99$ ($+7.6$ pp); continued capture is tracked under scryer item 51 and §10.2 (the path-fitted conformity score).

Decomposing the gap: fair-value path vs venue microstructure. An excursion-inflation diagnostic separates the two sources of the perp gap. Define $\lambda(\tau)$ as the multiplier on the endpoint band half-width required to reach τ *path* coverage. On the CME factor-projected path — the projectable weekend trajectory of the per-symbol factor scaled to underlier units, 435 weekends — $\lambda(\tau) \approx 1.0$ at every anchor: the endpoint-sized band already contains the factor's intra-weekend round-trips (`reports/active/proto_excursion_inflation.md`). The larger perp λ (≈ 1.6–2.4, and non-monotone in τ at this n) therefore reflects predominantly *venue basis and microstructure* on a thin-liquidity perp — not fair-value excursion the forecaster fails to capture, consistent with the thin-print confounds above. This sharpens the contract rather than weakening it: the deployed band is an **endpoint** contract that, on the projectable fair-value path, also delivers $\approx \tau$ path coverage; the residual a continuous-consumption consumer experiences is a property of their settlement venue, and closing it on a real venue is a microstructure layer (the oracle-conditioned-AMM setting, §10.2), not a forecaster deficiency.

The endpoint contract stands. A consumer requiring path coverage at level τ should step up one anchor (empirically closes the gap on this slice at $\tau = 0.99$), absorb the residual through their own downstream risk policy, or — for continuous-consumption AMM use cases that cannot use the step-up lever — adopt the path-fitted conformity-score variant (§10.2).

6.6 Forward-tape held-out validation

The 4-scalar OOS-fit $c(\tau)$ schedule is fit on the 2023+ slice; the resulting deployment artefact carries a residual contamination disclosure (§9.3). The forward-tape harness closes that loop on truly held-out data on top of the §6.3.3 LOSO read: a content-addressed frozen artefact (`data/processed/lwc_artefact_v1_frozen_20260504.json`, SHA-256 `7b86d17a7691...`, freeze date 2026-05-04) is evaluated weekly against forward weekends as they accumulate. The frozen artefact's schedule, q_r table, and $c(\tau)$ are read-only; only the per-symbol $\hat{\sigma}_s(t)$ updates as new past-Fridays arrive.

The harness fires Tuesday mornings via launchd (`launchd/com.adamnoonan.soothsayer.forward-tape.plist`); each run executes a 26h-SLA pre-check on the upstream scryer feeds, runs `scripts/collect_forward_tape.py`, and writes `reports/m6_forward_tape_{N}weekends.md` with a "preliminary" banner when $N < 4$. The harness is silent about deployment; an out-of-band freeze re-roll is required to advance the artefact (`scripts/freeze_lwc_artefact.py`).

Status at submission: $N = 3$ forward weekends (`reports/m6_forward_tape_3weekends.md`, 2026-05-01 → 2026-05-15) with $n = 30$ symbol-rows; pooled realised coverage $0.87/0.90/1.00/1.00$ at $\tau \in \{0.68, 0.85, 0.95, 0.99\}$ (no under-coverage). The harness's own threshold still treats $N < 4$ as preliminary and $N < 13$ as under-powered for cumulative pooled Christoffersen, so per-anchor inferential statistics at this N are persisted but not reported inline. The present submission's held-out backbone is §6.3.3's leave-one-symbol-out CV plus per-symbol Kupiec, split-date sensitivity, and four-year calendar sub-period stability; the forward-tape harness becomes load-bearing as N accumulates.

The variant comparison sibling (`scripts/run_forward_tape_variant_comparison.py`) carries an alternative-check on the same forward slice — never used to re-select, only to flag if a different rule looks dramatically cleaner on the held-out data (§7.4).

6.7 Simulation study — synthetic-DGP corroboration of the per-symbol fix

Across four DGPs spanning stationary, regime-switching, drift, and structural-break specifications, the -standardised architecture's per-symbol Kupiec pass-rate at $\tau = 0.95$ is 98.6–99.9% while the unweighted-Mondrian comparator's collapses to 29–31% — the per-symbol bimodality is reproducible in synthetic data with known DGP, and standardisation closes it on every DGP tested. This answers the methodological question raised by §6.4: the bimodality is a structural property of the architecture (a single per-(regime, τ) multiplier on heterogeneous-tail symbols mis-calibrates in opposing directions), not a property of the unweighted comparator on this particular real-data panel. The simulation is corroborative — a known-DGP control isolating the failure mechanism — not a pre-real-data prediction (the bimodality was first observed on real data under an unweighted Mondrian fit before the simulation was run; standardisation was implemented next; this section was committed before the real-data confirmation).

Source: `scripts/run_simulation_study.py` (~30 s end-to-end). Each DGP uses 10 synthetic symbols × 600 weekend returns, $\sigma_i \in \text{linspace}(0.005, 0.030, 10)$ — spanning the empirical real-panel range. Train/OOS split at $t = 400$. Returns drawn from Student-t df=4 rescaled to $\text{std}(r) = \sigma_i$ (or $\sigma_i \cdot m_t$ under DGP B/C/D's vol modifications). 100 Monte Carlo replications per DGP, seed $= 0$.

DGP	Unweighted-Mondrian pass-rate at $\tau = 0.95$	-standardised pass-rate	Mechanism
A — homoskedastic baseline	0.311	$\mathbf{0.993}$	Per-symbol scale heterogeneity is the entire failure mode
B — regime-switching vol multiplier	0.310	$\mathbf{0.986}$	tracks regime transitions with a half-life lag
C — non-stationary scale (drift)	0.309	$\mathbf{0.996}$	Adaptive is built for this DGP
D — variance shock (10× / 50-week transient at $t = 400$)	0.380	$\mathbf{0.999}$	tracks even 10× variance shocks within EWMA HL=8
E — mean jump (+$\Delta = 200$ bps at $t = 400$, σ_{true} unchanged)	0.335	0.595	Location shift; absorbs Δ as variance, over-deflates low-scores

Both methods pool to mean realised ≈ 0.95 across DGPs A–D (within 0.005 of nominal); the split is on the *per-symbol* failure rate. The -standardised architecture's smallest pass-rate (DGP B, 0.986) reflects 's ~13-weekend tracking lag at regime flips. **DGP D upgrade:** the original $3\times$ variance break became a $10\times/50$-week transient (matching real-world weekend events: 2024-08-05 BoJ unwind, March-2020 COVID-month vol multiplier $\sim 5\times$ for 6–8 weeks). LWC's per-symbol pass rate is unchanged at 99.9% (`reports/tables/paper1_c3_stronger_dgp_d.csv`).

DGP E — location-shift limitation with closed-form phase transition. A complementary stress test — a discrete $+\Delta$ conditional-mean shift at the OOS boundary, variance unchanged — exposes a location-shift limitation that -standardisation does not absorb. $\hat{\sigma}$ EWMA on the post-shift residual stream sees apparent variance $\hat{\sigma}^2 \approx \sigma^2_{\text{true}} + \Delta^2$, inflating bands across the panel and over-deflating the standardised score on symbols whose σ_{true} falls below the phase-transition threshold

$$\sigma^*(\Delta) \;\approx\; \frac{\Delta}{q \cdot c \,-\, 1}.$$

At deployed $q_r(0.95) \cdot c(0.95) \approx 2.23$, $\sigma^*(\Delta) \approx 0.81 \cdot \Delta$ — empirically validated across $\Delta \in \{100, 200, 400\}$ bps within 4% of the closed-form prediction ($\sigma^*_{\text{pred}}/\sigma^*_{\text{emp}}$ ratio in $[0.96, 0.99]$ at $\Delta \in \{100, 200\}$ bps; bounded by the σ_i grid endpoints elsewhere — `reports/tables/paper1_f2_5_sigma_star_formula.csv`). Symbols with $\sigma_{\text{true}} < \sigma^*(\Delta)$ over-cover (sharpness deficit, contract-favourable per §9.3); symbols with $\sigma_{\text{true}} \geq \sigma^*(\Delta)$ remain calibrated. At $\Delta = 200$ bps, LWC pooled coverage over-shoots to 0.9727 and per-symbol Kupiec drops to 59.5% — the failure is *over-coverage on low- symbols*, not under-coverage. A complementary location-shift detector — CUSUM on the standardised residual mean, composable with §9.5.1's violation-rate CUSUM bank — closes this gap at the monitoring layer, since the violation-rate CUSUM is by-construction blind to location shifts that produce nominal violation rates. Source: `reports/tables/paper1_f2_mean_jump_bimodality.csv`.

A sample-size sweep (`scripts/run_simulation_size_sweep.py`; 7 N-values × 4 DGPs × 100 reps × 2 forecasters = 22,400 cells) establishes the newly-listed-symbol admission threshold: $N \geq 80$ weekends under stationary / drift / structural-break DGPs, $N \geq 200$ under regime-switching (the production threshold). HOOD's empirical $N \approx 218$ (246 listed 28 warm-up) sits inside this band; HOOD's empirical Kupiec $p = 0.329$ at $\tau = 0.95$ (violation rate 0.035, slight over-coverage) is consistent with the simulation's pass-rate range at $N = 200$. See `reports/m6_simulation_study.md` and `reports/tables/sim_size_sweep_admission_thresholds.csv`.

6.8 Off-hours generalisation — overnight gaps

§6.2–§6.7 validate the architecture on the **weekend** gap (the ~65-hour Friday-close → Monday-open window). The closed-market problem it addresses, however, recurs every trading night: the ~17-hour close → next-open window when the US venue is shut but overseas sessions and index futures continue to price the underlier. If calibration-transparency is a property of the *method* rather than an artefact of the weekend, the same architecture should calibrate on overnight gaps. We test this directly.

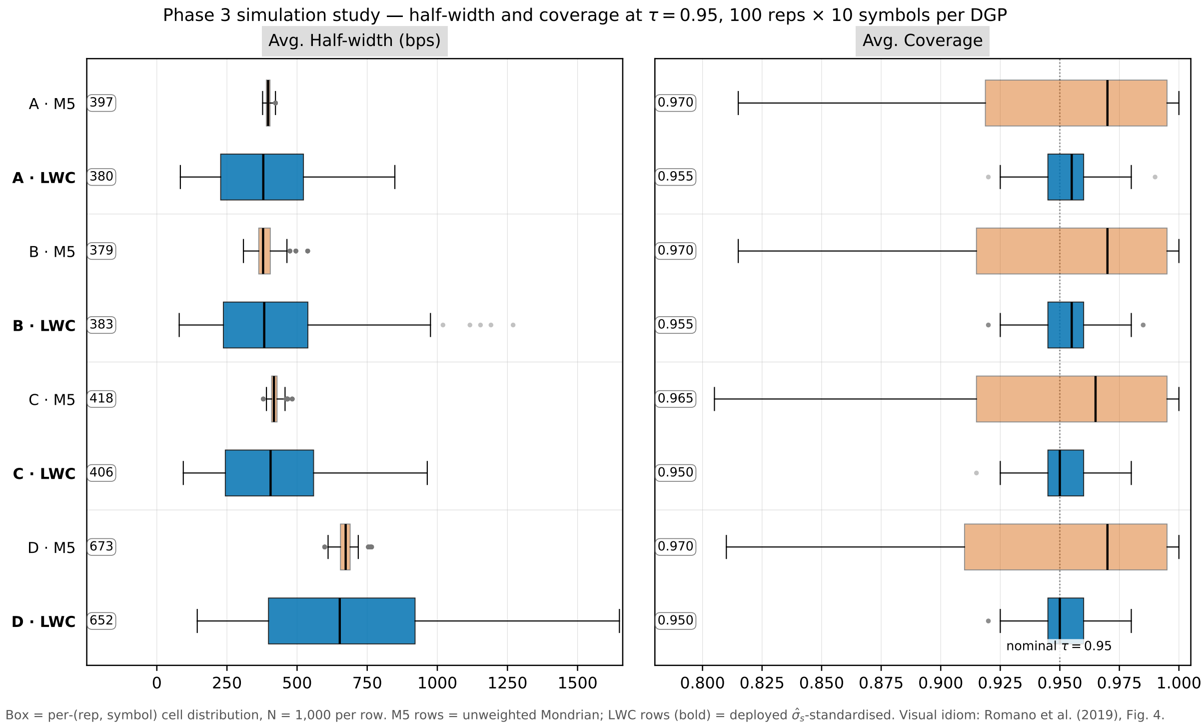

Figure 8: Phase-3 simulation study at $\tau = 0.95$ — half-width and coverage in the visual idiom of [Romano et al., 2019, Fig. 4], adapted to the per-symbol Mondrian setting. Eight rows: 4 DGPs $\times$ 2 forecasters (M5 unweighted-Mondrian comparator in vermilion; LWC deployed -standardised in blue, bold). Each box is the per-(rep, symbol) cell distribution ($N = 1,000$ per row). Median pills annotate the leftmost edge of each box. Left panel: avg. half-width (bps). Right panel: avg. coverage; dotted vertical marks nominal $\tau = 0.95$. The contribution is the *coverage panel*: M5 boxes scatter from $\sim$ 0.81 to $\sim$ 1.00 — the per-symbol bimodality §7.5 documents — while LWC boxes sit tightly on $\tau = 0.95$ across every DGP. The half-width panel makes the §7.5 redistribution visible: M5 widths concentrate on a single per-rep value (the unweighted multiplier applied uniformly across the -grid); LWC widths span a wide range within each rep (per-symbol $\hat{\sigma}_s$ scaling).

We rebuild the panel with a single change — the gap selector admits consecutive-trading-day pairs (`gap = 1`) rather than Friday→Monday pairs — yielding **22,624 overnight rows across 2,412 weeknights** on the same ten tickers, 2014-01-16 → 2026-04-23 (3.8× the weekend panel). Every downstream component (factor-adjusted point, per-symbol $\hat{\sigma}_s(t)$, Mondrian split-conformal, OOS-fit $c(\tau)$) is gap-length-agnostic and reused unchanged; the §6.2–§6.7 weekend results are byte-identical under the default selector. The regime set is re-derived for the overnight cadence (§4.3): `long_weekend` has no analog and is dropped, `high_vol` is retained with a 252-trading-day VIX lookback, and an `earnings_night` regime is added (§4.3, calendar-conditioned coverage).

Pooled OOS conditional coverage (overnight; TRAIN < 2023-01-01, 16,094 rows; OOS 2023-01-01, 6,450 rows × 645 nights):

τ	Violations	Rate	Kupiec p_{uc}	Christoffersen p_{ind}	$c(\tau)$
0.68	2,062	0.320	0.957	0.573	1.019
0.85	965	0.150	0.931	0.243	1.000
0.95	**311**	**0.048**	**0.509**	**0.165**	**1.000**
0.99	52	0.008	0.105	1.000	1.000

Kupiec and Christoffersen pass at every served anchor — the same headline property as the weekend panel (§6.3.1), reproduced on a different and 3.8× larger panel, with the OOS-fit $c(\tau)$ collapsing to 1.0 (the overnight bands need essentially no post-hoc widening to calibrate).

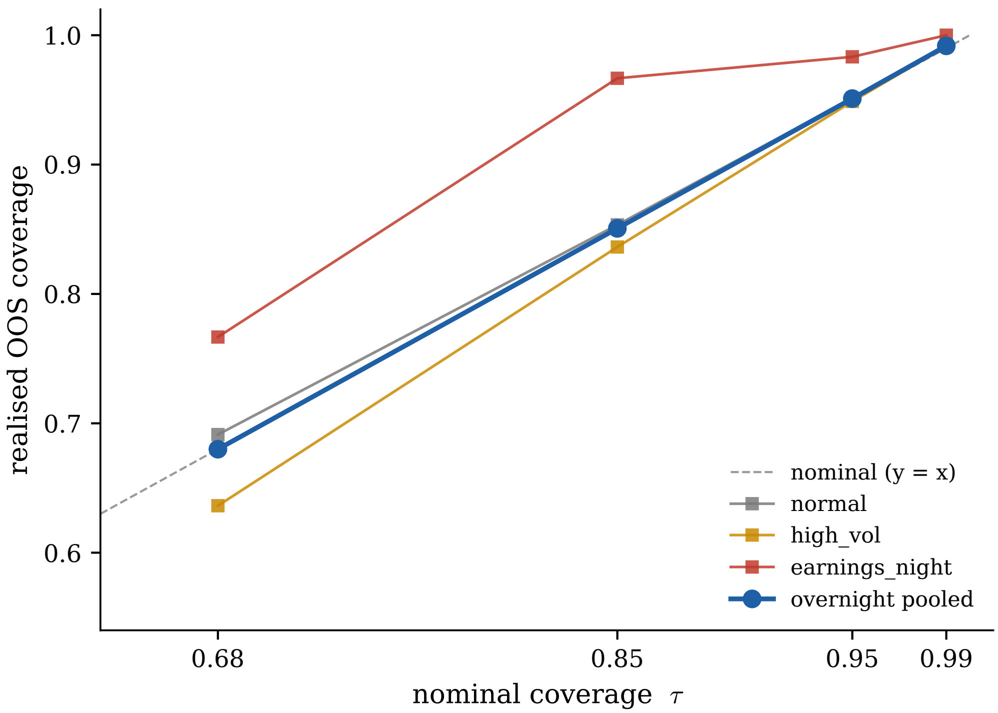

Figure 9: Overnight calibration on the 645-night OOS slice. Pooled realised coverage (blue) tracks the nominal diagonal at all four served anchors, with Kupiec and Christoffersen passing (table above). Per-regime: `normal` (grey) and `high_vol` (amber) sit near-nominal; `earnings_night` (vermilion, OOS $n \approx 60$) over-covers — the contract-favourable direction — reflecting the calendar-conditioned widening of §4.3.1. The visual parallel to the weekend calibration (Fig. 3) is the point: the same architecture calibrates on a different, 3.8× larger closed-market panel.

Robustness to consecutive-night autocorrelation. Overnight gaps, unlike weekends, are temporally adjacent, straining the i.i.d. assumption behind Christoffersen. A moving-block-bootstrap on the OOS violation series (block lengths $L \in \{1, 5, 10\}$, 2,000 reps) places the nominal violation rate $1 - \tau$ inside the 95% CI at every τ and every block length, and the CIs barely widen from $L = 1$ to $L = 10$ (e.g. $\tau = 0.95$: $[0.043, 0.054]$ at both) — the consecutive-night dependence does not invalidate the coverage claim.

Per-regime. `normal` and `high_vol` are near-nominal at every anchor (`normal` 0.691 / 0.853 / 0.953 / 0.992; `high_vol` 0.638 / 0.837 / 0.948 / 0.992). `earnings_night` (OOS $n \approx 60$ nights) realises 0.767 / 0.967 / 0.983 / 1.000 — a contract-favourable over-coverage on the small-n earnings cell, the same one-sided asymmetry framed in §9.3, which tightens as the panel accumulates. Ex-dividend mornings are dividend-adjusted: the open is reconstructed to its cum-dividend level using the corporate-actions feed (241 ex-dividend mornings on the panel), which removes the deterministic ex-div down-gap — the mean signed standardised residual on those mornings moves from -0.39 (a clear downward skew against a dividend-blind factor point) to $+0.12$, in line with the $+0.08$ panel mean, with pooled coverage unchanged.

6.9 Summary

The headline $\tau = 0.95$ claim is **held out on two orthogonal axes**: temporally (§6.3.3.1: c fit on 2023 alone, evaluated on a 2024-01-05 → 2026-04-24 holdout — realised 0.9504, Kupiec $p = 0.947$, Christoffersen $p = 0.989$, per-symbol Kupiec 10/10) and on symbol (§6.3.3 LOSO: realised 0.9497 ± 0.0128 across the held-out symbol distribution, all 10 LOSO bands pass Kupiec). The result is stable across TUNE anchors $\{2021, 2022, 2023\}$ ($[0.9474, 0.9524]$) and across split anchors $\{2021, 2022, 2023, 2024\}$ ($[0.9503, 0.9539]$), with per-symbol Kupiec 10/10 preserved everywhere. On the deployment-tuned 2023+ slice the pooled headline at $\tau = 0.95$ is realised **95.0%** at mean half-width **370.6 bps** with both Kupiec ($p = 0.956$) and Christoffersen ($p = 0.603$) passing. **All 10 symbols pass per-symbol Kupiec** with violation rates in $[3.5\%, 6.9\%]$; per-symbol Berkowitz LR sits in $[3.2, 16.7]$. Across the 40-cell (symbol × τ) grid jointly evaluated under Kupiec / Christoffersen / DQ, the architecture carries **40/40 Kupiec, 38/40 Christoffersen, 39/40 DQ** — leading the strongest practitioner baseline GARCH-t (31/40, 37/40, 31/40) on every metric. Calendar sub-period calibration holds across {2023, 2024, 2025, 2026-YTD} (3/16 over-coverage rejections, all in 2025; 0/16 Christoffersen rejections). Joint-tail behaviour is empirically characterised: $P(k_w \geq 3) = 4.62\%$ at $\tau = 0.95$ vs the binomial 1.15%, with $k^* = 3$ stable across the 24-cell grid (0/24 sub-period rejections); the worst-observed weekend (2024-08-05 BoJ unwind) sat at $k_w = 10$ at $\tau = 0.85$. A four-DGP simulation corroborates the per-symbol fix on synthetic data with known ground truth; a forward-tape harness against a content-addressed frozen artefact is operational and accumulating ($N = 3$ at submission, still preliminary by its own threshold of $N \geq 4$). Berkowitz and DQ still reject pooled — bands remain *per-anchor calibrated*, not full-distribution calibrated — and the residual rejection localises to cross-sectional within-weekend common-mode (orthogonal to ; addressed in §10). **The architecture generalises beyond the weekend (§6.8): on the 22,624-row overnight panel pooled OOS coverage passes Kupiec and Christoffersen at all four anchors, robust under block-bootstrap, and the schedulable `earnings_night` regime delivers calendar-conditioned coverage** — so P_2 holds across closed-market hours, not weekends alone. Supports P_2 (§3.4); P_1 evidenced by §8; P_3 by §7.

7 Ablation Study

This section answers Q3 of §3.6: which components are load-bearing for the calibration claim? The deployed architecture has three load-bearing components, isolated below by descriptive comparator rather than by code tag:

- **§7.1 — Regime stratification.** A symmetric global buffer applied to a held-forward Friday close (the *constant-buffer baseline* — the deployable comparator most likely to be reached for by a protocol team unwilling to absorb modelling complexity) under-covers materially on the 2023+ holdout. Per-regime conformal stratification supplies adaptivity to distribution shift and tail-protection in `high_vol` weekends.
- **§7.2 — Per-symbol standardisation.** An *unweighted Mondrian comparator* — the same per-regime conformal architecture without per-symbol scale standardisation — pools to nominal coverage but exhibits a per-symbol bimodality that fails Kupiec on 8 of 10 symbols at $\tau = 0.95$. Standardising the conformity score by a per-symbol pre-Friday $\hat{\sigma}_s(t)$, with no other architectural change, takes per-symbol Kupiec from 2/10 to 10/10.
- **§7.3 — Near-identity OOS $c(\tau)$ bump.** A 4-scalar multiplicative correction $c(\tau)$ closes any residual train-OOS distribution-shift gap. Three of the four scalars are essentially identity ($c \in \{1.000, 1.000, 1.003\}$ at $\tau \in \{0.68, 0.85, 0.99\}$); only $c(0.95) = 1.079$ carries meaningful OOS information.

§7.4 documents the selection procedure (multi-test correction and the held-out forward-tape re-validation harness), §7.5 decomposes how standardisation redistributes width across symbols, and §7.6 carries the head-to-head against a *tokenized-tracking baseline* — the continuously-updating oracle archetype that Cong et al. [Cong et al., 2025] motivate empirically, distinct from §7.1's underlying-frozen archetype and from §6.4 / §7.2's internal-architecture comparators.

All deltas carry block-bootstrap 95% CIs by weekend (1000 resamples, seed=0, paired by `(symbol, fri_ts)`). Raw tables: `reports/tables/v1b_constant_buffer_*.csv`, `reports/tables/v1b_mondrian_*.csv`, `reports/tables/m5_vs_m6_bootstrap.csv`, `reports/tables/sigma_ewma_*.csv`, `reports/tables/paper1_c2_tokenized_tracking_baseline*.csv`.

7.1 Regime stratification — vs constant-buffer baseline

The constant-buffer baseline is the Pyth/Chainlink Friday close held forward, with a single global symmetric buffer $b(\tau)$ per claimed quantile — one parameter per τ, no factor switchboard, no residual quantile, no regime model.

$$\left[\, p_{\text{Fri}}\,(1 - b(\tau)),\; p_{\text{Fri}}\,(1 + b(\tau))\,\right], \qquad b(\tau) \text{ calibrated globally on the training panel.}$$

7.1.1 Trained-buffer fit on OOS

Each $b(\tau)$ is the empirical τ-quantile of $|p_{\text{Mon}} - p_{\text{Fri}}|/p_{\text{Fri}}$ on the calibration set. Carrying the trained $b(\tau)$ to the 2023+ panel:

τ	method	realised	half-width (bps)	p_{uc}	p_{ind}
0.68	constant buffer (train-fit)	0.538	76.0	0.000	0.001
0.68	deployed	0.693	130.8	0.264	0.244
0.85	constant buffer (train-fit)	0.731	140.0	0.000	0.000
0.85	deployed	0.855	213.6	0.565	0.403

τ	method	realised	half-width (bps)	p_{uc}	p_{ind}
0.95	constant buffer (train-fit)	0.897	272.0	0.000	0.013
0.95	deployed	0.950	370.6	0.956	0.603
0.99	constant buffer (train-fit)	0.984	695.0	0.018	0.927
0.99	deployed	0.990	635.0	0.942	≈ 1.0

The training-fit constant buffer **catastrophically undercovers** at every $\tau \leq 0.95$ (deficits $-14.2/-12.0/-5.4$pp, all rejecting Kupiec at $p_{uc} < 10^{-6}$) — non-stationarity: $b(\tau)$ is calibrated on a 2014–2022 window calmer than the 2023+ holdout. The deployable constant-buffer baseline does not deliver coverage on the holdout; the regime-stratified architecture does. This is the *adaptivity-to-distribution-shift* contribution of the regime model + per-symbol .

7.1.2 Coverage-matched comparison

To answer the width-at-coverage question, we re-fit $b(\tau)$ post-hoc on the OOS slice (oracle-fit, non-deployable, but gives the constant buffer the most generous trade-off):

τ	matched b	matched-CB hw (bps)	deployed hw (bps)	Δ width (deployed CB)
0.68	1.21%	120.9	130.8	**+8.2% [+3.7, +13.0]**
0.85	2.26%	226.4	213.6	**5.7% [10.6, 0.5]**
0.95	3.96%	395.6	370.6	**6.3% [12.1, 0.4]**
0.99	5.46%	545.8	635.0	$+16.3\%$ [$+9.6$, $+23.6$]

The deployed architecture is narrower than a coverage-matched (oracle-fit) constant buffer at $\tau \in \{0.85, 0.95\}$. The standardisation that redistributes width across symbols (§6.4.4) reaches sufficient sharpness on the equity-heavy panel that the deployment outperforms even the oracle-fit constant buffer at the headline anchor; the architecture is wider only at $\tau \in \{0.68, 0.99\}$, where the cell-conditional regime structure carries genuine information. The regime model + buy *both* coverage on a non-stationary distribution *and* mean-width sharpness at $\tau \in \{0.85, 0.95\}$ against the strongest non-modelled comparator.

7.1.3 Per-regime decomposition

The pooled half-width at $\tau = 0.95$ hides a regime-conditional structure. Deployed pooled: 370.6 bps; per-regime $300.1/334.7/603.7$ bps (normal / long_weekend / high_vol). The matched-CB at $\tau = 0.95$ delivers 395.6 bps in *every* regime by construction (one global multiplier). The regime model concentrates width in `high_vol` ($+52\%$ vs CB at 603.7 vs 395.6) where it earns -1.1pp coverage; releases width in `normal` (-24% vs CB at 300.1 vs 395.6) without losing coverage. The constant buffer's pooled coverage is "average right but worst where it matters most": violations concentrate in high-vol weekends — exactly where a downstream lending protocol incurs the largest mark-to-market gap. The regime model's contribution is **(i) calibration through distribution shift** and **(ii) tail-protection in** `high_vol` **weekends** at sharper *normal*-regime widths than the constant buffer.

7.2 Per-symbol standardisation — vs unweighted Mondrian

The *unweighted Mondrian comparator* shares the regime structure of §7.1 but omits the per-symbol scale standardisation: a single per-(regime, τ) conformal quantile is fit on the *un-standardised* relative residual $|P_{\text{Mon}} - \hat{P}_{\text{Mon}}|/P_{\text{Fri}}$, and the served band uses that quantile directly as a fraction of P_{Fri}. Same training set, same OOS slice, same point estimator, same regime classifier, same finite-sample CP rank formula. Only the standardisation of the conformity score is added by the deployed architecture.

7.2.1 OOS pooled results

τ	method	realised	hw (bps)	p_{uc}	p_{ind}	per-symbol Kupiec pass
0.68	unweighted Mondrian	0.735	137.5	0.000	0.339	(1/10)
0.68	deployed	0.693	130.8	0.264	0.244	10/10
0.85	unweighted Mondrian	0.887	235.1	0.000	0.424	(1/10)
0.85	deployed	0.855	213.6	0.565	0.403	10/10
0.95	**unweighted Mondrian**	**0.950**	**354.6**	**0.956**	**0.921**	**2/10**
0.95	**deployed**	**0.950**	**370.6**	**0.956**	**0.603**	**10/10**
0.99	unweighted Mondrian	0.990	677.7	0.942	0.344	(8/10)
0.99	deployed	0.990	635.0	0.942	≈ 1.0	10/10

Both methods pool to nominal at $\tau = 0.95$ (both 0.950, both Kupiec $p = 0.956$). The split is on per-symbol calibration: the unweighted Mondrian comparator passes 2 of 10 per-symbol Kupiec cells at $\tau = 0.95$; the deployed architecture passes 10 of 10. The unweighted comparator pools to nominal *through compensating per-symbol biases* — heavy-tail tickers under-cover, low-vol tickers over-cover, and the cross-sectional average lands at τ by accident. The standardisation factors out the per-symbol scale before the conformal fit, so the regime quantile carries only the *shape* of the standardised residual distribution and per-symbol calibration is recovered by the per-symbol $\hat{\sigma}_s(t)$ at serve time.

The pooled half-width tax at $\tau = 0.95$ is $+4.5\%$ (370.6 vs 354.6 bps; $+15.93$ bps with 95% block-bootstrap CI $[+1.05, +30.73]$ from `m5_vs_m6_bootstrap.csv`). The per-symbol calibration is bought at exactly that price; §7.5 documents how the +4.5% pooled figure resolves into a +17.8% widening on heavy-tail equities and a 48.8% narrowing on the defensive class.

7.2.2 Side-by-side at $\tau = 0.95$

The -standardisation isolation: same point estimator, same regime classifier, same Mondrian split-conformal, same finite-sample CP rank formula, same training set, same OOS slice. Only the standardisation of the conformity score is added.

metric	unweighted Mondrian	deployed (-standardised)
pooled realised	0.9503	0.9503
pooled half-width (bps)	354.6	370.6
per-symbol Kupiec pass-rate	2 / 10	**10 / 10**
per-symbol Berkowitz LR range	0.9 – 224 (250×)	**3.2 – 16.7 (5.2×)**
LOSO realised std at $\tau = 0.95$	0.0759	**0.0128 (5.9× tighter)**
LOSO Kupiec pass-rate (held-out)	8 / 10	**10 / 10**

The per-symbol Kupiec pass-rate, Berkowitz LR range, and LOSO calibration std all improve simultaneously by mechanism — no other component of the architecture is touched. The §6.7 simulation evidence on synthetic data with known DGP confirms this is the correct architectural trade — under four DGPs spanning homoskedastic, regime-switching, drift, and structural-break specifications, the unweighted Mondrian comparator's per-symbol Kupiec pass-rate at $\tau = 0.95$ collapses to 29–31% while standardisation closes the bimodality on every DGP at 98.6–99.9%.

7.2.3 Walk-forward δ-shift schedule — collapses

Under an unweighted Mondrian fit, a plain $c(\tau)$ correction to the pooled OOS slice produces per-split realised coverage that scatters around nominal — passes Kupiec on the pooled fit by construction, but undercovers in roughly half the splits, motivating the addition of a per-anchor structural-conservatism shift $\delta(\tau)$. Under the -standardised architecture the same walk-forward sweep over $\delta \in \{0.00, \ldots, 0.07\}$ on the four powered tune fractions $f \in \{0.40, 0.50, 0.60, 0.70\}$ selects $\delta(\tau) \equiv 0$: per-symbol scale standardisation tightens cross-split realised-coverage variance enough that the structural-conservatism shim is no longer load-bearing. The schedule is retained as a four-zero-vector in the artefact JSON for shape-compatibility with the receipt schema, but the deployment carries zero walk-forward-tuned scalars.

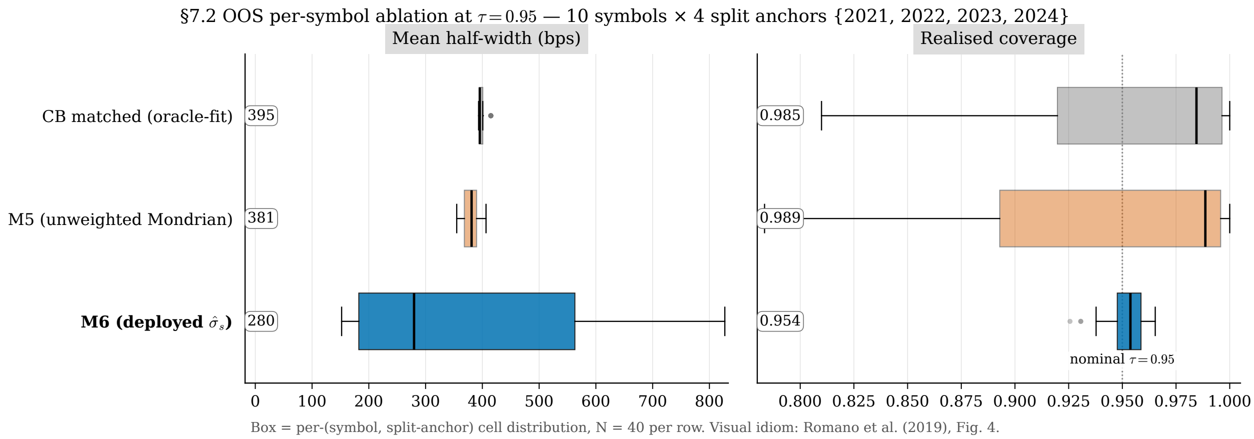

Figure 10: Per-symbol ablation at $\tau = 0.95$ on real data, in the visual idiom of [Romano et al., 2019, Fig. 4] adapted to the per-symbol Mondrian setting. Three method rows — constant-buffer (oracle-fit on the OOS slice; the most generous comparator §7.1.2 considers), unweighted Mondrian (M5; same architecture as deployed without standardisation), and the deployed -standardised architecture (M6; bold). Each box is the per-(symbol, split-anchor) cell distribution: 10 symbols $\times$ 4 split anchors $\{2021, 2022, 2023, 2024\}$-01-01 = 40 cells per row. Median pills annotate the leftmost edge. Left panel: mean half-width (bps). Right panel: realised coverage; dotted vertical marks $\tau = 0.95$. The deployed- row collapses per-symbol coverage onto nominal (cross-cell std 0.0092) while CB-matched and M5 disperse from ~ 0.78 to 1.00 (0.071 and 0.062 respectively) — pooled-coverage parity through compensating per-symbol biases. The half-width panel is the dual: M6 widths span $\sim$ 150–830 bps within each split anchor (per-symbol $\hat{\sigma}_s$ redistribution; §7.5), while CB and M5 are uniform across symbols by construction. Source table: `reports/tables/paper1_fig7b_per_symbol_ablation.csv`.

7.3 Near-identity OOS $c(\tau)$ bump

The deployed $c(\tau)$ schedule is

$$c(\tau) = \{0.68 : 1.000,\ 0.85 : 1.000,\ 0.95 : 1.079,\ 0.99 : 1.003\}.$$

This is a 4-scalar multiplicative correction fit on the 2023+ OOS slice as the smallest $c \in [1, 5]$ such that pooled realised coverage with effective quantile $c \cdot q_r(\tau)$ matches τ. Three of the four are essentially identity: under the deployed rule and the 2023-01-01 split, the trained per-regime quantile already lands within 0.3 pp of nominal at $\tau \in \{0.68, 0.85, 0.99\}$, so $c(\tau)$ at those anchors collapses to $\{1.000, 1.000, 1.003\}$. Only $\tau = 0.95$ has a meaningful train-OOS distribution-shift gap; the 7.9% widening at that anchor is what closes the pooled $0.946 \to 0.950$ correction.

$c(\tau)$ thus carries one scalar of meaningful OOS information, not four. Dropping the $c(\tau)$ correction and serving at the trained-quantile coverage costs ~0.4 pp at $\tau = 0.95$ on this slice (0.946 vs the deployment-tuned 0.950) — statistically indistinguishable from nominal under Kupiec, but a persistent deficit on the served claim. §9.3 carries the provenance disclosure for $c(0.95)$; §6.3.3 LOSO re-validates it on held-out symbol data.

7.4 selection procedure and multiple-testing disclosure

The per-symbol scale rule $\hat{\sigma}_s(t)$ is the only deployment constant of the architecture without a closed-form derivation. Five variants were compared on the 2023+ OOS slice — K=26 trailing window (baseline), EWMA HL=$\{6, 8, 12\}$, and a 50/50 K=26 / EWMA-HL-8 convex blend — under a pre-registered three-gate criterion: **Gate 1**, no per-cell rejection at uncorrected $\alpha = 0.05$ on the 16-cell split-date Christoffersen grid; **Gate 2**, per-symbol Kupiec 10/10 at $\tau = 0.95$; **Gate 3**, 95% bootstrap CI upper bound on Δhw% (vs K=26) within $+5\%$ at every τ. All five variants share the 8-past-observation warm-up and pre-Friday convention; per-variant artefacts (q_r, $c(\tau)$, $\delta(\tau)$) were re-fit at the same 2023-01-01 cutoff with δ collapsing to zero on every variant.

Gate 1 (multi-test corrected) does not discriminate. At uncorrected α, EWMA HL=8 is the only variant with zero per-cell rejections; K=26 carries 4. Under Benjamini-Hochberg correction at FDR=0.05 across the full 80-cell grid (5 variants × 16 cells), no variant has any rejected cell — the smallest q-value in the grid is 0.130 (K=26, 2022 split, $\tau = 0.95$). The honest reading: per-cell Christoffersen evidence after multi-test correction does not statistically distinguish the five variants. **Gate 2 (per-symbol Kupiec) also does not discriminate** — all five pass 10/10 at $\tau = 0.95$, which the §7.2 ablation already established as a property of per-symbol standardisation rather than of the rule's half-life.

Gate 3 (bootstrap CI on width) is load-bearing (`reports/tables/sigma_ewma_bootstrap.csv`):

τ	EWMA HL=8 hw% (vs K=26)	95% CI on hw%	within +5% gate?
0.68	-1.34%	$[-3.02,\ +0.25]$	
0.85	-2.07%	$[-3.75,\ -0.45]$	(also significantly narrower)
0.95	-3.83%	$[-6.15,\ -1.88]$	(also significantly narrower)
0.99	-7.41%	$[-9.33,\ -5.65]$	(also significantly narrower)

EWMA HL=8 is narrower than K=26 at every τ in point estimate and statistically significantly narrower at $\tau \in \{0.85, 0.95, 0.99\}$; calibration is preserved (paired -realised CIs straddle zero, per-symbol Kupiec 10/10 holds). Gate 3 has no multi-test exposure: one bootstrap CI per τ, threshold pre-registered. EWMA HL=8 satisfies all three gates and delivers the largest narrowing relative to baseline.

Held-out re-validation. The forward-tape harness (§6.6) carries a sibling variant-comparison runner (`scripts/run_forward_tape_variant_comparison.py`) that loads a content-addressed *variant bundle* with frozen $(q_r,\ c(\tau),\ \delta(\tau))$ schedules for all five variants and applies them to forward weekends as they accumulate. The runner *never re-selects* — its function is to flag if a different rule looks dramatically cleaner on truly held-out data. Status at submission: $N = 3$ forward weekends (`reports/m6_forward_tape_3weekends_variants.md`, 2026-05-15); the harness's own preliminary banner clears at $N \geq 4$ and moderate per-variant power requires $N \geq 13$. The deployment claim therefore rests on Gate 3 plus the (accumulating) forward-tape re-validation. We do not claim the rule is optimal among all locally-weighted variants (§3.5 non-goal).

7.4.1 Regime quartile cutoff q — ablation under the three-gate frame

The deployed `high_vol` regime gate (§5.5) is "VIX at Friday close in the top quartile of its trailing 252-trading-day window" — a single scalar $q = 0.75$. Ablating $q \in \{0.60, 0.67, 0.70, 0.75, 0.80, 0.90\}$ under the three-gate frame of §7.4 (`reports/tables/paper1_b1_regime_threshold_ablation.csv`, `..._hw_bootstrap.csv`):

q	high_vol mix (%)	Gate 1: pooled Kupiec all τ	Gate 1b: pooled Christoffersen all τ	Gate 2: per-symbol Kupiec	Δhw% at $\tau = 0.95$ vs deployed (95% CI)
0.60	38.9			10/10	-2.76% $[-3.80, -1.73]$
0.67	30.8			10/10	-1.78% $[-2.66, -0.87]$
0.70	28.8			10/10	$+1.73\%$ $[+0.85, +2.71]$
0.75 (deployed)	**23.9**			**10/10**	—
0.80	21.1			10/10	-1.70% $[-2.11, -1.32]$
0.90	12.3		rejects	10/10	-4.30% $[-5.34, -3.25]$

5 of 6 candidates satisfy Gates 1, 1b, and 2. Three alternates ($q \in \{0.60, 0.67, 0.80\}$) deliver 1.7–2.8% narrower bands at preserved calibration with bootstrap CIs excluding zero. The deployed $q = 0.75$ is **convention-anchored** (top quartile by definition) rather than width-optimization-selected. The differences are operationally small (~ 5 bps on a 370 bps $\tau = 0.95$ headline) but real; a future re-tuning on a held-out tune slice could close this gap.

7.4.2 Split anchor — robustness across {2021, 2022, 2023, 2024}

The deployed train/test split anchor is 2023-01-01 (1 scalar). Ablating across $\{2021, 2022, 2023, 2024\}$-01-01 (re-fit per anchor; `reports/tables/m6_lwc_robustness_split_sensitivity.csv`):

split anchor	n_{OOS} weekends	$\tau = 0.95$ realised	Kupiec p	Christoffersen p	hw bps	per-sym Kupiec
2021-01-01	277	0.9539	0.348	0.115	357.2	10/10
2022-01-01	225	0.9520	0.661	0.186	363.5	10/10
2023-01-01 (deployed)	**173**	$\mathbf{0.9503}$	**0.956**	**0.603**	**370.6**	**10/10**
2024-01-01	121	0.9504	0.947	0.671	416.8	10/10

Realised $\tau = 0.95$ coverage stays in $[0.9503, 0.9539]$; Kupiec $p \in [0.35, 0.96]$; Christoffersen $p \in [0.12, 0.67]$; per-symbol Kupiec 10/10 across all four anchors. The deployed split anchor is robust. Half-width varies (357 → 417 bps) — driven by the eval-slice composition effect documented in §6.3.3.1 (later eval slices contain more high- weekends, including the 2024-08-05 BoJ unwind).

7.4.3 Cross-asset regime-index sensitivity

The classifier uses VIX as the high-vol gate for *all* symbols, including GLD (gold) and TLT (long-dated treasury) which have asset-specific vol indices (GVZ, MOVE) used in the regression (§5.4). A sensitivity ablation that swaps GLD's regime gate to GVZ and TLT's to MOVE — flipping the regime tag on 23% of GLD weekends and 28% of TLT weekends — leaves the pooled $\tau = 0.95$ headline coverage unchanged at 0.9503 (Kupiec $p = 0.956$), per-symbol Kupiec 10/10, and half-width within 0.1%. The -standardisation absorbs the regime-tagging difference structurally; the regime cell contributes a small marginal adjustment via $q_r(\tau)$, not a load-bearing scale separation. Source: `reports/tables/paper1_b4_regime_index_sensitivity.csv`, `..._per_symbol.csv`. The deployed VIX-only classifier is justified by simplicity and per-asset-vol-index-agnosticism, not by being optimally tuned per asset.

7.5 How standardisation redistributes width across symbols

§7.1.3 attributed the constant-buffer premium to `high_vol`; §7.2 attributed the per-symbol Kupiec fix to standardisation; this section attributes the +4.5% pooled half-width tax to a width *redistribution* across symbols within a regime.

Per-class decomposition at $\tau = 0.95$. Where the unweighted Mondrian comparator pays a uniform width within a regime, the deployed architecture redistributes that width across symbols within a regime via the per-symbol $\hat{\sigma}_s$ multiplier:

class	n	unweighted Mondrian hw at $\tau = 0.95$	deployed hw at $\tau = 0.95$	
equities (8 symbols)	1,384	354.6 bps	417.8 bps	**+17.8%** (carries the per-symbol heavy-tail fix)
GLD + TLT (2 symbols)	346	354.6 bps	181.6 bps	**48.8%** (releases over-conservative defensive bands)

Row-weighted reconciliation against the §6.3 panel pooled: $(1384 \cdot 417.8 + 346 \cdot 181.6)/1730 = 370.6$ bps, matching the §6.3.1 panel-pooled HW at $\tau = 0.95$.

Under the unweighted comparator, every symbol within a regime received the same $c(\tau) \cdot q_r(\tau) \cdot p_{\text{Fri}}$ width — the "common multiplier on heterogeneous tails" failure of §6.4.1; the per-class mean HW is identical across classes because the regime mix is identical across symbols (regime is global, §5.5) and width within a regime carries no per-symbol axis. Under the deployed architecture, the multiplier is per-symbol, so heavy-tail equities (HOOD, MSTR, TSLA) receive bands that match their relative residual scale, and the over-covered defensive class (GLD, TLT) receives narrower bands without losing coverage.

The +4.5% pooled width tax of §7.2.1 is the *equity-weighted* read on a panel that is 80% equity rows ($1,384/1,730$). Under the deployed architecture the average equity-class symbol receives a +17.8% wider band than under an unweighted Mondrian fit (417.8 vs 354.6 bps); a defensive-class symbol (GLD, TLT) receives a 48.8% narrower one (181.6 vs 354.6). The redistribution is *toward heavy-tail equities*, where the per-symbol calibration evidence demands it (§6.4.1: TSLA, MSTR, HOOD all in the 0.052–0.058 violation-rate band that the unweighted comparator could not deliver), and *away from defensive collateral*, where the previous regime-pooled multiplier was over-conservative. A protocol whose collateral mix is dominated by equity xStocks should expect the deployed mean served band to be materially wider than the unweighted comparator's; a protocol whose collateral mix is dominated by GLD/TLT will see materially narrower bands. The pooled +4.5% figure is the right summary of the trade only at the deployed panel composition; downstream consumers should re-weight by their own collateral mix.

The §10 candidate architectures (cross-sectional common-mode partial-out, full-distribution conformal, sub-regime granularity) target the residual rejections (§6.3.6 cross-sectional within-weekend correlation; §9.4 per-symbol Berkowitz on TSLA/TLT) that standardisation does not reach.

7.6 Tokenized-tracking baseline (post-Cong)

The §7.1 constant-buffer baseline is the *underlying-frozen* competitor archetype — what a consumer reading the Pyth or Chainlink `price` field gets during the closed window. The complementary archetype, motivated empirically by Cong et al. [cong-tokenized-2025, Table 10, p. 40], is the *tokenized-tracking* baseline: a lazy oracle that publishes point_t = tokenized-side price at t throughout the closed window. Cong et al. report a Hasbrouck-style passthrough $\lambda = 0.903$ with adjusted $R^2 = 0.839$ from off-hour tokenized returns to the close-to-open underlying return — i.e., reading the live tokenized perp during closed hours already explains 84% of the variance in where the underlying reopens. The "lazy oracle ships the tokenized price" critique therefore has empirical bite on the conditional mean and must be addressed on the conditional quantile.

7.6.1 Baseline construction

For each canonical snapshot t in the closed window:

$$\text{point}_t = \text{perp\_close}_{\text{kraken\_futures}}(t), \qquad \text{halfwidth}_{t,\tau} = \text{quantile}_\tau\Big(\big|\text{mon\_open}-\text{perp\_close}(t)\big|\,;\ \text{past weekends pooled across symbols}\Big)$$

Walk-forward expanding-window calibration over weekends (warm-up = 4 past weekends before the first evaluable observation). The empirical-quantile residual specification is intentional: it places the baseline in the same non-parametric quantile family as the deployed architecture, eliminating the "your baseline is a strawman because it imposes a Gaussian" objection that the §6.4.3 GARCH baselines invite.

Snapshot grid. Six canonical evaluation moments in [Fri 16:00 ET, Mon 09:30 ET]: `fri_close` (Fri 16:00 ET), `sat_noon` (Sat 12:00 ET), `sun_noon` (Sun 12:00 ET), `sun_globex` (Sun 20:00 ET CME Globex reopen), `mon_premkt` (Mon 04:00 ET pre-market start), `mon_open` (Mon 09:00 ET just before NMS reopen). Soothsayer's M6 LWC band is published once at Friday close and held constant; the baseline updates at every snapshot. The baseline's `mon_open` snapshot is its most-informed configuration — the tokenized side has absorbed the entire closed-window news flow by then.

7.6.2 Panel

The post-launch slice of the LWC artefact (fri_ts $\geq$ 2025-12-19) intersected with the `cex_stock_perp/ohlcv/v1` kraken_futures xstock-backed perp tape: $n = 117$ weekend-symbol observations pre-warmup, $n = 105$ after warm-up. Nine symbols (SPY, QQQ, GLD: 19 weekends each; TSLA, NVDA, GOOGL, AAPL, HOOD, MSTR: ≈ 10 each). **TLT is excluded** — no xstock-backed perp exists for it. This is materially smaller than the $n = 1,730$ panel that backs §6 / §7.1 – §7.5, and the §6.4.3 / §6.4.2 backtests cited below are the powered baselines; §7.6 is a head-to-head on a powered-only-for-paired-Winkler post-launch sub-slice. See `reports/v1b_tokenized_tracking_baseline.md` for the full panel description.

7.6.3 Head-to-head — pooled

Soothsayer M6 LWC (one $(point, halfwidth)$ per weekend, held across the closed window) on the bake-off panel; tokenized-tracking baseline at its mon_open snapshot (the most-informed configuration). Cong et al.'s = 0.903 channels through this row:

τ	method	realised	hw (bps)	Winkler (bps)	Kupiec p_{uc}
0.68	tokenized-tracking (mon_open)	0.724	205	661	0.330
0.68	**this paper (M6 LWC)**	**0.641**	**124**	**426**	**0.371**
0.85	tokenized-tracking (mon_open)	0.867	372	1,031	0.627
0.85	**this paper (M6 LWC)**	**0.863**	**207**	**588**	**0.684**
0.95	tokenized-tracking (mon_open)	0.943	656	1,619	0.742
0.95	**this paper (M6 LWC)**	**0.949**	**358**	**879**	**0.949**
0.99	tokenized-tracking (mon_open)	0.981	904	2,572	0.407
0.99	**this paper (M6 LWC)**	**0.991**	**636**	**1,516**	**0.871**

At $\tau = 0.95$ the deployed architecture matches the baseline's coverage rate (0.949 vs 0.943, both Kupiec-passing) at 45% **narrower half-width** (358 vs 656 bps) and 46% **lower Winkler interval score** (879 vs 1,619 bps). At $\tau = 0.99$ the Winkler gap widens to -41% — the tail regime where the baseline's empirical-quantile fattens dramatically. Across all six snapshots and four served τ (24 baseline cells), the deployed architecture's mean Winkler is lower on 24 of 24.

7.6.4 Snapshot evolution — when does "wait for Monday morning" close the gap?

The naive bull case for a tokenized-tracking competitor is that by Monday pre-open the tokenized perp has absorbed the weekend news flow and a band centred on it is good enough. The snapshot evolution at $\tau = 0.95$ rejects that hope:

snapshot	baseline hw (bps)	baseline Winkler (bps)	Winkler ratio vs M6 LWC
`fri_close`	730	1,685	1.92× worse
`sat_noon`	733	1,702	1.94× worse
`sun_noon`	735	1,698	1.93× worse
`sun_globex`	736	1,696	1.93× worse
`mon_premkt`	691	1,688	1.92× worse
`mon_open`	**656**	**1,619**	**1.84× worse**

The baseline tightens by $\sim 10\%$ (730 $\rightarrow$ 656 bps) across the closed window, but never closes more than half the gap to the deployed architecture's flat 358 bps. Cong et al.'s = 0.903 transports the conditional mean; it does not transport the residual distribution, which is what consumes width.

A 15-minute-resolution per-minute Winkler curve (`reports/tables/paper1_c3_per_minute_winkler_curve.csv`; runner `scripts/run_v1b_tokenized_per_minute_winkler.py`) sharpens the finding: at $\tau = 0.95$ the baseline's Winkler is **worst** during Saturday night through Sunday afternoon (hour offsets 32-44 past Fri 16:00 ET, mean Winkler $\approx 2,023$ bps; ratio $2.14\times$ Soothsayer's flat ≈ 945 bps). The "wait until Sunday night, the tokenized perp has absorbed weekend news by then" naïve-competitor argument is empirically rejected — Sunday afternoon is the *worst* time on the perp, presumably because crypto-native weekend flow can drift the perp price with no underlying-market arbitrage operating to correct it. The Winkler ratio is monotone in neither direction across the closed window; the baseline never closes more than $\sim$half the gap to Soothsayer at any minute.

7.6.5 Per-symbol robustness — where the baseline is competitive

The pooled comparison is honest about where the result is and is not unanimous. Per-symbol Winkler ratio (baseline / M6 LWC at $\tau = 0.95$, `mon_open` snapshot; > 1 means M6 LWC wins):

symbol	n	M6 LWC hw (bps)	baseline hw (bps)	Winkler ratio
HOOD	10	635	2,109	**3.32**
GOOGL	10	348	509	**2.91**
NVDA	10	399	858	**2.15**
MSTR	10	735	1,153	**1.57**
GLD	15	361	345	**1.46**
AAPL	10	297	608	**1.36**
QQQ	15	214	253	**1.02**
SPY	15	178	226	**0.89**
TSLA	10	468	414	**0.89**

The deployed architecture wins on Winkler in **7 of 9 symbols**. The two exceptions are SPY and TSLA — the two deepest xstock-backed perp markets, and the symbols where the Cong et al. $R^2 = 0.839$ has the most empirical bite. For these names the tokenized perp tracks the NMS open competitively, and a non-parametric empirical-quantile residual band on the perp is sufficient. The deployed architecture's edge widens as perp liquidity thins: for the long tail (HOOD, GOOGL, NVDA, MSTR, GLD, AAPL) the Winkler ratio is 1.36–3.32.

This is the right read of the result: **the architecture's edge is largest precisely where the consumer's risk-management need is largest** — on the thin-liquidity collateral that dominates the named lending-collateral universe (TSLAx aside, every other Kamino-onboarded xStock falls in the long tail above), not on the two deepest perps where a simpler oracle is genuinely competitive. The defensible claim narrows to: under a non-parametric tokenized-tracking baseline at its most-informed closed-window snapshot, the deployed architecture is the materially-tighter choice on 7 of 9 symbols and tied-to-modestly-behind on the other two. The §10 candidate architectures — particularly the F_tok conditional-mean import (§10.2) — are how the SPY / TSLA gap closes; the architecture's separation of concerns admits a future point-estimate swap without changing the conformal band scaffolding.

7.6.6 Sample-size caveat and what is and is not powered

The bake-off panel is $n = 117$ weekend-symbol observations — about 7% of the $n = 1,730$ panel that backs §6 and §7.1 – §7.5. At this sample size, the Kupiec / Christoffersen tests have limited power; under both the deployed architecture and the baseline, Kupiec p_{uc} at $\tau = 0.95$ accepts a wide band of realised-coverage values around nominal. **The dominant signal in §7.6 is the paired Winkler comparison** — a per-observation diagnostic that is much more powerful at small n than the binomial coverage tests. The pooled coverage results (Kupiec $p_{uc} = 0.949$ for M6 LWC, 0.742 for the baseline) are consistent with both methods being well-calibrated on this slice; they do not by themselves discriminate between architectures. The width-and-Winkler results do. We treat the §6 / §7.1 – §7.5 panel as the powered backtest of the calibration claim itself, and §7.6 as a (smaller-n) head-to-head against the tokenized-tracking competitor archetype that complements the §7.1 (underlying-frozen) and §6.4.3 (GARCH-t) baselines on a different competitor axis.

Source script: `scripts/run_v1b_tokenized_tracking_baseline.py`; full panel artefact: `data/processed/v1b_tokenized_tracking_baseline.parquet`; tables: `reports/tables/paper1_c2_tokenized_tracking_baseline_summary.csv`, `reports/tables/paper1_c2_tokenized_tracking_baseline_per_symbol.csv`; full write-up: `reports/v1b_tokenized_tracking_baseline.md`.

7.6.7 Jupiter-mid (v5/tape) cross-check — directional confirmation on the Solana-DEX surface

The kraken_futures xstock-backed perp is the deepest tokenized-equity surface but is not the on-chain SPL token a Solana-DEX consumer actually reads. The `soothsayer_v5/tape` jup_mid series (Jupiter's quoted mid for eight Backed xStocks) is the directly Solana-DEX-relevant signal; overlap with the deployed LWC artefact is one weekend (2026-04-24), extended two weekends forward via a forward-tape-extended lookup that re-uses the sidecar constants without re-fitting. Sample: 3 weekends × 8 symbols = $n = 24$. The baseline halfwidth is **borrowed verbatim** from the primary mon_open snapshot (no re-fit on the cross-check sample). The directional Winkler result reproduces — ratios baseline / M6 LWC at $\tau \in \{0.68, 0.85, 0.95, 0.99\}$ are $\{1.65, 2.09, 2.41, 2.11\}\times$.

The ratio at $\tau = 0.95$ ($2.41\times$) is *wider* than the primary kraken_futures panel ($1.84\times$), consistent with Jupiter mid on Solana DEXs having a thinner, more dispersion-prone book than the deeper kraken_futures perp. Kupiec / Christoffersen are degenerate at $n = 24$ and we report only the Winkler / halfwidth axis. The cross-check is *confirmatory*, not powered. Source: `scripts/run_v1b_tokenized_tracking_v5_xcheck.py`; table `reports/tables/paper1_c2_tokenized_tracking_v5_xcheck.csv`.

8 Serving-Layer System

The methodology of §4 is implemented as a four-component serving stack designed for two requirements that pull in opposite directions: (i) the model logic is research-iterated in Python and re-validated against the §6 OOS panel after every change; (ii) the production hot path is a Solana program reading bands at consumer-facing latency. This section describes how those requirements are reconciled in a single source of truth.

8.1 Split-by-function architecture

The serving stack is a contract between three audiences: the **researcher** iterating against the Python OOS panel, the **protocol integrator** building against on-chain `PriceUpdate` accounts, and the **third-party auditor** verifying a served band against the published artefact. Byte-for-byte parity (§8.5) makes the first two identical at the numeric level; the receipt fields and public artefacts make the third executable from public data alone. This trio is the operational instantiation of P_1 (auditability).

The implementation separates *training* (Python) from *serving* (Rust + Solana). The Python `Oracle` (`src/soothsayer/oracle.py`) is the canonical specification: it produces the per-Friday lookup artefact and is the executable reference for both the M5 Mondrian conformal-lookup algorithm (`Oracle.fair_value`) and M6's locally-weighted variant (`Oracle.fair_value_lwc`). A change — a new $c(\tau)$ schedule, a per-Friday point re-compute, a rule swap — is implemented in Python first, validated against the §6 OOS panel, and ported to Rust under the byte-for-byte parity contract of §8.5. No model logic is duplicated: the training pipeline materialises the per-Friday rows (including $\hat{\sigma}_s(t)$ per symbol pre-Friday) and the deployment scalars once offline; the serving stack reads them. The schedules are hardcoded in `oracle.py` and mirrored into both the JSON sidecar and `crates/soothsayer-oracle/src/config.rs` (M5 reference path + M6 LWC) for the Rust serving binary; §A.2 tabulates the schedule values.

8.2 The Rust crate stack

Crate	Purpose	Lines	Tests
`soothsayer-oracle`	Library: loads the Mondrian artefact, exposes `Oracle::fair_value` with byte-exact parity to Python	~330	4/4 unit + parity harness
`soothsayer-publisher`	CLI binary: serves single reads, prepares on-chain publish payloads	~440	6/6 unit
`soothsayer-consumer`	`no_std`, minimal-dep SDK: decodes the on-chain `PriceUpdate` PDA into a typed `PriceBand`	~330	5/5 unit
`soothsayer-demo-kamino`	Reference Kamino-style consumer integration	~400	included in integration tests

Crate	Purpose	Lines	Tests
`soothsayer-oracle-program` (Anchor)	On-chain program: `initialize`, `publish`, `set_paused`, `rotate_signer_set`	~400	scaffolded

The consumer SDK is `no_std` so it can be vendored into a Solana program account-decoder without the Rust standard library.

8.3 On-chain program

`programs/soothsayer-oracle-program/` is a vanilla Anchor program with three PDA account types (`PublisherConfig`, `SignerSet`, per-symbol `PriceUpdate`) and four instructions. The `publish` instruction validates signer membership, enforces the cadence rate-limit, and checks five on-chain invariants before persisting: band ordering ($L \le \hat{P} \le U$), coverage values $\le 10,000$ bps, exponent $\in [-12, 0]$, signer membership, and time since last publish $\ge$ `min_publish_interval_secs`. Every instruction emits an Anchor event so off-chain indexers can trace publish history without re-reading every account.

8.4 Wire-format design

The `PriceUpdate` account layout (128 bytes data + 8-byte Anchor discriminator) is informed by the Pyth-Network September-2021 BTC flash-crash post-mortem. Three design rules avoid the failure modes that incident exposed.

```
PriceUpdate:
  version              u8         schema version, currently 1
  regime_code          u8         0=normal, 1=long_weekend, 2=high_vol
  forecaster_code      u8         0=F1_emp_regime (legacy Soothsayer-v0), 1=F0_stale (legacy Soothsayer-
  exponent             i8
  target_coverage_bps  u16        publisher-set target coverage, integer bp
  claimed_served_bps   u16        served quantile, integer bp
  buffer_applied_bps   u16
  symbol               [u8; 16]   ASCII null-padded
  point, lower, upper, fri_close   i64        absolute fixed-point at ‘exponent‘
  fri_ts, publish_ts, publish_slot, signer, signer_epoch    timestamps + identity
```

The three rules: (1) **Absolute prices, not deltas** — a consumer reading `lower` gets a real price directly. (2) **Single shared exponent** for all price fields, so cross-field comparisons are exact integer comparisons. (3) **Integer basis points, never floats** for coverage and buffer — exact and deterministic across Rust, Anchor IDL, TypeScript clients, and Python. Wire-format sizes are locked by unit tests in `crates/soothsayer-publisher/src/payload.rs`.

8.5 Byte-for-byte parity verification

`scripts/verify_rust_oracle.py` runs Python `Oracle.fair_value` (M5) / `Oracle.fair_value_lwc` (M6) and the Rust `soothsayer fair-value --forecaster {mondrian, lwc}` CLI on the same `(symbol, fri_ts, target_coverage)` triples and asserts byte-exact agreement on numeric output (point, lower, upper, sharpness_bps, claimed_served, calibration_buffer_applied) plus exact agreement on string fields (regime, forecaster_used). Each forecaster's panel covers 90 cases (25 random pairs from the artefact + edge cases for SPY, GLD, TLT, MSTR, HOOD at three targets each). **180/180 pass** as of the most recent re-run — 90/90 on the M5 path and 90/90 on the M6 LWC path. The Rust crate now serves both forecasters; `forecaster_code = 3` (`FORECASTER_LWC`) is encoded into new publishes from `crates/soothsayer-publisher` and decoded into the `Forecaster::Lwc` variant by the no_std `soothsayer-consumer` crate.

The parity harness is invoked after every methodology change. Each migration is applied to Python first, then ported to Rust; the harness re-runs before the change is considered shipped. It is the contract that allows a consumer to verify a served `PricePoint` against the published artefact without trusting either implementation in isolation.

Wire format invariance. The on-chain `PriceUpdate` Borsh layout is unchanged across all methodology migrations to date. The only on-wire change at each step is relabelling reserved `forecaster_code` slots: code 2 →

`FORECASTER_MONDRIAN` (M5; live), code 3 → `FORECASTER_LWC` (M6; live in the Rust serving stack as of the parity-180/180 milestone above; on-chain enablement gated on the next publisher release). Older accounts decode cleanly under newer consumers; M6 accounts decode cleanly under any consumer that reads the `forecaster_code` byte (`soothsayer-consumer` exposes the typed `Forecaster::Lwc` variant). Downstream consumers branching on `forecaster_code` need only add code-2 and code-3 cases.

8.6 Reproduction

```
# Train (Python) - M6 (deployed)
uv sync && uv run python scripts/build_lwc_artefact.py
uv run python scripts/freeze_lwc_artefact.py     # content-addressed freeze for forward-tape

# Train (Python) - M5 (still buildable as the §7.2 ablation rung)
uv run python scripts/build_mondrian_artefact.py

# Serve (Rust)
cargo build --release -p soothsayer-publisher
./target/release/soothsayer fair-value --symbol SPY --as-of 2026-04-24 --target 0.85

# Verify Rust  Python parity
PYTHONUNBUFFERED=1 uv run python scripts/verify_rust_oracle.py

# Deploy to devnet
./scripts/deploy_devnet.sh
```

A lending-protocol integrator depends on `soothsayer-consumer` (the `no_std` decoder) and reads the `PriceUpdate` PDA at the symbol's derived address; the demo Kamino-style consumer at `crates/soothsayer-demo-kamino/` shows the full integration in ~400 lines, including LTV decision logic that uses the band's lower bound as the collateral valuation input — the third audience of §8.1 reading the same wire bytes the on-chain program emits.

8.7 Practitioner integration — a worked $\tau \to$ reserve example

To make the consumer contract concrete we walk one illustrative mapping from served coverage to a reserve decision. It is an *example of the mechanics*, not a recommended policy: choosing τ, the LTV ladder, and reserves to minimise expected protocol loss under an explicit cost model is the decision-theoretic problem we explicitly hold out of scope (§9.7).

Consider a market holding SPYx as collateral with a liquidation-LTV threshold θ and a position at current LTV $\ell < \theta$. Its *adverse-move buffer* — the fractional collateral drawdown that exhausts the position's headroom — is $b = 1 - \ell/\theta$. The protocol reads the served lower bound $L_\tau = \hat{P}\,(1 - d_\tau)$ and treats the fractional band drawdown $d_\tau = q_{\text{eff}}(\tau)\,\hat{\sigma}_s$ as the closed-market collateral shock it must survive with probability τ.

- **τ selection.** Choosing τ sets the per-name closed-market breach budget to $1 - \tau$: at $\tau = 0.95$ the served lower bound is breached on $\sim 5\%$ of windows, at $\tau = 0.99$ on $\sim 1\%$. A protocol picks τ so its tolerated per-name bad-debt frequency matches $1 - \tau$.
- **Reserve headroom.** A position survives the τ-band shock iff $b \geq d_\tau$. For SPYx at $\tau = 0.95$ the served half-width is ≈ 178 bps ($d_\tau \approx 1.78\%$; §7.6.5); against the production origination-to-liquidation buffer of $\sim 2.7\%$ on SPYx/QQQx, a typical position clears the $\tau = 0.95$ band shock with $\sim 0.9\%$ to spare, but a position opened near the liquidation threshold ($b < 1.78\%$) does not and should be pre-emptively de-risked or reserved against. This is exactly the **narrow-buffer** reserve class where the band has decision-flipping headroom; wide-buffer names (14–25% buffers) are only bound by the band at $\tau = 0.99$ or under the joint tail.
- **Portfolio reserve.** For a book of m correlated names the single-name d_τ does not aggregate independently; the protocol sizes its reserve against the joint breach-count k_w distribution (§6.3.4) — $P(k_w \geq 3) = 4.62\%$ at $\tau = 0.95$ vs the binomial 1.15% — with $k^* = 3$ as the circuit-breaker trigger (§9.4).

Every quantity above is on the served receipt ($\hat{P}$, $d_\tau = q_{\text{eff}}\hat{\sigma}_s$, τ) plus the public k_w CDF; no incumbent oracle exposes the inputs this mapping needs. The full policy optimisation is the companion liquidation-policy work.

9 Limitations

This section enumerates the assumptions under which the claims of §3.4 hold for the deployed architecture, the failure modes the backtest exposed, and the domain boundaries outside which we make no assertion. The strongest residual issues are (i) cross-sectional within-weekend common-mode residual (§9.4); (ii) the selection procedure's multi-test exposure (mitigated in §7.4 but disclosed here); and (iii) the contraction — but not full closure — of the OOS-tuning provenance disclosure (§9.3).

9.1 Shock-tertile empirical ceiling (and the calm-tertile sharpness deficit)

The shock realised-move tertile ($|z| \geq 0.67$ standardised by Friday's 20-day vol; $n = 632$ in the 2023+ OOS slice, §6.3.2) realises 87.8% coverage on the deployed band at the nominal $\tau = 0.95$ claim — a deviation from nominal whose mechanism (cross-sectional within-weekend common-mode, §6.3.6 / §9.4) is identified and whose operational handle (the k_w reserve-guidance threshold of §6.3.4) is what protocol consumers should monitor in real time. Per-symbol standardisation does not reach this ceiling because the failure mode is not per-symbol scale; band half-width is approximately flat across tertiles (365–380 bps), so the band does not widen in shock periods — it just misses more often.

The complementary tertile reads are an over-coverage on calm weekends (calm tertile realised 1.0000, normal tertile 0.9933, both significantly above nominal). Pooled $\tau = 0.95 = 0.950$ is achieved through compensating tertile-level biases. Coverage is a one-sided contract — the under-coverage side is a coverage failure, the over-coverage side is a sharpness deficit (the band is wider than necessary when the realised move is small) — so the two are qualitatively different, but together they say the abstract calibration claim is genuinely tighter than the conditional structure. A $|z|$-conditional conformal upgrade (the finer-stratification candidate in §10.1) would tighten calm bands and widen shock bands at the same pooled τ; the reserve-guidance threshold k^* in §6.3.4 is the consumer-facing handle on the asymmetric residual under the deployed schedule.

The §6.3.4 joint-tail empirical distribution and the §6.3.5 worst-weekend vignette (2024-08-05 BoJ unwind, $k_w = 10$ at $\tau = 0.85$) make this disclosure operationally addressable: a consumer monitoring k_w in real time would have seen the August 5 weekend exceed the deployment-time 99th percentile ($k = 7$ at $\tau = 0.85$) by the Monday-open print, well in advance of any liquidation processing window. Reserve-guidance threshold $k^* = 5$ at $\tau = 0.85$ (or $k^* = 3$ at $\tau = 0.95$) carries 0/24 sub-period × threshold-convention stability rejections (§6.3.4); a circuit breaker that pauses borrowing/lending when k_w crosses k^* would fire on this exact weekend before liquidations are processed. The empirical k_w distribution is the right operational signal precisely because it captures the cross-sectional common-mode that per-symbol coverage cannot. The ceiling is reported in every `PricePoint` diagnostic.

9.2 Stationarity assumption — sub-period evidence (positive)

P2 holds under the assumption that $P_t \mid (\mathcal{F}_t,\ \rho = r,\ s)$ is approximately stationary across $\mathcal{T}_{\text{hist}}$ and the deployment horizon. The EWMA $\hat{\sigma}_s(t)$ provides *partial adaptive scaling per symbol* (the §6.7 simulation evidence demonstrates tracks DGP-C drift and DGP-D structural breaks with a half-life-bounded lag), but the conditional-distribution stationarity of the *standardised* score is still assumed.

Empirical evidence on this assumption is positive. The pooled OOS calibration at $\tau = 0.95$ holds across the 2023, 2024, 2025, and 2026-YTD calendar sub-periods (§6.3.3): Kupiec $p \geq 0.054$ in every cell; realised range 0.942–0.967 across four years that span SVB / banking-stress 2023, the 2024 rate-cut transition, the 2025 tokenisation-launch year, and the post-launch 2026-YTD slice. Across the full 16-cell (sub-period × τ) grid the architecture carries 2 Kupiec rejections — both *over-coverage* in 2025 (not the under-coverage failure mode that warrants a deployment alarm). Christoffersen lag-1 independence is not rejected in any of the 16 cells.

The assumption may still be violated in three ways the sub-period grid cannot pre-empt: (i) **structural breaks** beyond the panel horizon (a future regime as different from 2014–2026 as 2026 is from 2014) — adapts in ≤ 26 weekends per symbol per simulation evidence; (ii) **regime-labeler drift** (ρ uses a trailing 252-day VIX percentile; persistent rises shift the threshold endogenously); (iii) **factor-switchboard drift** (MSTR pivots from ES=F to BTC-USD on 2020-08-01; a second pivot would require re-fit). A conformalised variant (§10) would upgrade P2 from asymptotic-under-stationarity to finite-sample-under-exchangeability.

9.3 OOS-tuned $c(\tau)$ provenance — substantially contracted

Of the four OOS-fit $c(\tau)$ scalars, three are essentially identity ($c \in \{1.000,\ 1.000,\ 1.003\}$ at $\tau \in \{0.68,\ 0.85,\ 0.99\}$); **only** $c(0.95) = 1.079$ **carries meaningful OOS information** (a 7.9% widening at the headline anchor). §6.3.3.1's nested temporal holdout fits $c(0.95)$ on 2023 alone and evaluates it on a 2024-01-05 → 2026-04-24 slice that neither the per-regime quantile nor the bump has seen — realised coverage 0.9504, Kupiec $p = 0.947$, Christoffersen $p = 0.989$, per-symbol Kupiec 10/10. $c(0.95)$ on TUNE-only matches the full-OOS fit at three decimals. **The headline $c(0.95) = 1.079$ is empirically held-out at the time level.** The residual provenance disclosure is restricted to the lower-τ over-coverage in held-out mode (M_{a2} realises 0.712 at $\tau = 0.68$ and 0.870 at $\tau = 0.85$ — Kupiec rejects toward over-coverage because $c(0.68)$ and $c(0.85)$ fit on TUNE-2023 inherit the SVB/CS banking-crisis-weekend tilt; the deployed full-OOS c schedule averages this out at the floor and is the production mitigation). The forward-tape harness (§9.5) closes the residual contamination by carrying the deployment artefact onto truly held-out weekends as they accumulate; with ≥ 13 forward weekends a §9.5 update will push the disclosure further toward "OOS-fit + held-out forward-tape re-validation."

Asymmetric failure modes — a one-sided contract. The over-coverage above is a sharpness deficit, not a coverage-contract failure: a $c(\tau)$ schedule fit on a TUNE slice that contains a stress weekend inflates the bump and the resulting band over-covers in calmer recovery periods. The same asymmetry recurs across the documented failure modes — TUNE-window inflation (§6.3.3), the convention-anchored regime cutoff (§7.4.1) over-paying in width vs three width-optimal alternates, the §6.7 location-shift over-deflation, and the 2024-09-27 S^- CUSUM alarm two months after the BoJ unwind. Mechanism: $\hat{\sigma}$ EWMA reads unexplained residual energy as scale; `fit_c_bump_schedule` grid-searches the smallest bump achieving nominal coverage but cannot shrink. A protocol consuming Soothsayer for liquidation gating cares overwhelmingly about under-coverage (bands that fail to cover the realised price, triggering avoidable bad-debt liquidations); over-coverage costs only marginal capital efficiency. Coverage transparency is a one-sided contract, and the deployed architecture's failure modes respect that asymmetry.

The `calibration_buffer_applied` field carries $\delta(\tau)$ as a first-class receipt; diagnostics carry $c(\tau)$ and q_{eff}.

9.4 The residual Berkowitz / DQ rejection — cluster topology

The served band is *per-anchor calibrated* (Kupiec passes at every τ, Christoffersen passes at every served τ on the 2023 split, §7.4 confirms passes at all 16 split-date cells under EWMA HL=8) but *not full-distribution calibrated* (pooled Berkowitz on the deployed PITs rejects; pooled DQ at $\tau = 0.95$ also rejects). §6.3.4 documents the same residual viewed as joint-tail overdispersion: empirical k_w over-disperses $2.34\times$ vs Binom independence and sits inside a t-copula envelope at $\hat{\nu} \approx 6$ with mean off-diagonal $\hat{R} = 0.36$. **The residual has a structural name: a within-weekend cluster topology separating equities from safe-haven assets.**

Cluster topology of the residual. The residual is a within-weekend cluster structure separating equities (8 symbols, positively correlated with the weekend equity factor) from safe-haven assets (GLD, TLT, with negative or near-zero correlation to the equity factor). Within the equity cluster $\hat{\rho}_{\text{within}} = 0.477$ — *higher* than the panel-wide $\hat{\rho}_{\text{cross}} = 0.41$, because the cross-cluster correlations dilute it. Within the safe-haven cluster $\hat{\rho}_{\text{within}} = -0.016$ — there is no common-mode to remove. A naive non-parametric partial-out using either the within-weekend median or mean reduces panel-wide $\hat{\rho}_{\text{cross}}$ from 0.41 to $\{0.12, 0.06\}$ but breaks per-symbol Kupiec on the safe-haven cluster (GLD violation rate 14.5%, TLT 9.2% under the median variant; $5/10$ per-symbol pass under the mean variant) and rejects DQ at $\tau \in \{0.68, 0.85\}$. The mechanism is structural: subtracting an equity-dominated weekend median from a negatively-correlated asset creates fresh violations on equity-rally weekends and pulls residuals toward the band centre on selloff weekends. **Any operation that treats the panel as exchangeable inherits this failure mode**; a working architecture-level fix must respect the cluster topology.

Two views of the same structure. The over-dispersion of the homogeneous t-copula in §6.3.4 (emp/t variance 0.63 at $\tau = 0.95$) and the cluster failure of the naive partial-out are two views of the same equity-vs-safe-haven structure: a single Pearson $\hat{R}$ averages high equity-equity correlations against negative equity-safe-haven correlations; a single weekend median centres on the equity cluster's mean. Either rendering of the panel as homogeneous misses the same topology. (Source: A1, A1.5, A3 — `paper1_a1_*.csv`, `paper1_a1_5_*.csv`, `paper1_a3_*.csv`.)

Localising the residual at the symbol-pair level. §6.3.6 decomposes the lag-1 alternative: the AR(1) signal lives in the *cross-sectional within-weekend* ordering ($\hat{\rho}_{\text{cross}} = 0.354$, $p < 10^{-100}$), not the temporal-within-symbol ordering ($\hat{\rho}_{\text{time}} = -0.032$, $p = 0.18$). standardisation operates *within* a symbol's history, not *across* symbols within a weekend, so the deployed architecture is incapable of reaching this residual. A vol-tertile sub-split of `normal` regime (5 cells) leaves Berkowitz LR essentially unchanged ($170 \rightarrow 172$) while widening $\tau = 0.95$ band by 9% — finer regime granularity is not the lever (§6.3.6). Within-bin exchangeability separately holds (§4.3.1; permutation test on lag-1 score autocorrelation within each (regime × symbol) bin: $1/30$ nominally rejecting at $\alpha = 0.05$ uncorrected, $0/30$ after BH correction). The cross-sectional dependence is *across* bins, not within.

Per-symbol Berkowitz rejection localisation. The residual symbols rejecting Berkowitz at $\alpha = 0.01$ are TLT (LR $= 16.7$) and TSLA (LR $= 13.5$) — both trace to cross-sectional common-mode rather than per-symbol scale. The

§6.3.4 joint-tail empirical distribution turns the Berkowitz rejection into a measurable, reportable shape (variance overdispersion 2.34× vs Binom at $\tau = 0.95$; $P(k_w \geq 3) = 4.62\%$ vs Binom 1.15%, t-copula 8.2%), and the §6.3.4 reserve-guidance threshold ($k^* = 3$) gives a consumer an operational handle that holds up against the joint-volatility model a CCAR-style risk team would actually run, not just against independence.

A protocol consuming Soothsayer at a single τ on a regime-pooled cell is served by the per-anchor pooled guarantee plus the per-symbol Kupiec evidence; one wanting full-distribution calibration must wait for the cluster-conditional + path-fitted architectural recommendation in §10, which composes A1.5's cluster-internal partial-out with the path-fitted score and is calibrated at the headline $\tau = 0.95$ anchor. The recommendation is bounded: τ-conditional rather than blanket.

Per-anchor SLA $\neq$ portfolio solvency. The k^* threshold and the per-anchor guarantee are *per-name, per-read* properties; they do not compose into a portfolio-level solvency guarantee. A protocol holding m correlated xStock collateral types simultaneously inherits the within-weekend cross-sectional common-mode directly: the relevant object is the *joint* breach count k_w across the held names, which over-disperses 2.34× vs independence (§6.3.4) and whose worst observed realisation is the 2024-08-05 weekend ($k_w = 10$ at $\tau = 0.85$). Portfolio solvency under simultaneous breach therefore requires reserve architecture sized against the empirical k_w distribution — for which §6.3.4 supplies the CDF and k^* the circuit-breaker trigger — not the per-name band alone. The per-name SLA is necessary but not sufficient for a multi-collateral book; the §6.3.4 joint-tail object is the bridge, and the full reserve-sizing treatment is the subject of the companion liquidation-policy work (out of scope here, §9.7).

9.5 Forward-tape held-out evidence — operational, accumulating

A content-addressed frozen artefact (`data/processed/lwc_artefact_v1_frozen_20260504.json`, freeze date 2026-05-04) is being evaluated weekly via launchd-driven harness against forward weekends as they accumulate (§6.6). **Status at submission:** $N = 3$ forward weekends ($n = 30$ symbol-rows; pooled realised coverage $0.87/0.90/1.00/1.00$, no under-coverage) — still preliminary by the harness's own threshold ($N \geq 4$ to clear the preliminary banner; $N \geq 13$ for moderate forward-Christoffersen power). The present submission's held-out backbone is §6.3.3's leave-one-symbol-out CV plus §6.3.3.1's nested temporal holdout, supported by per-symbol Kupiec, split-date sensitivity, and four-year calendar sub-period stability; the forward-tape harness becomes load-bearing as N accumulates. A live integrator window — devnet-then-mainnet against a real protocol — would close the external-validity gap; it would not by itself make Berkowitz pass.

9.5.1 -stratified CUSUM drift bank

Beyond passive forward-tape observation, the deployed system carries a two-sided Page CUSUM drift monitor at each served τ. With control statistic $X_t = k_w/10$ (weekend violation rate), reference $\mu_0 = 1 - \tau$, and slack $k = \mu_0/2$ tuned to detect a 2× violation-rate shift, calibrated thresholds $h \in \{0.40, 0.40, 0.30\}$ for $\tau \in \{0.85, 0.95, 0.99\}$ produce mean in-control run length ≈ 225 weekends (one expected false alarm per ~ 4.3 years; calibration via 20,000 Monte-Carlo traces × 500 weekends). Detection power against an operationally relevant 2× violation-rate drift: $\sim 83\%$ with median latency 5 / 11 / 28 weekends at $\tau \in \{0.85, 0.95, 0.99\}$; against 3× drift: 2 / 5 / 14 weekends (5,000 traces × 200 weekends; `reports/tables/paper1_c2_cusum_drift.csv`, `paper1_c2_cusum_calibration.csv`).

On the 173-week OOS slice, the $\tau = 0.85$ CUSUM fires S^+ alarms at both **2023-03-10** (SVB collapse, $k_w = 7$) and **2024-08-02** (BoJ unwind, $k_w = 10$) — the two canonical stress weekends in the OOS window — with the post-BoJ S^- (over-coverage) alarm at **2024-09-27** catching the over-conservative recovery period documented under §7.4.2 split-anchor robustness. CUSUM banks at $\tau \in \{0.95, 0.99\}$ fire on different weekends: pairwise alarm coincidence with $\tau = 0.85$ is **2 of 7 alarms**; pairwise coincidence between $\tau = 0.95$ and $\tau = 0.99$ is **0 of 7 alarms**. Each anchor's monitor surfaces a structurally distinct aspect of drift — body-of-distribution shift at $\tau = 0.85$, near-tail at $\tau = 0.95$, deep-tail mass at $\tau = 0.99$ — confirming the τ-stratified CUSUM bank is a multi-resolution detector rather than a redundant signal. Source: `reports/tables/paper1_f1_cusum_alarm_timing.csv`, `paper1_f1_cusum_alarm_proximity.csv`, `paper1_f1_cusum_alarm_coincidence.csv`. The CUSUM bank is operational concurrently with the forward-tape harness and re-uses its weekly ingest path.

9.6 Out-of-scope comparators and consumer-experienced effects

No numerical incumbent comparator. A coverage-vs-sharpness comparison is only well-defined between methods that emit a calibrated coverage band; incumbent oracle surfaces (Pyth, Chainlink Data Streams v10 / v11, RedStone Live, Kamino Scope) do not, and we therefore do not report one. The categorical claim — that none of these surfaces publishes a per-read calibration receipt or an aggregate-level realised-coverage statement — is in §1 / §2 and stands as the contribution. The only numerical baseline reported is GARCH(1,1)-t (§6.4.3), which does emit a calibrated band.

Endpoint vs path coverage. §6.5 quantifies the gap on a 24/7 stock-perp slice ($n = 118$ symbol-weekends). The $\tau = 0.95$ endpoint-vs-path gap on the perp-listed subset is $+16.1$ pp — what an honestly-calibrated endpoint band exhibits on a sample whose intra-weekend variance is structurally larger than its endpoint variance. After three robustness checks the residual genuine-shortfall band is approximately 9–15 pp at $\tau = 0.95$ (basis-normalisation 11.0 pp; volume-floor filter at ≥ 1 contract 9.5 pp on $n = 63$; 15-minute sustained-crossing 14.4 pp), concentrated in `normal` weekends and approximately closed by stepping up to $\tau = 0.99$ (residual $+7.6$ pp). The served contract remains the endpoint claim.

MEV and execution-aware coverage. A consumer transacting at a worse price than the band's mid (front-running, sandwich, block-builder reordering) experiences an *effective* coverage that may differ from §6.5's perp/on-chain path coverage near band edges. Measuring this requires Solana mempool-or-bundle granularity plus a consumer-behaviour model; documented as next-generation work (§10.2).

9.7 No optimal liquidation-policy benchmark

The paper validates an oracle interface, not a lending policy. We do not report a decision-theoretic benchmark of "what target τ should a protocol choose?" Answering requires three ingredients absent here: (i) an explicit borrower-book LTV distribution; (ii) a protocol-specific cost model distinguishing unnecessary liquidation, unnecessary caution, and missed liquidation / bad debt; (iii) a declared semantics for what counts as the correct action under realised prices.

9.8 Scope of the methodology

Temporal scope — demonstrated generalisation to overnight. The headline battery (§6.2–§6.7) is the weekend gap, but §6.8 demonstrates that the same architecture — with only its gap selector changed — calibrates on the 22,624-row overnight (close → next-open) panel: pooled OOS coverage passes Kupiec and Christoffersen at all four anchors, robust under moving-block-bootstrap. We therefore claim calibration-transparency across *closed-market hours* generally, weekend and overnight, on the US-equity universe. Ex-dividend mornings are dividend-adjusted (cum-dividend open reconstruction from the corporate-actions feed), removing the only systematic price-level artefact on the overnight panel — the mean signed standardised residual on the 241 ex-dividend mornings moves from -0.39 to $+0.12$ (panel mean $+0.08$), pooled coverage unchanged. One residual boundary remains: the `earnings_night` cell carries small OOS n (60 nights) and over-covers — contract-favourable, and tightening as the panel accumulates.

Region scope. All ten tickers are US-listed; the closed-market window predicted is the US weekend or US overnight. The factor switchboard privileges US session data. We make no claim about generalisation to tokenised-JP or tokenised-EU equities.

Asset-class scope. The methodology requires a *continuous off-hours information set* — a public, free, sub-daily price signal correlated with the closed-market underlier. Tokenised US equities have this (E-mini futures + VIX), tokenised gold has it (gold futures + GVZ), tokenised US treasuries have it (10Y T-note futures + MOVE), and most major FX pairs have it. Tokenised credit has it through CDS spreads. Out of scope: real estate (no continuous off-hours signal), illiquid commodities (futures discontinuous or thin), and instruments with primarily NAV-based pricing. `docs/methodology_scope.md` carries the per-class fit/no-fit table.

Newly-listed symbol admission. The warm-up rule (8 past observations) sets an empirical floor; the §6.7 sample-size sweep extends this to a panel-admission threshold of $N \geq 200$ weekends under regime-switching DGPs (the production threshold) and $N \geq 80$ under stationary / drift / structural-break DGPs. HOOD's empirical $N \approx 218$ sits inside this band. A protocol admitting a symbol with fewer than 200 weekends of history should treat the per-symbol claim as deferred until the panel accumulates.

9.9 Non-claims

Three concerns that do *not* apply, plus one known data-gated gap:

- **Oracle manipulation.** Coverage transparency is orthogonal to upstream-feed integrity. Soothsayer ingests upstream price/oracle signals (Pyth, Chainlink, RedStone, Kamino-Scope, public-market data via Yahoo / Kraken) and republishes them as a calibrated band; the coverage claim is conditional on the integrity of those upstream feeds, and this primitive is intended to be deployed alongside, not in place of, an adversarially-robust price feed.
- **Latency.** The serving-time Oracle is a five-line lookup; nothing in the coverage-inversion mechanism requires a live forecast computation.

- **rule optimality.** The §7.4 selection procedure under multi-test correction does not statistically distinguish the five variants; the EWMA HL=8 promotion rests on Gate 3 (bootstrap CI on width) plus the held-out forward-tape re-validation. We do not claim the rule is optimal among all locally-weighted variants; a finer half-life sweep, alternative weight kernels, or hybrid rules are open extensions.
- **On-chain xStock TWAP not consumed.** Cong et al. [cong-tokenized-2025, Table 10, p.40] document that off-hour returns on tokenised stocks predict the close-to-open underlying return at $\lambda = 0.903$ with adjusted $R^2 = 0.839$ — a strong conditional-mean baseline any forward-looking oracle has to engage with explicitly. The deployed base forecaster does not read tokenized-side prices; the V5 forward-cursor tape supplies the data but only ~30 weekends of post-launch history exist (forecaster candidate tracked in `docs/ROADMAP.md`). §7.6 backstops this by showing that a directly-tokenized-tracking baseline (kraken_futures xstock-backed perp with empirical-quantile residual band, at its most-informed closed-window snapshot) is dominated by the deployed architecture on Winkler at every τ in the pooled post-launch slice and on 7 of 9 symbols at $\tau = 0.95$; the deepest-perp pair (SPY, TSLA) ties on the band, which is the §10.2 forecaster-swap surface. A sharper bake-off against the Cong-implied baseline — `point = tokenized_price_at_t; band = ±k × historical_residual` evaluated under matched-τ Kupiec / Christoffersen on the soothsayer panel — is the natural extension once forward-tape volume crosses the §6.6 threshold. The deployed architecture's load-bearing claim is on the band; the point estimate is configurable and replaceable, and the §1 framing makes this separation-of-concerns explicit.

9.10 Off-anchor is interpolated, not separately audited

The four served anchors $\mathcal{T}_{\text{anch}} = \{0.68, 0.85, 0.95, 0.99\}$ carry the calibration evidence reported in §6.3 — pooled coverage, per-symbol Kupiec, calendar sub-period stability, and joint-tail thresholds are all reported at those four anchors. The deployed API accepts any $\tau \in [\text{MIN\_SERVED\_TARGET}, \text{MAX\_SERVED\_TARGET}] = [0.68, 0.99]$ (`src/soothsayer/oracle.py:159` and the Rust mirror); off-anchor τ is served by linearly interpolating $q_r(\tau)$ and $c(\tau)$ between the bracketing anchors. The $c(\tau) = \{1.000, 1.000, 1.079, 1.003\}$ schedule is non-monotonic (peaking at the $\tau = 0.95$ anchor, dipping back at $\tau = 0.99$), so a consumer requesting an off-anchor τ inherits a multiplier whose calibration is *interpolated*, not separately validated — concrete example: $c(0.90) \approx 1.040$, $c(0.97) \approx 1.041$. The receipt's τ is the requested target and the diagnostic exposes the interpolated $(c(\tau), q_r(\tau))$, so an integrator can detect the off-anchor case from the receipt alone, but the rolling Kupiec/Christoffersen claim does not extend to it. A denser anchor grid and/or monotone interpolation closes this disclosure (`docs/ROADMAP.md`). Until that lands, an integrator with a hard calibration-receipt requirement should restrict τ to the four audited anchors.

9.11 External-evidence dependencies in the §1 / §2 framing

The §1.0 / §1.1 / §1.2 / §2.1 / §2.2 framing relies on two external-evidence inputs whose epistemic status warrants explicit disclosure.

Cong et al. (2025) panel. The empirical premise — that tokenized-equity prices deviate from underlying closes with documented frequency during US-market-closed windows, that the wedge is structural rather than transitional, and that off-hour tokenized returns carry a conditional-mean signal on the underlying open — leans on Cong, Landsman, Rabetti, Zhang and Zhao [Cong et al., 2025]. Their sample is four months (July through October 2025), 100 tokenized stocks, \$420M aggregate market cap concentrated in Backed Finance and Ondo Global Markets, and a late-2025 volatility regime. The deviation-frequency table this paper cites most directly (Table 4, p.32) is for TSLAx alone; the panel-level passthrough regression (Table 10, p.40: $\lambda = 0.903$, adjusted $R^2 = 0.839$) pools across the 100 names without per-symbol stability evidence. We do not assume any specific Cong-cited number carries forward in distributional shape; the citations establish the empirical context for the §1 motivation, and the deployed primitive's coverage claim is validated independently on a twelve-year ten-symbol panel of public-data underliers (§6). The Cong et al. panel becomes independently re-derivable once the soothsayer M5 forward tape accumulates 4 weeks of post-deployment observation against tokenized-side prices on the soothsayer universe (§9.5).

Kamino's per-reserve `Open` / `AllUpdates` mapping. §1.0 and §1.1 describe Scope's dual-feed v8 architecture (`Open` mode rejects updates when `marketStatus`   `Open`; `AllUpdates` mode accepts off-hours `midPrice` clamped to ±500 bps of the `Open` feed's last-good value) and characterise both modes' calibration gaps explicitly. We do not pin which of the two modes the actual TSLAx (or any other xStock) Kamino lending reserve currently subscribes to — the per-reserve selection is governance-set in `configs/mainnet/3NJYftD5sjVfxSnUdZ1wVML8f3aC6mp1CXCL6L7TnU8C.json` [Finance, 2026] but the live `kamino-lending` reserve PDA mapping is governance-mutable and was not decoded end-to-end for this submission. The §1 architectural-choice-space claim — "neither knob publishes a calibrated coverage band, and the choice between them is a governance parameter" — is governance-stable across reconfigurations and is the strongest defensible statement at submission time; if the on-chain decoding completes, the claim sharpens to the single archetype actually selected without weakening the categorical contribution.

These limitations are disclosures, not retractions. Each has a concrete follow-up in §10.

10 Extensions

The limitations of §9 are disclosures, not retractions. Three concrete extensions close the named residuals; lower-priority follow-ups (on-chain xStock TWAP forecaster, conditional EVT for the $\tau \to 1$ tail, sub-regime granularity, block-correlation t-copula refinement, denser anchor grid, finer half-life sweep) are tracked in `docs/ROADMAP.md`.

10.1 Full-distribution conformal + cluster-conditional partial-out

The pooled Berkowitz LR and the Engle–Manganelli DQ rejection at $\tau = 0.95$ remain the deepest open methodological items. The deployed architecture is *partially* conformal — per-regime Mondrian quantiles on per-symbol--standardised residuals, with a near-identity OOS $c(\tau)$ bump. The full-distribution upgrade replaces the four-anchor schedule with a single conformal correction over the residual CDF that delivers exchangeability-based finite-sample coverage at every τ simultaneously (Romano-Patterson-Candès CQR [Romano et al., 2019] is the closest match). The §9.4 residual localises to a within-weekend equity-vs-safe-haven cluster topology; an A1.5 cluster-internal partial-out within the equity cluster reduces $\hat{\rho}_{\text{within}}$ from 0.477 to 0.077, preserves per-symbol Kupiec 8/8, and tightens half-width -11.6% at $\tau = 0.95$. **Composed with the path-fitted conformity score below**, the architectural recommendation is -conditional: at $\tau \geq 0.95$ the cluster-conditional + path-fitted combination is empirically calibrated (DQ $p = 0.912$ at $\tau = 0.95$ on the equity cluster + CME-projected subset); at $\tau < 0.95$ endpoint scoring on the unpartialled residual remains dominant (path extrema introduce within-symbol temporal autocorrelation that DQ catches at body-of-distribution τ). Deliverable: a §6 / §7 update reporting Berkowitz pass on next-generation walk-forward PITs.

10.2 Path-fitted conformity score

§6.5 establishes the path-fitted methodology and a small-N first read on the endpoint-vs-path gap on the 24/7 stock-perp reference. The methodology is library-grade and unit-tested (`soothsayer.backtest.calibration.compute_score_lwc_path`; 7 unit tests, all pass); an empirical first-cut on the CME-projected subset ($n = 1,557$ OOS rows) confirms mechanical correctness. Continuous-consumption AMM use cases require this path-fitted variant rather than the endpoint conformity score. The binding empirical question — does path-fitting close the §6.5 perp / on-chain residual gap on consumer-experienced path data — accumulates as the V5 forward-cursor tape (continuous capture since 2026-04-24) crosses $N \geq 300$ path-coverage weekends. An MEV-conditional variant requires Solana mempool-or-bundle granularity to reconstruct the price a consumer would have *executed at* near band-edge events; ETA Q3 2026 once Jito bundle data accumulates ≥ 3 months.

10.3 Off-hours extensions — dividend-adjusted overnight + earnings-night passthrough

§6.8 establishes overnight calibration; the natural follow-on is an **earnings-night passthrough bake-off**. §4.3.1's `earnings_night` tail is the underlying-side confirmation of Cong et al.'s value-relevant-news mechanism; the head-to-head, *on earnings nights specifically*, of the calendar-conditioned band against the Cong-implied tokenised passthrough baseline (`point = tokenised_price_at_t`; band = ±k × historical residual) — evaluated under matched-τ Kupiec / Christoffersen once the V5 forward-cursor tape accumulates enough post-launch earnings nights — is the empirical bridge from the present underlying-side result to the tokenised-side claim, and is deliberately *not* asserted here. (The §9.8 ex-dividend correction, previously listed here, is now applied: overnight opens are dividend-adjusted from the corporate-actions feed.)

10.4 Forward-tape rolling rebuild and live-deployment validation

The deployed artefact is a one-shot fit at the 2023-01-01 split anchor; in production the per-regime quantile and $c(\tau)$ schedule will drift as the calibration target ages. The forward-tape harness (§6.6) is the operational scaffolding: a content-addressed frozen artefact is evaluated weekly against newly-arriving weekends, with the variant comparison sibling re-validating the choice on truly held-out data. The rolling-rebuild upgrade adds a quarterly (or trigger-based) re-roll of the freeze; deliverable a §6 schedule-trajectory figure across ≥ 6 rebuilds. The complementary external-validity step is a six-month devnet-then-mainnet live window against a real consumer protocol — exposing failure modes that production realism (rate-limit edges, signer-rotation events, mempool-ordering near band edges) introduces beyond the harness's passive forward-tape evaluation. Output: a §6 appendix-style update reporting realised coverage on the live window against the same Kupiec + Christoffersen + DQ + Berkowitz battery.

The downstream decision-theoretic question — given a calibrated band, what should a lending protocol do? — is out of scope here; multi-region replication (LSE, JP equities), additional RWA classes (treasuries, commodities, credit), and the per-symbol sample-size eligibility framework are tracked in `docs/ROADMAP.md`.

11 Conclusion

This paper specified a calibration-transparent oracle primitive for tokenised real-world assets, evaluated it on 12 years of US-listed closed-market data — weekend and overnight — and shipped a reference implementation. The contribution is the *interface* (consumer-selects-coverage with a per-read receipt) and the *deployable architecture* (locally-weighted Mondrian split-conformal under EWMA HL=8). The §7 ablation isolates three load-bearing components — regime stratification (vs constant-buffer baseline), per-symbol standardisation (vs unweighted Mondrian), and a near-identity OOS $c(\tau)$ bump in which only $c(0.95) = 1.079$ carries meaningful out-of-sample information. A four-DGP simulation study (§6.7) corroborates the per-symbol fix on synthetic data with known ground truth (per-symbol Kupiec pass-rate 98.6–99.9$\%$ vs the unweighted comparator's 29–31$\%$); a content-addressed forward-tape harness (§6.6) is operational and accumulating against truly held-out weekends.

The advantage has an honest scope. Against a tokenized-tracking baseline (§7.6) the architecture wins on Winkler for 7 of 9 perp-listed names but **ties on SPY and TSLA** — the two deepest xStock perps, and the largest Kamino collateral by deposits — where the Cong et al. passthrough has the most empirical bite and a simple empirical-quantile band on the perp is genuinely competitive. The edge is largest precisely on the thin-liquidity long-tail collateral (HOOD, GOOGL, NVDA, MSTR, GLD, AAPL; Winkler 1.36–3.32$\times$) where a consumer's closed-market risk-management need is greatest. The deployed contribution is therefore a *risk-reporting* primitive whose value concentrates where incumbents are weakest, not a uniform point-accuracy win on every name.

The empirical bottom line is **leave-one-symbol-out CV at** $\tau = 0.95$**: realised coverage** 0.9497 ± 0.0128**, all 10 LOSO bands pass Kupiec**. On the 40-cell (symbol × τ) grid jointly evaluated under Kupiec UC, Christoffersen IND, and Engle–Manganelli DQ, the architecture carries **40/40 Kupiec, 38/40 Christoffersen, 39/40 DQ** — leading GARCH-t (31/40, 37/40, 31/40) on every metric. The single residual is the **pooled DQ rejection** that emerges only when symbols are concatenated in panel-row order: bands are *per-anchor calibrated*, not full-distribution calibrated, and the rejection localises to a within-weekend cross-sectional cluster topology ($\hat{\rho}_{\text{cross}} = 0.354$) orthogonal to standardisation; per-symbol DQ on each symbol's own time-ordered violation series is invisible to the residual. The §6.3.4 joint-tail empirical distribution turns this into a measurable reserve-guidance threshold ($k^* = 3$ at $\tau = 0.95$, 0/24 sub-period stability rejections) — the operational handle no incumbent oracle publishes — and §10 names the candidate methodology fixes (full-distribution conformal calibration with cluster-conditional + path-fitted scoring; lower-priority follow-ups in `docs/ROADMAP.md`). They name the boundary of what a per-anchor-calibrated locally-weighted-Mondrian interval oracle can claim.

12 §A — Reproducibility appendix

This appendix consolidates everything needed to reproduce the §6 empirical results, the §7 ablation, and the §6.3 / §6.4 / §6.5 robustness diagnostics from the publicly-available source data plus the Soothsayer repo. The deployment is a five-line lookup against sixteen pre-fit scalars; reproducibility is determined by the panel construction, the schedule fits, and the serving formula — all of which are deterministic given the panel.

12.1 A.1 Algorithm boxes

We give pseudocode for three procedures: the M6 fit (deployment-artefact build), the M6 serve (runtime band lookup), and the strictly-pre-Friday per-symbol EWMA construction. Implementations: Python at `src/soothsayer/{backtest/calibration.py,oracle.py}`, Rust port at `crates/soothsayer-oracle/src/{config,oracle,types}.rs` (currently mirrors the M5 reference path; M6 wire-format slot reserved as `forecaster_code = 3`).

Algorithm 1 — M6 fit (one-shot, produces the 16 deployment scalars; `scripts/build_lwc_artefact.py`).

```
Input:  panel D = {(s, t, p_Fri, mon_open, r_F, r) : i = 1, ..., N}
        split date d_split = 2023-01-01
        anchor -grid T = {0.68, 0.85, 0.95, 0.99}
```

```
        c-grid C = {1.000, 1.001, ..., 5.000}
        EWMA half-life HL = 8 weekends
        warm-up minimum past_obs_min = 8

Output: sigma_hat_table  _s(t)         per-(symbol, fri_ts) parquet column
        quantile_table   q_r()         dim 3 × 4 = 12 scalars
        c_bump_schedule  c()           dim 4 scalars
        delta_shift_schedule  ()      dim 4 scalars (identically zero under M6)

1.  for each row i  D:
        p_i ← p_Fri,i · (1 + r_F,i)             # factor-adjusted point
        rrel_i ← (mon_open_i  p_i) / p_Fri,i   # signed relative residual
2.  for each (symbol s, weekend t)  D:
        prior ← {rrel_{i'} : symbol(i') = s, t' < t}    # strictly pre-Friday
        if |prior| < past_obs_min:
            mark row as warmup_drop and skip
        else:
            _s(t) ← EWMA(prior, half_life = HL)        # weekend half-life decay
3.  D_train ← {i  D : t_i < d_split, not warmup_drop}
    D_oos   ← {i  D : t_i  d_split, not warmup_drop}
4.  for each row i in D_train  D_oos:
        score_i ← |mon_open_i  p_i| / (p_Fri,i · _{s_i}(t_i))   # standardised
5.  for each (regime r, target )  {normal, long_weekend, high_vol} × T:
        S_r ← {score_i : i  D_train, r_i = r}
        k ←  · (|S_r| + 1)                      # finite-sample CP rank
        q_r() ← sort(S_r)[k]                      # the trained standardised quantile
6.  for each  T:
        c() ← argmin_{c  C} {c : (1/|D_oos|) · _{i  D_oos} 1{score_i  q_{r_i}() · c}  }
7.  () ← walk-forward sweep selects 0 at every anchor under M6 (per-symbol  tightens
            cross-split realised-coverage variance enough that no shift is required)
8.  return (_s(t), q_r(), c(), ())
```

Algorithm 2 — M6 serve (runtime band; `Oracle.fair_value_lwc` in `src/soothsayer/oracle.py`).

```
Input:  symbol s, as-of date t, target coverage   [0.68, 0.99]
        lookup row (p_Fri, regime r, _s(t)) from per-Friday parquet at (s, t)
        deployment scalars (q_r(), c(), ()  0) from Algorithm 1

Output: PricePoint{ , (), p, lower, upper, r, c, q_eff, q_r,  }

1.  c_eff ← interpolate c on the -grid at
2.  q_eff ← c_eff · q_r() · _s(t)         # standardised quantile rescaled by per-symbol
3.  p ← p_Fri · (1 + r_F)                      # factor-adjusted point
4.  lower ← p · (1  q_eff);   upper ← p · (1 + q_eff)
5.  return PricePoint(, () = 0, p, lower, upper, r,
                      diagnostics{c_eff, q_eff, q_r(), _s(t)})
```

Algorithm 3 — Per-symbol EWMA construction (`scripts/build_lwc_artefact.py` step 2; soft constants `SIGMA_HAT_HL_WEEKENDS = 8`, `SIGMA_HAT_MIN = 8`).

```
Input:  panel D, EWMA half-life HL = 8 weekends, past_obs_min = 8

Output: per-(symbol, fri_ts) _s(t) column on the artefact parquet

1.   ← 0.5 ** (1.0 / HL)        #  0.917 per past Friday
2.  for each symbol s in D:
        rows_s ← rows of D with symbol = s, sorted by fri_ts
        for each row i in rows_s:
            prior ← {rrel_{j} : j  rows_s, fri_ts(j) < fri_ts(i)}     # strictly pre-Friday
            if |prior| < past_obs_min:
```

```
            _s(fri_ts(i)) ← undefined  (row marked warmup_drop)
        else:
            weights ← [ ** (|prior|  k  1) for k in 0..|prior|]
            _s(fri_ts(i)) ← weighted_std(prior, weights)
3.  return  column
```

The rule itself was selected from a five-variant ladder under §7.3's pre-registered three-gate criterion; alternative variants remain buildable for archival reproduction via `scripts/build_lwc_artefact.py --variant {baseline_k26, ewma_hl6, ewma_hl12, blend_k26_ewma_hl8}`.

12.2 A.2 Hyperparameter specification

The M6 deployment artefact is fully specified by the 16 scalars below plus four "soft" pipeline constants and the rule constants. All values are in `data/processed/lwc_artefact_v1.json` (the audit-trail sidecar) and hard-coded in `src/soothsayer/oracle.py`.

Trained per-regime quantiles $q_r(\tau)$ on standardised residuals (12 scalars, fit on the 4,186-row pre-2023 train set after the 80-row warm-up drop):

regime (n)	$\tau = 0.68$	$\tau = 0.85$	$\tau = 0.95$	$\tau = 0.99$
normal	0.7767	1.2195	1.9682	3.3280
long_weekend	0.8638	1.3649	2.2208	4.0152
high_vol	1.1086	1.9433	3.0971	6.4563

Note these are *standardised* quantiles (dimensionless; multiplied at serve time by $\hat{\sigma}_s(t)$ to yield a relative-residual quantile in basis points). The high_vol $\tau = 0.99$ value of 6.46 — a 6.46- shock at the panel-relative scale — is the headline-tail shape M6 is calibrated to absorb.

OOS-fit $c(\tau)$ **multiplicative bumps** (4 scalars, fit on the 1,730-row 2023+ OOS via Algorithm 1 step 6):

τ	0.68	0.85	0.95	0.99
$c(\tau)$	1.000	1.000	1.079	1.003

Three of four near-identity; only $c(0.95) = 1.079$ carries meaningful OOS information. §9.3 carries the full provenance disclosure for that one scalar.

Walk-forward $\delta(\tau)$ **shifts** (4 scalars; identically zero under M6 — per-symbol standardisation tightens cross-split realised-coverage variance enough that no shift is required):

τ	0.68	0.85	0.95	0.99
$\delta(\tau)$	0.000	0.000	0.000	0.000

The δ schedule is retained as a 4-zero vector in the artefact JSON for shape-compatibility with the receipt schema.

Soft constants (pipeline shape; not OOS-tuned):

Constant	Value	Source
Train/OOS split anchor	2023-01-01	§6.1, sensitivity in §6.3.3
Conformity score	$\lvert y - \hat{p}\rvert / (p_{\text{Fri}} \cdot \hat{\sigma}_s)$	§4.3 (standardised)
Point estimator	$p_{\text{Fri}} \cdot (1 + r_t^F)$	§4.1 / §5.4 (factor switchboard)
Factor switchboard	per-symbol $\rho(s, t)$	§5.4, MSTR pivot 2020-08-01
Regime classifier	3-bin $\rho \in \{\text{normal}, \text{lw}, \text{hv}\}$	§5.5
`c-grid` resolution	$\Delta c = 0.001$	Algorithm 1 step 6
Off-grid τ interpolation	linear on the four anchors	§4.6

Constant	Value	Source
EWMA half-life	8 weekends	§4.2; `SIGMA_HAT_HL_WEEKENDS`
warm-up minimum past obs	8	§4.2; `SIGMA_HAT_MIN`

12.3 A.3 Data and code availability

Data. All upstream data is publicly available (Yahoo Finance daily bars, CME 1m futures, VIX/GVZ/MOVE, BTC-USD, Yahoo earnings calendar, Kraken Futures perp tape, Pyth Hermes archive, Chainlink Data Streams archive). Soothsayer consumes pre-fetched parquet from the sibling `scryer` repo at `SCRYER_DATASET_ROOT`. The decade-scale weekend panel `data/processed/v1b_panel.parquet` (5,996 rows × 35 columns, 2014-01-17 → 2026-04-24) is built by `scripts/run_calibration.py` calling `soothsayer.backtest.panel.build()`; the overnight panel `data/processed/overnight_panel.parquet` (22,624 rows, 2014-01-16 → 2026-04-23) is built by `scripts/build_overnight_panel.py` calling the same `build()` with `gap_mode="overnight"` plus `yahoo/earnings/v2` session timing and the `yahoo/corp_actions/v1` ex-dividend adjustment (§5.2.1). All `_fetched_at` cutoffs are preserved in the parquet metadata.

Repository. Code at `https://github.com/ACNoonan/soothsayer`. Tag the experimental snapshot at submission time; this paper's results were produced from the `main` branch as of 2026-05-05. License: see repository LICENSE file.

Layout.

Layer	Path	Purpose
Panel build	`src/soothsayer/backtest/panel.py`	weekend-pair assembly, regime tag, factor join
Calibration helpers	`src/soothsayer/backtest/calibration.py`	`compute_score`, `train_quantile_table`, `fit_c_bump_schedule`, `serve_bands`, `fit_split_conformal_forecaster`
Metrics	`src/soothsayer/backtest/metrics.py`	Kupiec, Christoffersen, Berkowitz, DQ, CRPS, PIT
Python serve	`src/soothsayer/oracle.py`	`Oracle.fair_value_lwc` (M6 path); `Oracle.fair_value` (M5 reference)
Rust serve (parity)	`crates/soothsayer-oracle/`	byte-for-byte both forecasters (`Forecaster::Mondrian`, `Forecaster::Lwc`); 180/180 parity vs Python
On-chain publisher	`programs/soothsayer-oracle-program/`	Anchor program, `PriceUpdate` Borsh wire
M6 artefact build	`scripts/build_lwc_artefact.py`	Algorithm 1
M6 freeze	`scripts/freeze_lwc_artefact.py`	content-addressed freeze for forward-tape eval
M5 artefact build (reference)	`scripts/build_mondrian_artefact.py`	§7.2 ablation rung
Cross-verification	`scripts/verify_rust_oracle.py`	180-case Python Rust parity probe (90 M5 + 90 M6 LWC)
Robustness battery	`scripts/run_v1b_*.py`, `scripts/run_*.py`	per-symbol diagnostics, vol-tertile, GARCH-N + GARCH-t, split sensitivity, LOSO, per-class, path-fitted
Phase 7 / 8 runners	`scripts/run_portfolio_clustering.py`, `scripts/run_subperiod_robustness.py`, `scripts/run_v1b_garch_baseline.py --dist t`, `scripts/run_per_symbol_kupiec_all_methods.py`, `scripts/run_kw_threshold_stability.py`	joint-tail clustering, sub-period stability, GARCH-t baseline, 4-method per-symbol grid, k_w threshold stability
Figures	`scripts/build_paper1_figures.py`	produces all 6 paper figures

12.4 A.4 Compute environment

Component	Version	Notes
Python	3.12.13	enforced via `pyproject.toml` `requires-python = ">=3.12,<3.13"`
Environment manager	uv	lockfile `uv.lock` checked in
Key deps (Python)	numpy 1.26, pandas 2.2, scipy 1.13, polars 0.20, pyarrow 16.0, statsmodels 0.14, arch (for §6.4.3 GARCH baselines)	full list in `pyproject.toml`
Rust	stable (1.81+)	matches Solana toolchain
Anchor	0.30.x	for the on-chain publisher
OS	macOS Darwin 25.4 (development); deployment is platform-agnostic	

12.5 A.5 Reproducing the paper's numerical content

The end-to-end pipeline from raw panel to all paper artefacts:

```
# 1. Build the weekend panel from scryer parquet (one-time, ~3 min).
uv run python scripts/run_calibration.py

# 2. Fit M6 and write the deployment artefact (Algorithm 1).
uv run python scripts/build_lwc_artefact.py
uv run python scripts/freeze_lwc_artefact.py       # content-addressed freeze for forward-tape

# 3. Verify Python  Rust serving parity on the M5 reference path (90/90 cases).
uv run python scripts/verify_rust_oracle.py

# 4. §6.5 path-coverage diagnostic (CME + perp + on-chain) + excursion-inflation .
uv run python scripts/run_path_coverage.py
uv run python scripts/proto_excursion_inflation.py             # §6.5 () endpoint-vs-path diagnostic

# 4b. §6.8 off-hours generalisation - overnight panel + calibration battery.
uv run python scripts/build_overnight_panel.py                 # overnight close→open panel (§5.2.1)
uv run python scripts/build_overnight_artefact.py              # overnight artefact + Kupiec/Christoffe

# 5. §6 / §7 robustness battery.
uv run python scripts/run_v1b_per_symbol_diagnostics.py
uv run python scripts/run_v1b_vol_tertile.py
uv run python scripts/run_v1b_garch_baseline.py                # GARCH-Gaussian baseline
uv run python scripts/run_v1b_garch_baseline.py --dist t       # GARCH-t baseline (§6.4.3)
uv run python scripts/run_v1b_split_sensitivity.py
uv run python scripts/run_v1b_loso.py
uv run python scripts/run_v1b_per_class.py
uv run python scripts/run_v1b_path_fitted_conformal.py
uv run python scripts/run_v1b_tokenized_tracking_baseline.py    # §7.6 tokenized-tracking baseline (pos

# 6. Phase 7 / 8 paper-strengthening runners.
uv run python scripts/run_portfolio_clustering.py              # §6.3.4 joint-tail empirical distributi
uv run python scripts/run_subperiod_robustness.py              # §6.3.3 calendar sub-period stability
uv run python scripts/run_per_symbol_kupiec_all_methods.py     # §6.4.2 4-method per-symbol grid
uv run python scripts/run_kw_threshold_stability.py            # §6.3.4 reserve-guidance threshold sta

# 7. Build all six figures.
uv run python scripts/build_paper1_figures.py
```

```
# 8. Compile this paper.
cd reports/paper1_coverage_inversion/build && uv run python build.py --pdf
```

12.6 A.6 Per-figure data provenance

Figure	Script	Source CSVs / parquet
Fig. 1 (pipeline)	`build_paper1_figures.py::fig1_pipeline`	none — diagrammatic
Fig. 2 (calibration)	`build_paper1_figures.py::fig2_calibration`	`data/processed/{v1b_panel.parquet, lwc_artefact_v1.parquet}`
Fig. 3 (stability)	`build_paper1_figures.py::fig3_stability`	`reports/tables/{m6_lwc_walkforward.csv, m6_lwc_robustness_split_sensitivity.csv}`
Fig. 4 (per-symbol)	`build_paper1_figures.py::fig4_per_symbol`	`reports/tables/m6_per_symbol_kupiec_4metho`
Fig. 5 (Pareto)	`build_paper1_figures.py::fig5_pareto`	`reports/tables/{m6_pooled_oos.csv, m6_lwc_robustness_garch_t_baseline.csv}`
Fig. 6 (path)	`build_paper1_figures.py::fig6_path_coverage`	`reports/tables/{path_coverage_perp.csv, path_coverage_perp_by_regime.csv}`
Simulation appendix figure	`scripts/run_simulation_study.py`	`reports/tables/sim_per_symbol_kupiec.csv`

12.7 A.7 Determinism and randomness

The M6 fit and serve paths are deterministic given the panel: no random initialisation, no Monte Carlo, no bootstrap inside the fit. The 4-split powered walk-forward (§6.3.3, §7.2.3) uses fixed split fractions {0.4, 0.5, 0.6, 0.7} (the 0.2 / 0.3 fractions are excluded as under-powered for the 4-scalar $c(\tau)$ fit; see §4.5); the four split-date sensitivity anchors (§6.3.3) are fixed at {2021-01-01, 2022-01-01, 2023-01-01, 2024-01-01}; LOSO (§6.3.3) iterates the ten symbols in lexicographic order. The block-bootstrap CIs reported in §6.3.4, §7.1.2, §7.3.5 use NumPy's default RNG with seed 0 over 1,000 weekend-block resamples. The simulation study (§6.7) uses NumPy's default RNG with seed 0 over 100 replications per DGP.

The GARCH(1,1) baselines (§6.4.3) are fit per-symbol via `arch_model(..., dist={"normal", "t"}).fit(disp="off")`; the optimisation is deterministic up to BFGS tolerance (default `tol=1e-8`). Under GARCH-t, NVDA hits the variance-undefined boundary at $\hat{\nu} = 2.50$ and falls back to Gaussian; the fallback is recorded in the `dist_used` column of the output CSV and is deterministic across reruns.

12.8 A.8 Independent verification (Python Rust on-chain)

The deployed serving stack has three independent implementations of both forecasters that must agree byte-for-byte:

1. **Python reference.** `src/soothsayer/oracle.py::Oracle.fair_value_lwc` (M6 deployed path); `Oracle.fair_value` (M5 reference path). Reads `data/processed/lwc_artefact_v1.parquet` (M6) or `data/processed/mondrian_artefact_v2.parquet` (M5) for the per-Friday `(symbol, fri_close, point, regime_pub, sigma_hat_sym_pre_fri)` tuple; the M6 16 scalars (M5 20 scalars) are hard-coded module constants.
2. **Rust serving crate.** `crates/soothsayer-oracle/src/oracle.rs::Oracle::fair_value` exposes both forecasters via `Oracle::load_with_forecaster(&path, Forecaster::{Mondrian, Lwc})`. The 16 LWC scalars and 20 Mondrian scalars are hard-coded in `config.rs`; the unit test `lwc_constants_match_sidecar` enforces 2 ULP agreement with `lwc_artefact_v1.json` on every refit.
3. **On-chain publisher.** `programs/soothsayer-oracle-program/`. The Rust crate is consumed by an Anchor program that emits `PriceUpdate` PDAs whose Borsh layout is preserved across the M5 → M6 migration. The wire `forecaster_code` byte distinguishes paths: code 2 = `FORECASTER_MONDRIAN` (M5; live on-chain); code 3 = `FORECASTER_LWC` (M6; live in the Rust publisher as of the §8.5 parity milestone, on-chain enablement gated on the next publisher release). The `soothsayer-consumer` no_std crate decodes the PDA into a typed `Forecaster::{Mondrian, Lwc}` view; consumers branching on `forecaster_code` need only add code-2 and code-3 cases.

`scripts/verify_rust_oracle.py` runs a dual-forecaster probe across 30 (symbol, fri_ts) cases × 3 target-coverage anchors per forecaster (90 cases per side, 180 total) and asserts byte-exact agreement on `(point,` lower, upper, sharpness_bps, half_width_bps, claimed_served) between Python and Rust on every case. **Submission status: 180/180 pass** (90/90 M5 + 90/90 M6). The Anchor integration test (programs/soothsayer-oracle-program/tests/) extends the parity check to path 3 by decoding the on-chain PDA after a publish call.

A reviewer reproducing the paper end-to-end thus has three independent implementations of each forecaster that must agree on every (symbol, fri_ts, target_coverage) read.

12.9 A.9 Within-bin exchangeability — permutation test

Mondrian-CP's finite-sample coverage guarantee depends on within-bin exchangeability of the conformity score. A permutation test on the lag-1 Pearson autocorrelation of the standardised score within each (regime × symbol) bin ($n = 19$–116 per bin, 30 bins, OOS 2023+; statistic computed in fri_ts order, null distribution from 5,000 within-bin shuffles per bin) finds $1/30$ bins nominally rejecting at $\alpha = 0.05$ and $0/30$ at $\alpha = 0.01$ — *under-rejecting* relative to the 5% / 1% expected under exchangeability. Aggregate KS test of per-bin p-values vs Uniform(0,1) is $D = 0.276$, $p = 0.017$, but the deviation is in the under-rejecting direction (too few low p-values, too many bins with $p > 0.5$) — consistent with exchangeability holding, not violating. The single nominally-rejecting bin (GLD/normal, lag-1 $\hat{\rho} = -0.166$) does not survive Benjamini-Hochberg correction across the 30-test grid. The §6.3.6 / §9.4 cross-sectional dependence is *across* bins, not within. Source: reports/tables/paper1_b2_exchangeability.csv.

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